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DEVELOPMENT OF MULTI-BRANCH DENTAL PRACTICE MANAGEMENT SYSTEM WITH POINT-OF-SALE AND BUSINESS ANALYTICS FOR DR. WENNIE A. SAMPER DENTAL CLINIC

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DEDICATION

This work is for those who have been with us throughout the development of this study.

First, to the parents and families who have been with us all the way and have been an ever-supportive presence. They not only provided all the necessary resources and facilities but also encouraged us and loved us when we needed it most. They taught us to dream big and go for our goals. Without them, nothing would have happened.

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ABSTRACT

Title: Development Of Multi-Branch Dental Practice Management System With Point-Of-Sale And Business Analytics For Dr. Wennie A. Samper Dental Clinic

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Keywords: Dental Practice Management, Multi-branch Systems, Point-of-Sale (POS), business Analytics, Electronic Medical Records, Healthcare Informatics, Clinic Workflow Automation

This study presents the development of a comprehensive, web-based multi-branch dental practice management system designed to modernize the operations of Dr. Wennie A. Samper Dental Clinic by replacing inefficient, error-prone manual processes with an integrated digital solution. Developed using the Agile Scrum methodology, the system centralizes nine core modules including Electronic Medical Records, automated appointment scheduling, Point-of-Sale (POS) billing, and a business analytics dashboard to streamline cross-branch coordination and data-driven decision-making. Rigorous evaluation via black box, usability, performance, and security testing yielded perfect scores, confirming that the platform is highly functional and secure for immediate deployment. By automating administrative workflows and providing real-time operational insights, the system effectively eliminates documentation loss and billing conflicts while establishing a foundation for future enhancements such as teleconsultation and predictive analytics.

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CHAPTER 1
INTRODUCTION

This chapter included the project context, purpose and description, objectives, significance of the project, scope and limitation, and a project dictionary and end notes. It provided an overview of this study, which focused on the Development of multi-branched dental practice management system with point-of-sale and business analytics for dr. Wennie a. Samper dental clinic.

1.1 Project Context

The care of dentistry has developed into more than just a repair of the caries to be important in the general health of the body, where it is related to the treatment of diseases such as cardiovascular disease and diabetes management [5]. Nonetheless, numerous dental facilities, particularly in the developing areas, were still using the old paper-based administrative practices [1]. Dr. The example of Wennie A. Samper Dental Clinic which was established in 2019 in the Sta. Justina branch and grew to Iriga and San Pascual in 2020 and 2023 illustrates this challenge wherein the clinic was expanding to more locations and had to deal with growing numbers of patients using manual processes. These paper processes were very inefficient with lost documents, missed appointments and billing mistakes taking place [4]. Many small dental clinics could not access affordable customized solutions because the service providers had not tried to implement electronic medical records as well as automated solutions before due to their size [1]. This caused efficiency hindrance through technology gap, augmented administrative burden, and finally revenue profitability [3]. To address these issues, the Dr. Wennie A. Samper Dental Clinic system wanted to introduce new ways to approach the dental practice management by consolidating patient records, billing, and business analytics into one user-friendly system. It was made easy to use just like any modern application and strong enough to handle various functions, including patient charting and financial reporting. The modern patients of the dentist demanded fast and effective services and professional assistance, which demanded powerful technological assistance [1]. The

Dr. Wennie A. Samper Dental Clinic system also helped to lessen the chances of error due to the creation of real-time patient files and automatic schedule synchronization, which led to the increased efficiency of the clinic.

The issue of financial management was still a major problem in most clinics [3]. The clinic owners were likely to take financial wrong steps without having a clear picture of their income and expenses. This was countered by the integrated Point-of-Sale (POS) system which was a financial dashboard that tracked the real-time transactions offering the greatest transparency [1].

Business analytics also contributed to the improved effectiveness of operations, as it offered actionable insights. The trends that could be detected in the clinics included the most frequent treatment methods, busiest times of the day, and the trends in annual revenue [3]. This information was accessible to the clinics, and this enabled them to make knowledgeable data-driven decisions as opposed to resorting to guesswork [3].

Effective communication was important in a smooth-running clinic. When lack of information sharing between the various departments caused inefficiencies: doctors could not access the records of patients, administrative employees had difficulties with scheduling conflicts, and receptionists could not properly address patient requests [5]. The Dr. Wennie A. Samper Dental Clinic system averted these problems, by harmonizing the records, appointments and billing into a single system that will enable all staff members to interact with one another in a smooth manner [1][3].

The issue of security was also prominent. Paper records were susceptible to losses, copying and unauthorized access and there were other digital systems that were inadequate in the control of sensitive patient information [4]. To deal with this, the Dr. Wennie A. Samper Dental Clinic system introduced enterprise-level security measures such as a strong authentication system, role-based access control, to ensure that data protection practices and patient confidentiality are observed [1][3].

Many people were now anticipated to make payments using digital methods like mobile wallets and credit cards, and no longer necessarily using cash [1]. The combined POS system enabled free flows of transaction and automated financial calculations whereby the clinic owners had clear and precise financial details without need to create advanced accounting skills [3].

Finally, patient satisfaction was not merely an act of delivering dental treatment, but it was the whole experience. Patient dissatisfaction may come up because of long wait time, scheduling problems and ineffective service delivery [1]. The Dr. Wennie A. Samper Dental Clinic system enabled dental professionals to concentrate on the real issue, which is to provide the patient with the highest level of care and establish the long-term relationships [5].

1.2 Purpose and Description of the Project

The main aim of the project was the creation of the Multi-Branched Dental Practice Management System with the Point-of-Sale and Business Analytics to Dr. Wennie A. Samper Dental Clinic to completely transform the system of the clinic. This new system would see the paper-based administrative processes which were only fragmented to be replaced with a comprehensive and smart digital platform that was able to effectively bring together key functions like patient management, billing, business analytics and internal coordination into one user friendly solution. It touched upon the major problems of the past like record management by hand, late billing, schedule clashes, inadequate business insights in the form of manual data processing and handling by human hand and boosted the operational effectiveness, reduced human errors and improved patient service through Electronic Medical Records (EMR) and Electronic Dental Chart (EDC) to ensure that the past history and treatments are safely and instantly tracked. The next addition of the system features was the POS module that made the billing easier, offered the possibility of many payments methods processing, and provided the correct financial reporting functions that significantly assisted in avoiding delays and collections leakages.

At the same time, the Business Analytics Dashboard provided the administrators with actionable data regarding the demographic and income parameters to allow them strategic decision making in accordance with the goals of the clinics. Its features of security and reliability were authentication, encryption, and role-based access control that guaranteed privacy and compliance of data, and automated notification of appointments and follow-ups enhanced communication and effectively lowered no-show rates in patients. The system's comprehensive reporting capabilities enabled real-time monitoring of key performance indicators across all clinic branches, facilitating immediate identification of operational bottlenecks and revenue opportunities while its intuitive design reduced staff training time significantly and its scalable architecture ensured seamless accommodation of future growth and technological advancements.

The uniqueness of the system is that it was able to develop one, fully integrated digital platform which has radically changed how Dr. Wennie A. Samper Dental Clinic operates because it has replaced fragmented and paper-based administrative systems. This solution is more than mere automation it is a one-of-its-kind multi-branched practice management, Electronic Medical Records (EMR), point-of-sale billing, and an extensive business analytics system in one easy-to-use interface. This type of integration does not only mean no longer having to deal with manual errors and scheduling problems, but also timely and practical data regarding revenue and service utilization to administrators that makes the clinic a technology-savvy institution that has set the stage of strategic digital implementation in the field of dental healthcare.

1.3 Objectives of the Project

Generally, this study aimed to develop and implement a Multi-Branch Dental Practice Management System with Point-of-Sale and Business Analytics for Dr. Wennie A. Samper Dental Clinic. Specifically, this study aimed to achieve the following:

1. Identify the challenges encountered in managing dental clinic operations, particularly in:

    1.1 Patient Registration and Record Management

1.2 Appointment Scheduling and Treatment History Tracking

1.3 Billing, Invoicing, and Payment Processing

1.4 Inventory and Supply Management

1.5 Communication and Workflow Coordination

1.6 Monitoring Clinic Performance and Revenue Trends

2. Develop a web-based platform with the following core modules:

   2.1 Patient Management Module

   2.2 Electronic Dental Module

   2.3 Appointment Scheduling Module

   2.4 Billing and Point-of-Sale (POS) Module

   2.5 Inventory Management Module

   2.6 Business Analytics and Reporting Module

   2.7 Role-Based Access and Security Module

   2.8 Communication and Workflow Module

   2.9 System Administration Module

3. Conduct a black box testing by executing test cases based on system requirements using the following criteria:

   3.1 Usability Testing

   3.2 Performance Testing

   3.3 Security Testing

   3.4 Security Testing

4. Deployed the system in a live clinical environment to ensure full functionality and usability for both dental practitioners and administrative staff.

1.4 Significance of the Study

The results of this study were deemed significant to the following:

Clinic Owners and Managers. The present research helped in assisting Dr. Wennie A. Samper Dental Clinic and other owners of dental clinics through automating patient registration, billing, and scheduling appointments and incorporating business analytics to make decisions. Business analytics module was useful in monitoring the trend on revenues, monitoring service performance, and determining growth opportunities, and real-time reporting and visual dashboards assist in making informed decisions and strategic planning. The system minimized human error, manual workload, and streamlined service delivery resulting in a more efficient clinic management and better patient care due to automation of main processes and built in business analytics to foster informed strategic planning.

Administrative Staff. The system with consolidated appointment calendar, automated invoicing, and role-based access facilitated the work of clinic receptionists and billing personnel in a smoother way. It minimized the presence of redundancy and enabled them to concentrate on serving quality services to the patients. Consolidated scheduling, automated invoicing, and controlled access provide the system with a smoother workflow, reduced redundancy, and higher emphasis on providing quality services to patients.

Patients. The patients also had the advantage of having a user-friendly portal, which enabled them to access their schedule of appointments, their payment history, and their personal records. Communication was also enhanced by use of reminders and real-time updates, which increased satisfaction and transparency in the system. The patients will gain by having better satisfaction, transparency concerning their corresponding schedules, payment history through a highly user-friendly portal, and efficient interaction in form of reminders.

Dental Practitioners. Rapid and systematic access to electronic medical record (EMR), treatment history, and appointment setups were beneficial to dentists. The system assisted them to perform clinical activities more effectively, lessened paperwork, and made proper record keeping of patient care and follow-ups. The system assists the dental practitioners to better handle the clinical workload, and the accuracy of the work is guaranteed by quick access to the structured electronic medical records and listings of appointments by schedule.

Healthcare Providers and Private Clinics. The dental and other medical clinics may use or modify the system developed during this study to improve their functions. The modular design enabled a flexible approach to customization on a variety of clinic workflows, which is why the model is implemented in digital healthcare management. The other clinics can consider or modify the pragmatic, adaptable quality of the model to digital healthcare management to enhance the operation further.

Researchers. This research was significant to the scientists because it brought information about the establishment of integrated dental practice systems. This study allowed seeing how technology can simplify the work of a dental clinic by analyzing the application of such features as patient management, billing automation, and business analytics to enhance the clinic and provide better care to patients. This work may lead researchers to consider how to maximize these systems and increase their effects on dental practice management to give knowledge about how technology can streamline the operations of the dental clinics and how it can improve patient care by researching on the application of the major integrated practice features.

Future Researchers. This study can be the basis of the innovation of dental practice management systems in the future. They may consider how to incorporate artificial intelligence, mobile, and cloud-based storage in order to have more scalable and intelligent clinical solutions. The current research will serve as a foundation of the further, more intelligent clinic solution research based on the implementation of the latest technologies, including the use of artificial intelligence, mobile platforms, and cloud-based storage.

1.5 Scope and Limitations of the Project

This study focused on developing a web-based Integrated Dental Practice System with Point-of-Sale (POS) Integration, Inventory Tracking, Communication Workflow, and Business

Analytics for the Dr. Wennie A. Samper Dental Clinic. The research addressed common challenges in dental clinic operations by identifying issues in patient registration and record management, appointment scheduling and treatment history tracking, billing and payment processing, inventory and supply management, communication and workflow coordination, and monitoring clinic performance and revenue trends. The study encompassed the development and implementation of nine core modules: Patient Management for handling electronic dental records (EDR), Electronic Dental Charting for tooth-specific treatment logging, Appointment Scheduling, Billing and POS for real-time invoice processing, Inventory Management for supplies and medicines, Business Analytics for tracking clinic performance, Role-Based Access and Security

Module, Communication and Workflow Module for task coordination, and System Administration for managing users, settings, and data backups. The system supported multi-branch operations, was accessible through desktop or laptop browsers, and underwent comprehensive black box testing including usability testing, performance testing, security testing, and unit testing based on system requirements.

The implementation was limited to the internal operations and workflow of Dr. Wennie A. Samper Dental Clinic and did not support integration with external platforms such as health insurance systems, national health databases, or third-party payment gateways beyond basic POS functionality. The business analytics were based solely on internal clinic data and did not include

AI-driven forecasting, machine learning algorithms, or external benchmarking capabilities. The Electronic Dental Chart utilized a tabular or form-based interface instead of an interactive 3D dental model or advanced visualization tools. Appointment scheduling did not include real-time patient queueing systems, SMS confirmation features, or integration with external calendar applications. The system did not include a dedicated mobile application, and all data entries (such as patient profiles, treatment history, billing records) required manual input by clinic staff rather than automated syncing from legacy systems or external sources. The study did not encompass insurance claim processing, AI-assisted treatment planning, telemedicine capabilities, or integration with dental equipment for automated data capture. Multi-language support, advanced reporting beyond basic analytics, and cloud-based deployment were outside the current project scope. The research was conducted within a specific timeframe and geographical location, focusing solely on one dental clinic, which may limit the generalizability of findings to other dental practice settings or healthcare environments.

1.6 Project Dictionary

The following were the technical terms relevant to this study, each defined both conceptually and operationally to ensure clear communication between the researchers and readers:

Appointment Scheduling. A process of creating, organizing, and managing appointments to optimize time and resource utilization [3]. In this study, it referred to the system module that enabled clinic staff to create, reschedule, and monitor patient appointments to minimize scheduling conflicts and errors.

Audit Log. A chronological record of system activities used to support monitoring, accountability, and compliance [1][3]. In this study, it referred to the system feature that automatically recorded user actions such as logins, record updates, and transaction activities for security and traceability purposes.

Authentication. A security process used to verify the identity of a user before granting system access [1][5]. In this study, it referred to the login mechanism that validated authorized users (dentists, receptionists, and administrators) through secure credentials and access protocols.

Billing. A system function that manages the generation and processing of financial charges and payments [1][3]. In this study, it referred to the module responsible for generating patient invoices, processing payments, and recording financial transactions within the clinic system.

Business Analytics. The use of data analysis techniques to evaluate organizational performance and support decision-making [3]. In this study, it referred to the system tools that generated reports on revenue, patient trends, and treatment statistics to support administrative and managerial decisions.

Data Privacy. The practice of safeguarding personal and sensitive information from unauthorized access or misuse [1][4]. In this study, it referred to the system's implementation of policies and controls to protect patient and clinic data in compliance with healthcare data protection regulations.

Database. An organized electronic repository designed for efficient storage, retrieval, and management of structured data [3]. In this study, it referred to the centralized storage system used to maintain patient records, treatment histories, appointment schedules, and billing information.

Dental EMR (Electronic Medical Record). A digital system used to document and manage patient medical and dental information [5]. In this study, it referred to the electronic module that stored patient histories, diagnoses, treatment records, and related clinical data in place of paper-based records.

Electronic Dental Chart. A digital chart that records tooth-specific procedures and clinical observations [5]. In this study, it referred to the system feature that allowed dentists to document procedures, treatment dates, and notes for individual teeth in an electronic format.

Encryption. A security technique that converts data into a coded format to prevent unauthorized access [1][5]. In this study, it referred to the method used to protect stored and transmitted patient and financial data through secure encoding mechanisms.

Performance Testing. A type of software testing that evaluates system speed, stability, and responsiveness under varying workloads [6]. In this study, it referred to testing conducted to verify that the system could handle multiple users and large data volumes without performance degradation.

Point-of-Sale (POS). A system used to process payments and complete sales transactions [1][3]. In this study, it referred to the integrated payment module that enabled clinic staff to accept payments, generate receipts, and record transactions linked to patient records.

Role-Based Access Control (RBAC). A security approach that limits system access according to predefined user roles [1][3]. In this study, it referred to the mechanism that assigned specific permissions to users based on their roles, ensuring controlled access to sensitive clinic data.

Security Testing. A software testing process used to identify vulnerabilities and assess system defenses against threats [6]. In this study, it referred to testing procedures conducted to verify the effectiveness of authentication, encryption, and access control features.

Self-Service Portal. An online interface that allows users to independently access services and information [1][5]. In this study, it referred to the patient-facing module that enabled users to view appointments, medical records, and billing details without staff assistance.

Usability Testing. A testing method that evaluates how easily users can learn and use a system [6]. In this study, it referred to assessing whether clinic staff and patients could navigate and operate the system efficiently with minimal training.

Unit Testing. A software testing technique that examines individual components of a system in isolation [6]. In this study, it referred to testing performed on each functional module, such as appointment scheduling and patient registration, to ensure accurate and reliable operation prior to system integration.

Notes

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CHAPTER 2
REVIEW OF RELATED LITERATURE AND SYSTEMS

This chapter discussed the analysis of the literature related to the study. It included a summary of local and foreign literature and systems, as well as the Synthesis of the Art, Gap Bridged by the Study, and Notes. The discussion was based on the information gathered from books, articles, software and other research materials.

2.1 Related Literature and Systems

The literature review that was applicable to the current study was presented in the following article. It encompassed both foreign and local literature regarding systems which furnished required information regarding the utilization of an Integrated Dental Practice System with Point-of-Sale and Business Analytics. The review assisted the researchers and the reasons why the researchers comprehend the motivation of their proposed system and the significance of researching the existing technologies.

2.1.1 Integrating an Automated POS System for Health Clinic

The incorporation of the Point-of-Sale (POS) systems and business analytics tools within the dental practices have changed the way the clinics conducted their financial transactions, patient records and revenue cycles. Most dental facilities were moving to automated payment systems and Data-driven decision-making tools to make the operation smoother, reduce errors, and improve patient experience. Few foreign and local studies had been conducted to determine the effect of these technologies on clinic efficiency, financial management and compliance with the regulations. The literature reviews below provided a detailed discussion of these studies and showcased how they would be applicable to the creation of an Integrated Dental Practice System with POS and Business Analytics.

In their article, The Necessity and Problems of Promoting POS in Dentistry, Smith and Nguyen [2023] investigated the influence of modern POS systems on increasing the level of accuracy in transactions and financing of dental facilities and how it can be used to track the financial performance of a clinic. They stressed that conventional cash based operations and manual accounts were the main causes of financial mismatch and business inefficiencies and could be eliminated by adopting digital POS systems. In their study, they discovered that contactless payment systems, mobile payments and e-invoicing had a great effect of shortening waiting time and increasing patient satisfaction [1].

In a business intelligence study, Kumar and Lee [2023] conducted authentic research to understand the role of POS-based business intelligence as a tool in business intelligence to enhance the sales monitoring and revenue forecasting at a dental clinic in the research titled Data Mart in Business Intelligence with Hefesto. They emphasized that automation of revenue tracking enabled clinics to uncover fraud, enhance transparency and financial decision-making. The study found that real time sales monitoring and predictive analytics were very beneficial to dental clinics as it allowed them to control their cash flow [2].

Artificial intelligence (AI) was also making a revolution in the current POS and business analytics systems. In their study article, Johnson and Williams, [2022] called it Digital Transform of the retail Point of Sale in the artificial Intelligence Era and they discussed the use of AI in dental POS technology. They talked about how AI-based POS systems automated billing, sorted transactions and forecasted revenues. In their research, they revealed that AI-based algorithms that detect fraudulent transactions could indicate suspicious activities, which could protect money and save under the rules and regulations [3]. In their study, How POS Technology Improves Financial Transactions in Aesthetic Dental Clinics, Miller and Davis [2023] targeted cosmetic dentistry clinics and the use of further developed POS in their facilities. They have highlighted that automated invoices, installment payment plans, and insurance integration played a great role in improving customer satisfaction and clinic revenue. Their research has found out that providing multi-channel payment systems like credit cards, digital wallets, and installment plans enhanced retention and operational efficiency among patients [4].

In addition to financial management, dental POS systems business analytics tools were very important in maintaining regulatory compliance and security of data. In their article Big Data and Digitalization in Dentistry: Ethical Issues and Regulatory Challenges, Garcia and Rodriguez [2022] talked about the ramifications of gathering and examining the data of patient transactions. In their study, they indicated that dental practices with POS-integrated systems of business analytics were required to have data security controls to follow the HIPAA (Health Insurance Portability and Accountability Act) and GDPR (General Data Protection Regulation) in Europe. They emphasized that patient financial and treatment records might become vulnerable to information leaks and privacy attacks unless these documents were adequately encrypted and controlled access to them was provided [5]. The significance of incorporation of safe, artificial intelligence based and predictive POS systems in contemporary dental centers was emphasized in these studies.

On the ground, scholars had investigated the transition of Philippine dental practices away towards manual record keeping and cash dealings to digital payment systems. In the article of their study, Santos and Cruz [2023] present the analysis of the increasing utilization of online payments and POS systems in Philippine dental practices: Philline Dental Services Market Outlook to 2026. Their study reported that clinics that had installed POS-integrated financial track tools experienced a percentage of 15- 20 percent improvement in patient retention rates since digital payments were convenient to patients and personnel [6]. Likewise, Reyes and Bautista [2023] came up with a web-based system of patient management at Seraphic dental clinic and incorporated features of POS, including automated billing, creation of invoices, and utilization of payment systems. In their article, Automated Dental Management System with POS Integration, it is established that the manual records keeping and cash basis caused financial inefficiencies and errors that were successfully removed by the POS-integrated system [7].

Moreover, several local researches had pointed out the importance of business analytics in POS facilitated dental clinics. In the article, which they called Utilizing Payment Analytics in Dental POS Systems, Garcia and Santos [2023] examined the role of real-time tracking of transactions and automated revenue reporting as far as cash flow management was concerned. Their research focused on the fact that predictive analytics allowed the clinic owners to anticipate the revenue trends in a month, which results in more appropriate financial planning. The researchers established that decision making based on data played an important role in running the operation of dental clinics, minimizing billing discrepancies, and maximizing profitability [8].

Balderas et al. (2014) have designed a Dental Information System and Scheduling SMS Technology Sorongon Dental Clinic. The aim of the system was to automate the day-to-day operations such as scheduling, patient record and payment processing. The use of SMS appointments and balance reminders helped the clinic to enhance patient interaction and decrease the number of no shows. The experiment proved that the automation of these processes resulted in more consistent reporting and the smooth running of the clinic activities [9].

Chua, Padrino, and Receno (2012) analysed Averia Dental Clinic in terms of system analysis. They traced the causes of operational inefficiencies, including delays in services and re-work expenses, which cost $93,550 per year. The clinic minimized the number of rework incidences and enhanced service delivery through adoption of standard procedures and service checklists. This research paper has placed significance of operational auditing and standard protocols in improving clinic operations and profitability [10].

Abregana et al. (2018) created a LAN-Based Dental Information Management System within Sevallos Dental Clinic. The system included patient information modules, dentist data module, inventory management module and accounting. It was easy to retrieve patient records, provided smooth updates and came up with several reports, such as income statements and treatment plans. The significance of this integration of information management systems highlighted the importance of technology to enhance the accessibility and decision-making of information in the dental practice [11]. The study by Calalo, Dy, and Riego (2017) was a system analysis of CMC Dental Clinic, operational processes, inventory, and appointment scheduling.

They established that cancelling of appointments resulted in a lot of revenue loss, with the opportunity cost estimated at $18,250 in one year. The clinic responded to the problem of low operation efficiency and a high rate of cancellations by adopting regular maintenance schedules, purchase order deadlines, and an increase in inventory control with the 5S housekeeping system. This paper has highlighted how operational structured operational strategies have affected financial performance within dental clinics [12].

The article by Mangmang (2018) examined the implementation and the creation of a POS system that included profitability measurements capabilities in retail business, which can be implemented to dental clinics. The system also had the features of notification of overdue payments and out of stock which made the financial monitoring and inventory control easier. The system enhanced operational efficiency and financial control and was developed with the use of Microsoft Visual Studio and SQL Server. This paper proved the role of POS systems with built-in analytics that may help to make informed decisions and improve operations in dental practices [13].

Altogether, it can be concluded that the adoption of both POS systems and business analytic tools had become the necessary measure towards digitalization and financial performance. Both external and internal research focused on the need to utilize automated payment processing, revenue prediction, AIdriven scam alert, and regulatory conformity in the contemporary clinic administration. With further development of technologies, the introduction of the wholly-integrated dental POS system with business, analytics capabilities would play a critical role in the improvement of patient experience, better management of the revenue cycle, and optimal clinic functioning. The present literature review has presented solid evidence as to the creation of such by showing its practical advantage, economic implications and future in the dental industry.

2.1.2 Digital Evaluation and Judging Systems in Campus Events

Jarvis Analytics made a major transition to data-driven management in dental practices when Henry Schein, a healthcare products and services provider, bought 80 percent of the company in 2021. Tools developed by Jarvis Analytics were aimed at assisting dental professionals to compare clinical and financial data to enhance decision-making in different areas of operation. These business intelligence tools were integrated, and practices were able to maintain performance metrics, including retention rates of patients, revenue trends, and success rates of treatments. As per research, this purchase demonstrated the way of embracing business analytics to achieve increased efficiency and profitability in dental practice [14].

An article published in the International Journal of Environmental Research and Public Health by Nguyen et al. [2020] talks about the application of big data in enhancing the accuracy of oral health care. Dental professionals would be able to give individualized care using the real-time data insights available after using business analytics tools to analyze large datasets, including patient demographics and clinical records. The study highlighted that this information-driven method did not only help improve patient care but also allowed dental practices to anticipate health trends and detect treatment gaps and subsequent enhancement of general patient care. With the use of big data analytics, dental practices would be able to shift their focus towards a more proactive oral health approach [15].

John C. Davis [2020] emphasized the importance of data analytics in the management of the financial metrics in the dental practice. Davis held that dental practices could maximize their financial performance by tracking patient billing, types of treatment, and costs of the operation through the use of analytics software. The study noted that tracking important financial measures enabled the practices to minimize inefficiencies, enhance cash flows, and maximize profitability. Moreover, with the insights, it was possible to facilitate practices to ease operations, enhance patient satisfaction, and make better decisions regarding resource allocation and staffing. This research highlighted the need to embrace the use of data-driven solutions in dental offices in a bid to achieve financial stability and growth [16].

IntelePeer [2020] discussed the use of predictive analytics in dental practices. The predictive analytics were based on using past data about the patient to predict what will happen in the future, including patient reservation rate and staff requirement. Dental practices can ensure the optimal use of staff by predicting the busy hours and the type of patients they will see, thus improving the waiting time of patients and the general efficiency of the operation. The research determined that predictive analytics were effective in improving the patient experience, as well as the service delivery of the clinic itself as it could better plan in terms of surging demand and resource usage [17].

Adit [2020] outlined how practice analytics software is used to identify the effectiveness of dental practice. Adit notes that some of the key performance indicators (KPIs) followed by data analytics tools included patient demographics, usage of treatment, and financial performance to offer useful information about the health of a practice. These insights allowed practices to track development, enhance patient retention, and simplify operations. Through regular assessment of these KPIs the practitioners were likely to notice those areas that needed to be improved, to suit the business strategies with patient needs and to benefit the business as much as possible. The paper has shown the way business analytics can be utilized to make strategic decisions and contribute to more efficient and effective dental practice management [18].

The article by mConsent (2019) discussed the application of analytics software to open up opportunities in the growth of their dental practices. The authors elaborated how business analytics tools had the potential to create a competitive advantage as they allowed practice to monitor the trend in treatment, the level of patient satisfaction and financial performance. The analysis of these data would help the dental practices to get a better sense of the needs of patients, enhance service delivery, and manage practices. This strategy contributed to the expansion of dental offices due to the use of quality decisions according to analytic data, which guaranteed the requirements of patient care and the achievement of operational objectives. The researchers concluded that dental practices that capitalized on business analytics were set in a better position to succeed in the competitive business environment [19].

DSO Pro [2020] offered an in-depth manual on the application of business analytics in the dental practice. The study shows that dental offices might be able to use data analytics to enhance operational efficiency and patient outcomes. Business analytics assisted the practices in streamlining appointment booking, tracking the treatment progress, and financial performance.

Dental practices might save time and money through the identification of inefficiencies and the management of workflow by analytics to improve care delivery to patients. This paper has highlighted that the adoption of business analytics tools was necessary to improve the performance and sustainability of the contemporary dental practices [20].

2.1.3 Integrating Communication Workflow in Health Services

Efficient Effective message delivery and simplified processes were imperative to the improvement of healthcare delivery in the current Hospital Management Systems (HMS). Alyami et al. [2023] and the American Hospital Association [2012] pointed out that poor communication may slow the treatment of patients and disrupt the clinical processes [24]. They emphasized the importance of integrating communication systems such as Cerner CareAware Connect, in a bid to enhance the flow of information and efficiency in the operations [25]. Backline Health [2023] also emphasized that the workflow of communication not only enhanced the care given to patients but also increased the efficiency of the workflow, which benefited clinicians and their environment [26]. This was backed by Stryker [2023], who introduced solutions that merged communication and workflows, which combined care teams, patients, and their families to improve healthcare experiences [27].

The Hospital Management System project by GeeksforGeeks [2024] was one of the real-life examples of application of these principles by incorporating communication and workflow modules to optimize performance and assist in making the right decisions at the right place at the right time [28]. Collectively, these papers have emphasized the importance of communication and workflow optimization in enhancing patient outcomes and hospital performance overall.

2.1.4 System Testing using Black Box Testing

Black Box Testing proved to be good in practical situations to test the functionality and reliability of the digital health platforms. This testing method was also applied in research by Yu et al. [9], where the iPICS (Interactive Prostate Cancer Information, Communication, and Support Program), which is a digital health system aimed at assisting patients and caregivers with prostate cancer, was being evaluated. The research revealed that the system was found to be mostly in compliance with its design needs enabling the users to easily search, bookmark and share contents. Nonetheless, there were certain differences in precision with integration of external material, which represent the scope of future development of search algorithms and the usability of such systems by older users. This study has highlighted the role of constant appraisal in the creation of digital health systems so that they can address the needs of users and be of high quality. On the same note, Alvaro et al. [10] used the Black Box Testing to a mini-Enterprise Resources Planning (ERP) system with emphasis on identifying bugs and their reduction. The results showed that the testing strategy proved to be effective in both the detection and prevention of software defects, which helps to improve the system reliability and its performance. Ayuningtyas et al. [11] also conducted a study in which they examined the Black Box Testing with the use of both performance and functional testing. They used systematic methods of testing in order to cover and provide quality assurance. This style helped in the process of making software development more efficient in terms of testing.

Taken together, these studies revealed the possible practical uses of Black Box Testing in different digital health platforms and software systems, as well as the importance of Black Box

Testing in that context in terms of functionality, reliability, and user satisfaction. In addition to functional testing, there is extensive system evaluation that involves a number of testing dimensions that are very important to dental clinic management systems. Unit testing puts the emphasis on the most minute aspect of the software design the unit or the software component upon which, following the code development, code review and verification of the code to correspond to component level design, unit testing case design commences. The usability tests examine the ease of use and technology acceptance of the application, and the researchers start the process by providing information to the participants about the features of the application and the purpose of the study. Performance testing test is used to measure system responsiveness, easy entry, and the availability of the necessary functionalities that are important in the effective practice of clinical activities. Security testing is part of the vulnerabilities like buffer overflows that have been used by malicious software, and the current frameworks offer common security models to all the applications to protect the data and maintain system integrity that the system largely adhered to its design requirements, allowing users to efficiently search, bookmark, and share content. However, some precision disparities were noted when integrating external content, indicating areas for improvement in search algorithms and usability for older users. This research underscored the importance of continuous evaluation in developing digital health systems to ensure they met user needs and maintained high-quality standards. Similarly, Alvaro et al. [10] applied Black Box Testing to a mini–Enterprise Resources Planning (ERP) system, focusing on bug detection and minimization. The findings demonstrated that the testing strategy was effective in identifying and reducing software defects, thereby enhancing system reliability and performance. In another study, Ayuningtyas et al. [11] explored the use of Black Box Testing with performance and functional testing methods. They applied systematic testing approaches to ensure comprehensive coverage and quality assurance. This approach contributed to more efficient testing processes in software development.

Collectively, these studies demonstrated the practical applications of Black Box Testing in various digital health platforms and software systems, emphasizing its role in ensuring functionality, reliability, and user satisfaction. Beyond basic functional testing, comprehensive system evaluation encompasses multiple testing dimensions critical to dental clinic management systems. Unit testing focuses effort on the smallest part of software design, the module or software component where after code development, review, and verification for correspondence to component level design, unit testing case design begins. Usability testing evaluates the application's ease of use and technology acceptance, where researchers begin by explaining features and study objectives to participants. Performance testing assesses system responsiveness, ease of data entry, and the presence of essential functionalities critical for effective clinical practice. Security testing addresses vulnerabilities such as buffer overflows which have been exploited by malicious software, with modern frameworks providing common security models for all applications to ensure data protection and system integrity.

2.2 Synthesis of the State-of-the-Art

The literature synthesis indicated significance of the reviewed studies in contemporary research. It linked the essential aspects of the research issues, thus broadening the horizon of the researchers to the context and the purpose of their research. The synthesis was to improve the knowledge of past research, not repeating it, but discussing similarities and differences in different topics of research. This also involved the evolution of an Integrated Dental Practice System comprising of Point-of-Sale and Business Analytics which involved both operational and analytical facets.

The latest developments in the medical field of dental care proved to bring clinical innovation and business intelligence together, which is based on the prior research that was mentioned in our earlier examination. This analysis of fundamental system architecture overall practice management solution was an indication of an increasing awareness of the combined healthcare provision requirements. As an example, Wu et al. laid the groundwork with cloud-based system architecture, and Lee et al. concentrated on business intelligence integration. This tendency has been, however, more recently demonstrated by the study of Chen et al. [2023] on mobile health monitoring and the creation of platforms of patient engagement by Hall [2021]. All these studies together indicated convergence of different streams of research to formulate overall solutions in the management of contemporary dental practice.

One of the trends was the involvement of different elements of operation. As an illustration, Kim secured protocols of payment processing, such as those developed, formed the basis of financial management systems developed by Taylor et al. [2021] subsequently to full financial analytics platforms. This development showed the isolation of functional domain coming together into integrated systems as continued by the Brown et al. [2022] supply chain optimization study and the financial management solutions to small practices by Torres et al. [2022].

Analytical abilities were also advanced, and there was the development of specific sophistication. The breakthrough of machine learning applications that provide the foundation of Singh et al. [2022] is grounded.

The data mining research of Martin et al. [2023]. This piece of work was practically used in the clinical decision support systems described by Gonzales et al. [2023] and business analytics implementation by Lopez et al. [2021].

One of the significant factors of system development was the cultural and regional adaptation. Indicatively, Santos et al.'s [2022] investigation into adoption patterns in informed Garcia et al.'s [2021] larger investigation of business intelligence adoption in the Philippine healthcare. Such regional modifications in turn led to the creation of specialized clinical information systems by Rivera et al. [2023], showing that global technologies needed to be localized in order to be effective in a region. The same tendency of adaptation was supported by Sanchez et al. [2022] mobile health applications and Hernandez et al. [2022] patient record management systems that demonstrated how general principles were changed to fit the concrete market situation. One of the main improvements which have been made in contemporary literature is the trend toward integrated system architecture, instead of considering POS systems and analytics as distinct units that need to be integrated afterward, more modern solutions promote the creation of these two components as inseparable parts of a single system.

Some of the key design principles that came out of this synthesis were user centric integration that did not add complexity to the system, data architecture that provided seamless integration between clinical, operational and financial areas, and implementation strategies that took into consideration the differences in technical expertise among dental staff. The synthesis showed that there are several fruitful paths to develop in the future such as analytics capacity development due to the introduction of artificial intelligence, patient outcome prediction, and the development of security processes. The overlapping of multiple bodies of research proposed a future in which dental practice would exist within highly advanced technological ecosystems that in some way comfortably integrated clinical services, business processes, and analytical knowledge, with the focus on regionalism and cultural sensitivity suggesting a dawning realization of how universal technologies are being decidedly contextualized to maximize their efficiency in a range of healthcare practice settings.

2.3 Gap Bridged by the Study

The available literature reviewed the application of Point-of-Sale (POS) systems in enhancing the accuracy of transactions and financial tracking in dental and healthcare facilities. The role of POS systems in providing good financial records was discussed by Nakano [1], and Dominguez et al. [2] wrote about the revenue tracking using the Hefesto methodology and POS-integrated business intelligence. Nonetheless, these works focused predominantly on sales and financial reporting and did not investigate the way POS systems may be integrated with the wider business analytics to assist in patient management, monitoring, and operational decision-making in dental clinics.

To fill this gap, the current research combined POS functionality and business analytics to offer real-time information on financial performance, patients services, inventory use, and the overall clinic operation using a centralized dental practice administration system.

The advantages of the digital transformation and POS technologies in the retail and aesthetic dental clinics were also noted in the existing literature by Santos and Bacalhau [3] and Miller and Davis [4], especially with respect to enhancing the efficiency of transactions and revenue management. Irrespective of these contributions, the research that was done on the overall financial performance and results did not dwell on the application of predictive analytics and data-driven tools in the context of real dental clinic operations.

To fill this gap, the current study has included analytics-based reports and trend analysis that is specific to dental practices and allows clinics to predict the demand of services, identify peculiarities of transaction patterns, and aid the process of informed financial planning based on real clinical and billing data.

It was also demonstrated in existing literature that most systems had separate components focusing on billing, clinical records and analytics leading to fragmented solutions that restricted meaningful insights. This division caused the dental clinics to find it hard to utilize data well in enhancing operations, planning treatment, or making strategies.

The current study overcame this shortcoming through creating a single platform that incorporated the electronic dental records, POS transactions, and business analytics, which offers a better comprehensive and practical picture of the performance within the clinic.

Available sources by Garcia and Rodriguez [5] found several ethical and regulatory issues in the dental data management process, especially in patient privacy and adherence to data protection laws and regulations but offered insufficient information on how these demands might be enacted in the context of combined POS and analytics systems. On the same note, Garcia and

Santos [8] investigated on payment analytics of dental POS systems without giving clear strategies to adhere to the regulations.

To accomplish this gap, the study established role-based access control, data encryption and audit logging to secure and compliantly handle financial and clinical data and at the same time provide analytics-based decision making.

According to the literature on the subject, local investigations by Santos and Cruz [6] as well as Reyes and Bautista [7] have mentioned the increasing implementation of POS and automated dental management systems in the Philippine dental clinics with the main emphasis on digital payments and receipt of bills. Nevertheless, such studies were silent on the effect that the business analytics could have on revenue forecasting, inventory, or strategic resource planning. Even more recent research by Ho et al. [12] and Wardhana et al. [18] focused on web based dental management systems but did not focus on the integration of POS data with overall business analytics, with similar fashion.

To fill this gap, the current paper has designed a multi-branch, analytics-based dental management system, which has used POS data to assist in revenue projections, inventory optimization, and performance evaluation which directly addressed the operational and financial requirements of dentistry in the Philippines.

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CHAPTER 3
TECHNICAL BACKGROUND

This chapter provided an in-depth discussion of the technical aspects relevant to the development and implementation of the proposed system. It included the system architecture, hardware and software requirements, and the technologies utilized to ensure optimal performance and functionality. By outlining these technical foundations, this chapter established the feasibility and operational framework of the project.

3.1 Overview of Current Technologies Used in the System

The given Integrated Dental Practice System of Dr. Wennie A. Samper Dental Clinic employed an ultimate range of modern technologies to provide a secure, efficient, and scalable system of running clinical operations management, billing, and business insights. The web-based system combined frameworks and tools in the software development, database management, data security, and analytics to meet the multifaceted requirements of dental clinics. Role-Based Access Control (RBAC) was also one of the fundamental technologies that were applied to the system, and it was used to administer permissions and user access within various modules of the system. The system allocated different roles to the administrators, dental practitioners, receptionists and finance personnel to make sure that every user only communicated with features which he/she was supposed to. This helped a great deal in preventing unauthorized access, assisted in remaining within the privacy rules, and made operations more efficient. The system has also been using data encryption protocols to encrypt sensitive patient records, billing data and business reports within the database. The use of both in-transit and at-rest encryption methods was done to maintain the confidentiality, integrity, and resistance of the data against cyber-attacks. In combination with a safe web authentication procedure, the system approved of the strong authentication procedures that involved the session expiry. To manage data, the system had a relational database management system (RDBMS) to manage structured data. It assisted with the real-time updates of patient records, appointment schedule, invoices, and inventory data. All critical actions by the users were recorded by automated audit logging mechanisms which gave a viable base on which the activity monitoring and compliance auditing of the same could be performed. In order to enable informed decision-making, the system incorporated business analytics engine, which applies embedded data visualization libraries and trend analysis tools. This allowed the clinic managers and owners to understand the level of revenue performance, patient demographics, trends in service demand and financial wellbeing. The combination of these technologies formed one unified and intelligent platform that improved the process of running the dental practice, enhanced data security, and allowed planning strategically through real-time analytics.

3.2 Resources

Here, the necessary resources in the creation and adoption of the Integrated Dental Practice System with Point-of Sale Integration and Business Analytics to Dr. Wennie A. Samper Dental Clinic were described. It contained information about the specifications of the hardware, software, system and what is required to achieve the best performance, security and scalability. This combination of well-chosen components helped to provide a holistic solution to the purpose of the project improving the overall patient care, enhancing the work efficiency, and maximizing the financial performance.

3.2.1 Hardware Specifications

This section outlines the physical computing resources and computing components that are needed in the deployment of the system in a production environment. The hardware requirements made sure that the system possesses adequate processing capability, memory as well as storage capacity to perform efficiently and achieve the expected user load when used live. These requirements are the bare minimum configuration to ensure that the system performs and is stable and delivers services.

Table 1 provides the key hardware requirements in the deployment of the system. All parts are defined with the minimum configuration that it needs to achieve maximum performance in its operational use.

Table 1. Hardware Specifications

HARDWARE TOOL	SPECIFICATION	PURPOSE
Processor	Intel Core i5 or higher	Deployment
Memory	GB RAM or higher	Deployment
Storage	512GB Solid State Drive (SSD)	Deployment
Input Device	Keyboard, and Mouse	Deployment
Output Device	Monitor	Deployment

Table 1 gives the hardware requirements that underlie the implementation of the system. The Intel Core i5 or more is a sufficient processor that provides sufficient computational power to operate the system and handle user requests effectively. The 8 GB RAM is enough to allow people to multitask and perform well on the system whereas the 512 GB SSD has enough storage space and can access data faster than the hard drives. The user can interact with the system using standard input devices like keyboard and mouse and a monitor can be the main output device to display the interface of the system and information to the users.

3.2.2 Software Specifications

This part determines the tools, programming and technologies used in the development stage of the system. The software requirements include operating system, programming and scripting languages, development frameworks and version control systems which are the technical base of developing, testing and maintaining the system. These tools allow the efficient development of code, version management, and collaborative programming to develop the project life cycle. The right choice of software tools provides the quality of code, maintainability, and compatibility with the current web development standards.

Table 2 describes the software tools and technologies that have been used in the system development process. Every component has a particular role in ensuring the development of a strong system that is sustainable.

Table 2. Software Specifications

SOFTWARE TOOL	SPECIFICATION	PURPOSE
Operating System	Windows 10 or 11	Development
Programming Language	PHP v8.1.0, Visual Studio Code	Development
Scripting Language	HTML 5, CCS 3, Bootstrap, and JavaScript	Development
Framework	CODEIGNITER	Development
Version Control	GitHub	Development

The software specifications provided in Table 2 indicate the development environment that was used when constructing the system. The operating system platform used in the development is windows 10 or 11. The basic server-side programming language is PHP version 8.1.0, and Visual Studio Code is the effective code editor with the development attributes. The front-end scripting technologies that comprise HTML5, CSS3, Bootstrap, and JavaScript were used to develop the responsive and interactive user interfaces. CodeIgniter is the PHP framework that offers a well-structured architecture and reusable modules to help in the development of an efficient system. GitHub is the version control software, which allows sharing the development process, following the code, and maintaining the version during the project.

Functional Requirements

Functional requirements provide the required capabilities that should be offered by the system to ensure effective functionality. In the case of the Integrated Dental Practice System, the functions are involved in enhancing the management of patients, simplifying operations at the clinic, and making transactions secure. These critical and well-defined main functions are well established to help Dr. Wennie A. Samper Dental Clinic to go about its daily affairs with precision and economy.

Table 3 shows the functional requirements of the Multi-Branched Dental Practice Management System that is going to be used in clinics of Dr. Wennie A. Samper.

Table 3. Functional Requirements

FUNCTIONAL REQUIREMENTS	DESCRIPTION
Patient Registration and Records Management	Allowed for secure registration and storage of patient information, including personal details, medical history, and treatment records.
Appointment Scheduling	Enable real-time booking, rescheduling, and cancellation of appointments, reducing scheduling conflicts and no-show rates.
Point-of-Sale (POS) Integration	Automated billing, invoicing, and payment processing while linking financial transactions with patient treatment records.
Role-Based Access Control (RBAC)	Assigned specific access rights to users based on their roles (Dentist/Admin, Staff, and Patients), protecting sensitive data and ensuring compliance.
Business analytics Board	Provided visual insights and reports on clinic performance, patient visits, and revenue trends to support informed decision-making.

Table 3 above outlines functional requirements that characterize the main capabilities required to manage clinic operations in an extensive manner. Patient Registration and Records Management has been implemented to maintain continuity of care through superior centralized storage of patient data. Appointment Scheduling system reduces bureaucratic work and lack of time in scheduling. POS Integration increases the accuracy and transparency of financial transactions by connecting the treatment records and the transactions. Role-Based Access Control provides the protection of data by limiting access according to the responsibilities of the users. The Business Analytics Dashboard allows making decisions based on the data and provides real-time information about the performance. Both combined features simplify the operations of a clinic, lessen the manual labor, and increase the quality of the services in Dr. Wennie A. Samper Dental Clinic.

Non-Functional Requirements

The non-functional requirements stipulated the general performance and quality of the system, so that it operates in an efficient manner as well as can meet the expectations of the users. An effective system needed to be not just able to fulfil the role it was developed to perform, but also to be efficient and safe and as less disruptive as possible.

Table 4 showed the non-functional requirements of the POS and Dental Practice System with Business Analytics.

Table 4. Non-Functional Requirements

NON-FUNCTIONALREQUIREMENTS	DESCRIPTION
System Reliability	Maintained consistent operation across all modules, ensuring that core features like appointment scheduling and billing were always accessible.
Ease of Use	Provided an intuitive interface for Dentist/Admin, Staff, and Patient.
Security and Privacy	Implemented encryption, authentication, and access control to safeguard patient data and sensitive clinic information.
Performance and Scalability	Delivered fast responsive times during peak hours and supported simultaneous access by multiple users without system shutdowns.
Data Backup and Recovery	Performed automatic backups and included data recovery options to prevent loss due to system failures or unexpected interruptions.

The non-functional requirements as detailed in Table 4 were necessary in ensuring that the overall quality of the system, its reliability, and user satisfaction were satisfied. System reliability ensured that there was constant operation of all modules which meant core functionality was always available. The system offered an easy user interface to all the types of users, which facilitated easy user experience and lowered the learning curve. The issue of security and privacy was addressed using encryption, authentication and access control to prevent unauthorized access to patient data. High-performance and scalability also made it possible to respond fast during peak hours and allow access by multiple users without impacting performance. The backup and recovery measures of data did automatic backups to ensure loss of data due to system failures is avoided. To provide an easy user experience, and to avoid page hanging, optimization was applied to the sites such as lazy loading, pagination of large data sets, asynchronous loading of data, debouncing of search operations, efficient memory management, and use of cache to minimize the server requests. All these non-functional requirements made the system reliable, secure, efficient, and resilient to offer a strong platform on which the clinic will perform optimally.

System Flowchart

A system flowchart refers to a graphical illustration that indicates the data and control flows of a given system through its components. It is an important resource in software development and testing, giving a visual account of the interaction of various components of a system and processing information. It can be used by developers and stakeholders to gain an idea of how a complex process works, areas of possible bottlenecks, and proper functionality of the system before its implementation, by visualizing the logical sequence of operations and decision points. Furthermore, these diagrams enable systematic debugging and maintenance by mapping out the entire system architecture in a single view, making it easier to identify where errors might occur or where improvements can be made. By providing a clear reference point for both current development work and future modifications, flowcharts help ensure consistency and reduce the time required to onboard new developers to existing projects. These diagrams serve as critically important and essential documentation tools that facilitate communication between technical and non-technical team members throughout the development lifecycle. Additionally, flowcharts support the testing phase by providing clear pathways for creating comprehensive test cases that cover all possible system scenarios. The standardized symbols and notations used in flowcharts ensure consistency across different projects and make it easier for new team members to understand existing systems quickly.

The figure below shows the entire workflow of the dental clinic management system, as seen by the dentist/admin. The functions of this flowchart include user authentication, appointments, treatment and photo uploads on the treatment documents, user management, and reporting features.

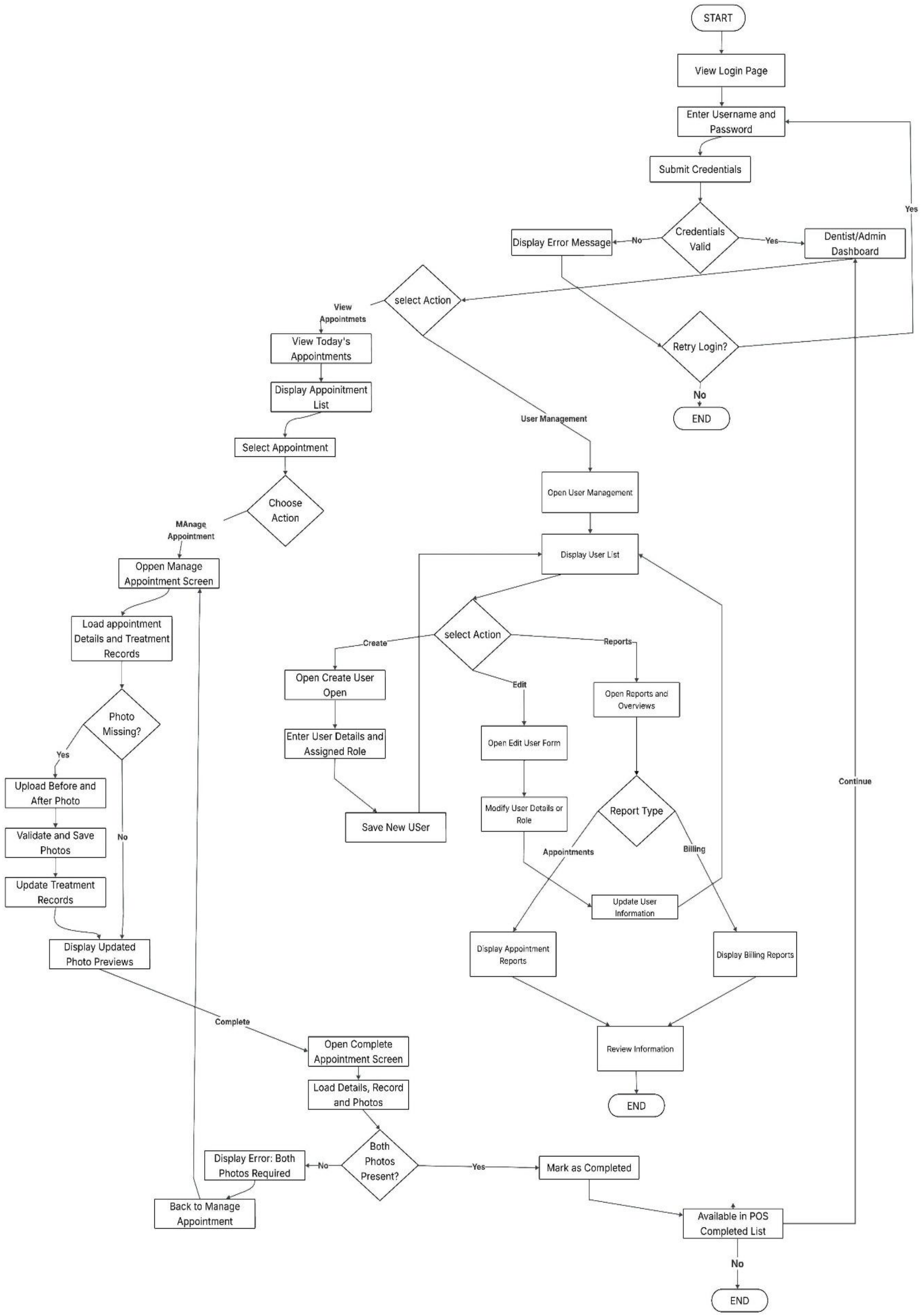

Figure 1. **System Flowchart (Dentist/Admin)**

The Dentist and administrative workflow start with a secure credential validation and a dashboard with which it is possible to navigate through appointment management and administrative capabilities. The dentists have their daily schedule, handle treatment records uploading the required before and after photos with validation checks and can finalize the appointment only when both photos are attached which automatically allows using POS billing to cut down on financial processing. The administrators are in control of user management, create accounts and assign roles, and access appointment and billing reports, to aid clinical and financial control. This system applies the data integrity by providing data checkpoints especially dual-photo requirement that guarantees quality assurance and proper documentation of the treatment during the clinical workflow. The workflow is integrated with an end-to-end appointment lifecycle management to ensure that dentists can see appointment lists, see the details of all the appointments, and edit appointment records with complete capabilities. Photo management functions give options of uploading with before and after labeling, validation to avoid incompleteness in documentation and update of existing treatment records. The administrative route is further extended to item and user list management, which helps to present detailed item information, allows updating inventory privileges, and administers a user account fully by creating, modifying, and assigning roles. Report generation capability gives the administrators an opportunity to filter the reports according to the type of report so that they can analyze specific parts of the appointments or billing data to assist in making strategic decisions or monitoring the operations of the dental practice management system.

Figure 2 shows a complete continuous flow diagram of the working process of members of the staff within the management system of the dental clinic. The diagram is drawn with typical flowchart symbols: the start and end terminals are presented as ovals, process steps are presented as rectangles, decision points are presented as diamonds, and reference of flow direction is presented as arrows. This combined design demonstrates the way in which all functions of the staff are interconnected with the help of one centralized dashboard.

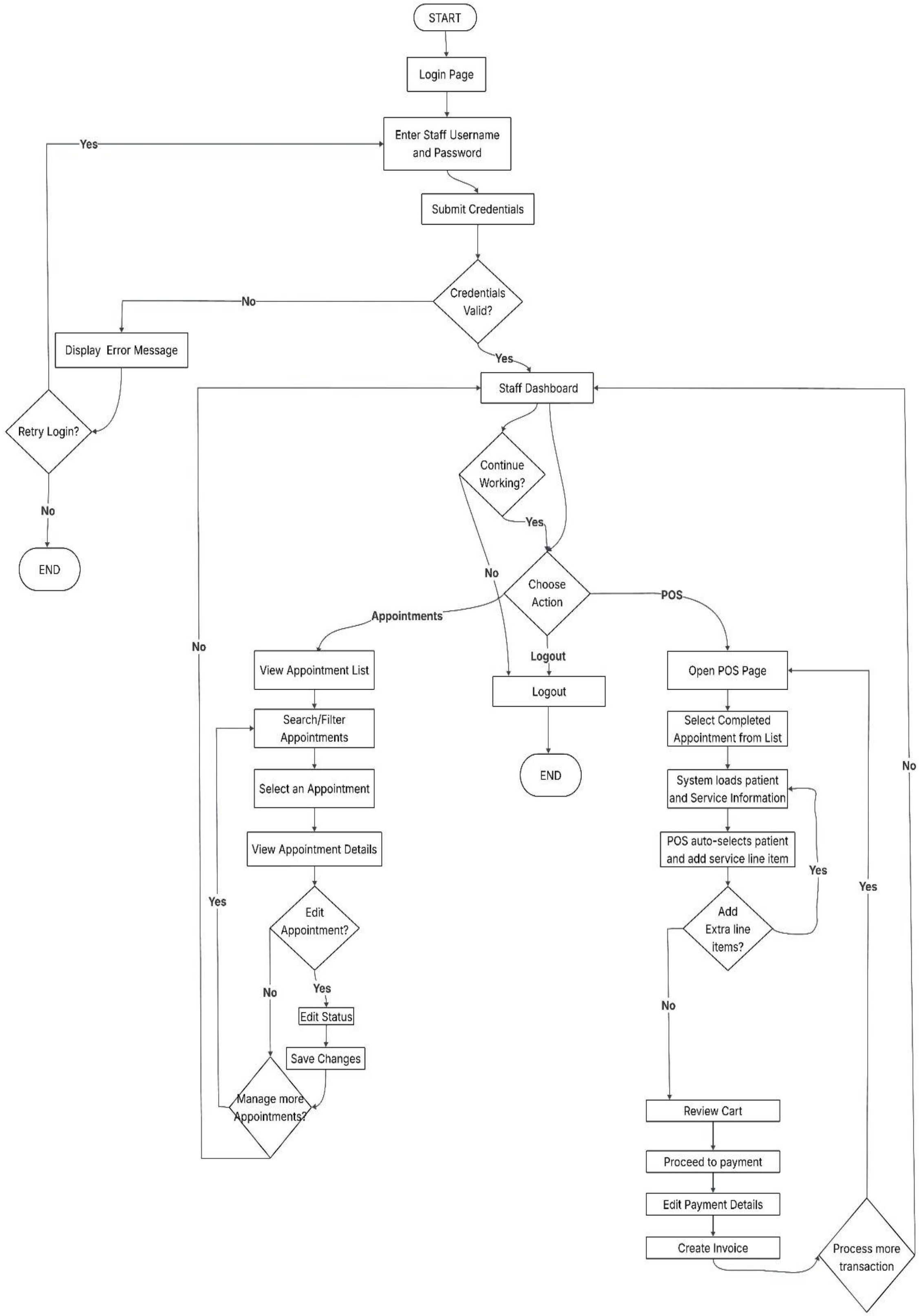

Figure 2. **System Flowchart (Staff)**

The flowchart demonstrates the effective work process that is based on the staff dashboard, which is planned to optimize daily processes within the dental clinic. Upon successful authentication of the logins, the staff members are shown their primary dashboard, which is the central point of access to the crucial functionalities. Through this dashboard, employees will have an option to record appointments, do Point of Sales (POS) transactions, or log out of the system safely. Appointment management module offers full functionality, such as the ability to search a particular appointment, sort the list of appointments by different criteria (date or the name of a patient) and see the details of an appointment.

The staff members may also choose to edit appointment details when needed and in this case the system can also do batch operation in terms of internal loops that know how to handle many appointments effectively without going back to the main dashboard every time an operation is performed. This serves to save time during the busy clinic time when there are many appointments to update or review at a time. The POS streamlines billing operations with patient and service information automatically loaded into the workflow completed appointments and minimizes manual data entry and possible errors. Staff can add other line items on demand, see the contents of the cart and move on to process payments with built-in invoice-generating features. Its continual design pattern reflects the real daily processes by its cyclical design with internal loops in each module facilitating effective batch processing which lets the staff members carry out several related activities without any unnecessary navigation.

In the meantime, the central dashboard acts as a tactical point of concentration, so any change between various functions can be made depending on the requirements of the clinics during the day.

Figure 3 illustrates the patient system workflow, encompassing registration, authentication, and dashboard functionalities. The flowchart maps user interactions from initial account creation through appointment booking, treatment history access, and billing management.

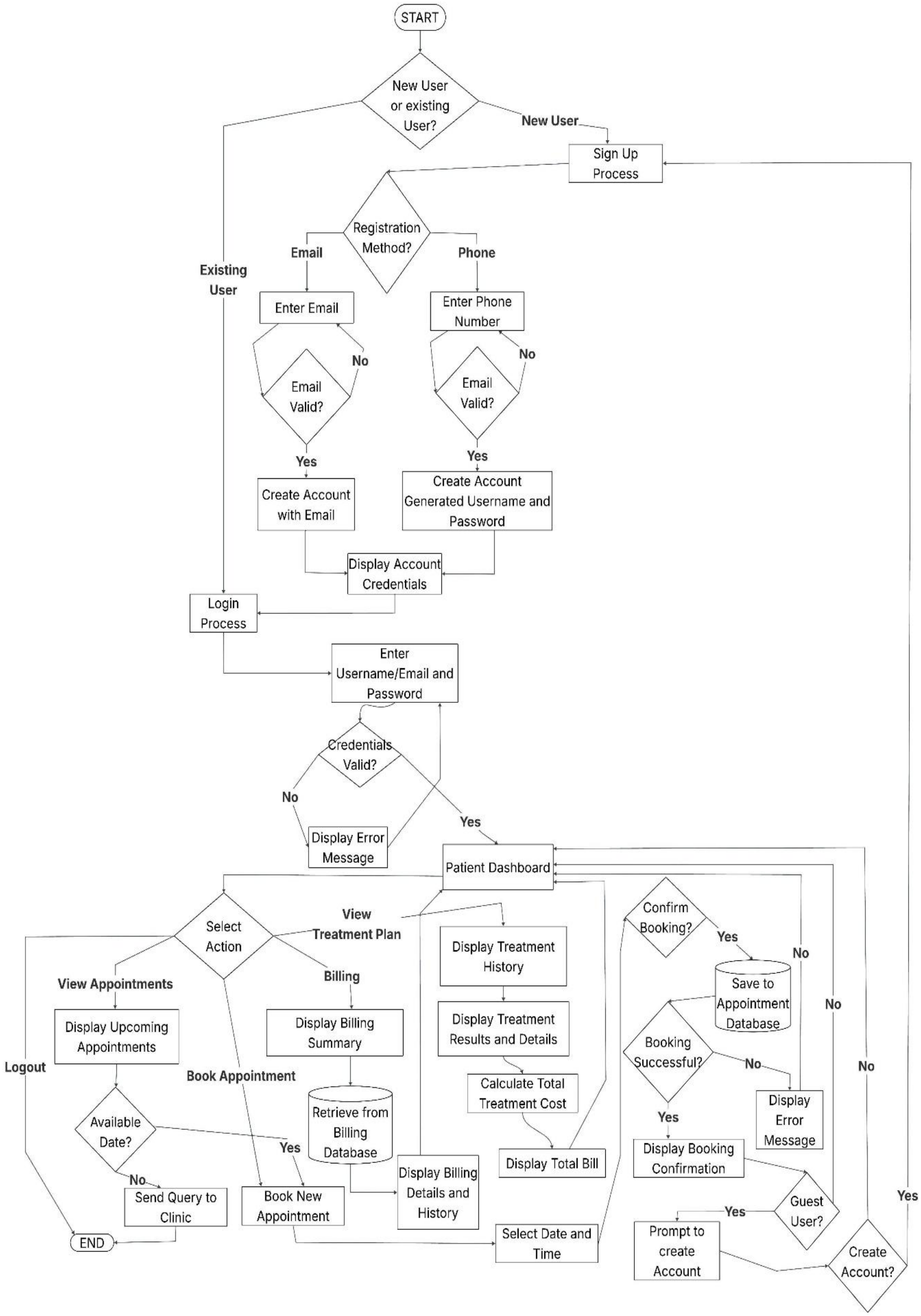

Figure 3. **System Flowchart (Patient)**

Patient flowchart is user-friendly with two possible registration routes: email authentication and phone authentication. New users are first validated prior to the creation of accounts, and existing users have access to the system by undergoing credential validation and error handling systems. After a successful authentication, patients can access a centralized dashboard that provides four main features, namely appointment view, treatment plan access, billing review, and booking a new appointment, and the booking section provides date availability checks and alternative query submission in case of unavailability. Effective bookings initiate database storage and create confirmation displays, and the guest users are provided with post-booking account creation messages to convert them to registered ones. The treatment perspective incorporates the calculation of billing to facilitate the breakdown of all costs, database interactions that are made at strategic points (appointment storage and billing retrieval) to ensure the persistence of selected data across sessions, and the logout feature is one way of offering a secure termination of the session.

The workflow architecture prioritizes on patient convenience and efficiency in the system by way of strategic decision points and automated processes. The booking of appointments uses smart date checks that avoid time clashes and suggest patients to an open spot on the schedule, whereas the request form option offers some degree of flexibility to the patients with special needs regarding their schedules that are not included in the regular schedule. The transparency of the treatment plan viewer is improved by the detailed view of the treatment history of the patient and the appropriate cost calculation, which could be found in the billing database, so that the patient is able to choose his dental treatment wisely. This ensures enable clinical and financial integration of information in a single interface, which will decrease confusion and will assist to make the patient more engaged in their treatment plan and its related costs.

System Architecture Diagram

An architecture diagram is a graphical representation of the structural organization of a software system in terms of components, relationships, and interactions. It is a necessary source of documentation to design and test strategies for the system.

Figure 4 shows the architecture design of the proposed system, which shows the structural components and how they interact with the rest of the system structure

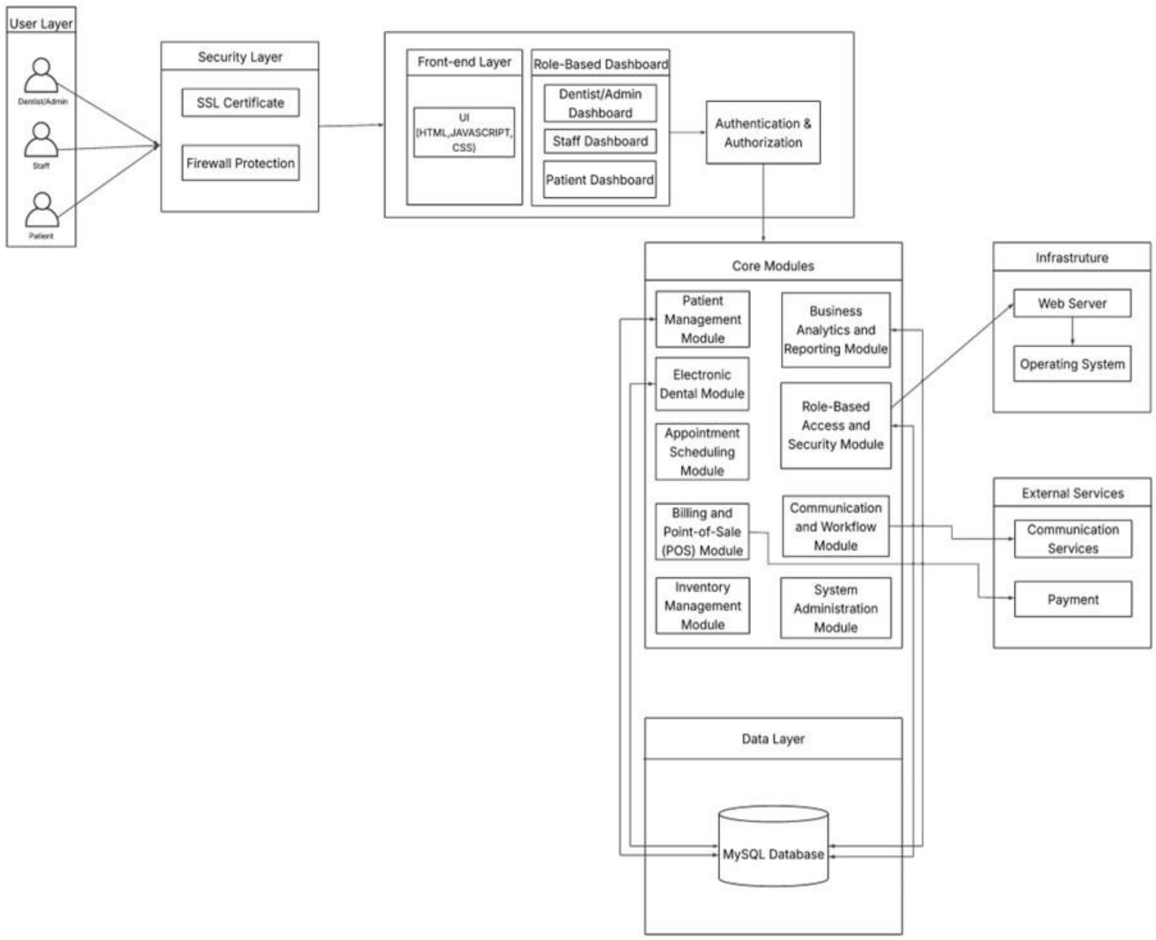

Figure 4. **Architectural Diagram**

This entire system architecture illustrates a multi-layered system of an enterprise-class Dental Practice Management System of six diverse layers. The top layer, the User Layer, acted as the support of three distinct user roles (Dentist/Admin, Staff, and Patient), who can access the system through the Security Layer with the use of the SSL certificate encryption and firewall assistance of the HIPAA-compliant data transmissions and network security. The role of such secured connections is fed into the Front-end Layer, which is written using HTML, JavaScript, and CSS technologies and has the capability of dynamically creating Role-Based Dashboards, which vary depending on the individual requirements and privileges associated with each role with all user requests being handled through an Authentication and Authorization module. The Core Modules layer is then interacted with by the authorized users, and it represents the central business area in which there are ten diverse modules placed side-by-side: the left column includes the medical and administrative modules like Patient Management Module (to manage demographics, medical records, and patient information), Electronic Dental Module (to manage digital dental records and processing), Appointment Scheduling Module (to schedule and reminders), Billing and Point-of-Sale Module (to make credit card payments and issues invoices) and Inventory Management Module (to track dental materials) whereas the right These base modules are bi-directionally related to the Data Layer which is on the bottom and which operates as having a centralized MySQL Database as the point of truth of all patient records and scheduled appointments along with financial transactions, inventory, and user system data. Lastly, the system has two significant infrastructure elements in the right-hand side, that is, the infrastructure component that provides services related to a Web Server running on an Operating System as a hosting platform, and an External Services component that provides Communication Services related to email and notifications as well as Payment Services.

3.2.3 Programming Environment

The programming environment consisted of the tools, libraries, and development frameworks used to build and maintain the system. This section highlighted the integrated development environment (IDE), version control systems, and other essential utilities.

Front End

The front part of the Youth Governance System has the role of providing a responsive, user-friendly, and interactive interface to users, both SK officials and KK members. It is created with standard web technologies that guarantee the usability in devices and the screen sizes. Such technologies supported different functions, including registration of members, events listing, seeing attendance, and profile management.

The Integrated Dental Practice System had its front-end dealing with offering a user-friendly interactive interface. It applied the latest web technologies to introduce the most important modules, including patient registration, making an appointment, POS invoicing, and analytics, in a user-friendly way. Front-end development was done using the following tools:

HTML. HyperText Markup Language structure the interface and formed the foundation of the web application. It organized input forms for patient registration, tables for viewing treatment history, and buttons for submitting or confirming appointments. HTML allowed the layout of interactive components, enabling users to effectively interact with system features.

CSS. Cascading Style Sheets governed the appearance of the interface of the system. It stipulated the elements of design like fonts used, colors, distance, and structure of the layout, which led to a professional and aesthetic atmosphere that could be placed in clinics. CSS made the system easier to read and understand as the entire system had the same style.

Bootstrap. It is a front-end toolkit that offered a responsive grid and pre-built components of the UI, such as modals, dropdowns, and buttons. It made the web application design easier and compatible with different devices such as desktops, tablets, smart phones and did not interfere with the functionality of the web application.

Back End

The back end was the heart of the system, where it processed data, did its authentication and system logic. It incorporated server-side programming and database management and security measures to make the system functional and more secure.

JavaScript. Improved the interactivity of the system to make it dynamic and real time. It favored elements like calendar to appointments, on-the-fly form validation and respondent filters of patient information. When applied to AJAX, it permitted. asynchronous updates, which enhance the responsiveness and the dynamism of user interfaces.

PHP. Request processing and backend logic were done in PHP. It managed data collection, submission of forms and system activities like the production of invoices and the updating of treatment records. The php was also used to support a smooth flow of information between the front end and the database so that any actions of the user would lead to correct manipulation of data.

CodeIgniter. The PHP framework was CodeIgniter which was used to develop the system. It was also based on the MVC (ModelViewController) design pattern and was known to be lightweight and fast. It handled user authentication, routing user requests, form validation and communicating with the database. CodeIgniter has been selected as simple and has inbuilt security support and is therefore an appropriate tool to build a strong web application.

MySQL. This was used as Relational Database Management System (RDBMS) in the storage and management of system data. It managed patient records, appointment history, user profiles, POS records and audit logs. MySQL made sure that the data is properly structured and easy to retrieve to facilitate effective operations in the system.

XAMPP. The local development environment which was employed when building the system was XAMPP. It incorporated Apache, MySQL, PHP, and other services needed to enable the developers to replicate the live environment on their machines. It facilitated testing, debugging and testing of the system before implementation to guarantee stable work in a practical environment.

3.3 Testing Plan

This section described the testing strategies and methodologies that were adopted to test. the functionality, performance and security of the Development of Multi-Branched Dental. Point-of-Sale and Business Analytics DR Practice Management System. Wennie A. Samper Dental Clinic. The testing plan also made sure every module which included patient records, booking appointments, point-of-sale and analytics was operable and as per the requirements of Dr. Wennie

A. Samper Dental Clinic. Various testing stages such as unit testing, integration testing, system testing and user acceptance testing were adopted to ensure that the overall reliability and usability of the system were accurate. This was to be done through each testing phase where one is to detect and resolve errors at various levels of the entire system architecture, ranging through individual component validation to the entire end-to-end workflow validation. The extensive testing strategy also involved the real-world usage conditions and the stress testing to make sure that the system could handle the peak load operation and data integrity of the multiple simultaneous users and in multiple branches.

Figure 5 presented the detailed testing plan to be followed throughout the development lifecycle.

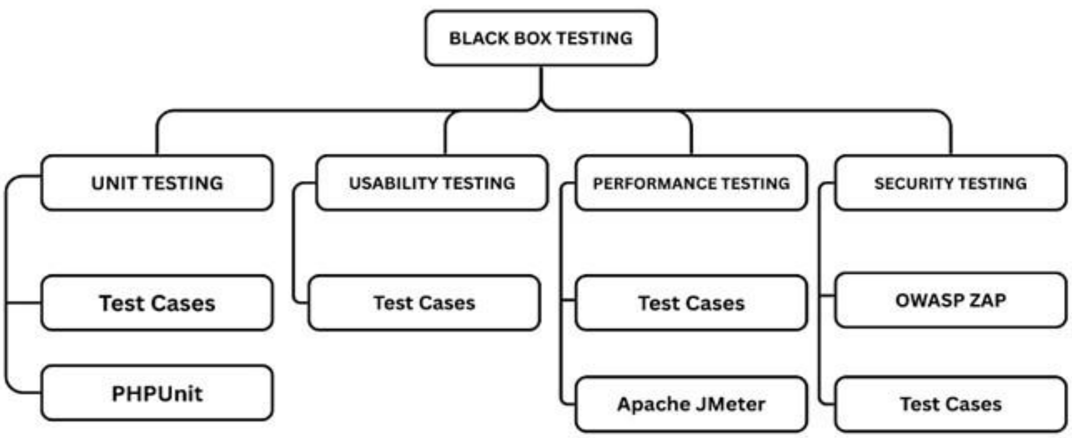

Figure 5. **Black Box Testing Plan**

Figure 5 This figure displays a detailed black box testing model that has four key types of tests namely: Unit Testing, Usability Testing, Performance Testing, and Security Testing. Unit Testing makes use of Test Cases and PHPUnit to check that code components work properly. Test Cases Usability Testing is done using Test Cases that are used to test the friendliness of the system to clinic employees and patients. Performance Testing involves use of the Test Cases and Apache JMeter to test the system in relation to speed, responsiveness and stability under different loads. OWASP ZAP and Test Cases are used in the security testing to find weaknesses and vulnerabilities and confirm the effectiveness of the authentication, encryption, and role-based access control measures in the dental practice system to prevent unauthorized entry and breach of patient data and the financial information and guarantee the functionality, ease of use and efficiency, and security of the system prior to its deployment.

3.3.1 Types of Testing

Software testing employed various approaches to ensure system reliability, functionality, and performance. Different testing types, including unit testing, usability testing, performance testing, and security testing, helped identify defects and optimize software quality throughout the development stages.

Unit Testing. Examined individual components in isolation to verify each function performed correctly according to specifications, helping identify and fix bugs early. This phase employed test cases and PHPUnit to validate individual code units before integration [10] [11].

Usability Testing. Focused on ease of use and user experience. Since the system served different roles (dentists/admin, staff, and patients), the interface needed to be intuitive. This stage evaluated layout readability, responsiveness, and accessibility of features. Test cases ensured users could accomplish tasks with minimal effort [1][2].

Performance Testing. Examined system responsiveness under various usage levels, including processing multiple invoices simultaneously and handling concurrent appointment requests. Apache JMeter simulated scenarios to test system throughput, identify bottlenecks, and optimize resource usage [3][4].

Security Testing. Ensured the system defended against vulnerabilities compromising sensitive information. IT specialists conducted access control and data handling tests using OWASP ZAP to simulate attacks like injection and cross-site scripting, ensuring robust protection against common threat vectors. These measures guaranteed compliance with data protection standards [5] [6].

3.3.2 Testing Tools and Framework

Testing automation, control, and implementation of test cases could not have been done without testing tools and frameworks. They offered an organized method of software validation which guarantees correctness, reliability and scalability during the life cycle of development.

Test Cases. Created essential modules like patient information management, point-of-sale operations, and business analytics functioned properly. Each test case included input, expected results, and pass/fail conditions for close monitoring and problem resolution. Test cases were applied across all testing types including unit, usability, performance, and security testing [7].

Apache JMeter. A performance testing tool that simulated large numbers of users to test system performance under stress. It evaluated the system's load handling capability, ensuring critical functions like invoice generation and report visualization-maintained performance during high demand. Apache JMeter worked with test cases to measure system throughput and detect performance bottlenecks [8].

OWASP ZAP. An open-source tool used to identify web application vulnerabilities. It was applied with test cases to test for threats like SQL injection, broken authentication, and cross-site scripting, ensuring the system complied with security best practices [9].

PHPUnit. An open-source testing framework developed for PHP applications. It was used with test cases to run automated unit tests that verified individual functions, methods, and code operated as intended, ensuring every system module worked properly independently before integration with other modules [10].

Notes

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CHAPTER 4
METHODOLOGY, RESULTS AND DISCUSSION

This chapter details the system's development using the Agile Scrum methodology, which is based on iterative phases including Sprint Planning, Execution (Designing & Development), Sprint Review, and Retrospective. Agile is an incremental and timeframe iterative approach [1]. This flexible, team-driven approach was crucial for successfully developing the Multi-Branched Dental Practice Management System with Point-of-Sale and Business Analytics for Dr. Wennie A. Samper Dental Clinic.

4.1 Sprint Planning

Sprint Planning adopted Agile Scrum in gathering, analyzing and writing down system requirements. The first problems observed by the team were the following: manual records management, inefficient billing processes, and poor data analysis.

Researchers collected requirements of clinic stakeholders through interviews, surveys and direct observation. They were divided into the functional requirements (patient registration, appointment scheduling, point-of-sale integration, and business analytics), and non-functional requirements (security, reliability, and usability). All the requirements were given priority depending on the effect they had on the clinic operations and user needs.

Frequent planning with the clinic personnel periodically was used to keep the requirement in line with the real work patterns. This cooperative strategy provided a strong base on which effective solutions to actual clinic issues could be developed, whereas feasibility reports facilitated the understanding that the suggested solutions would be operational within the current infrastructure of the clinic.

Figure 6 shows the Gantt Chart that maps out the project timeline, including sprint durations, task dependencies, and key milestones. This chart helped with resource allocation, tracking progress, and managing the schedule throughout the project.

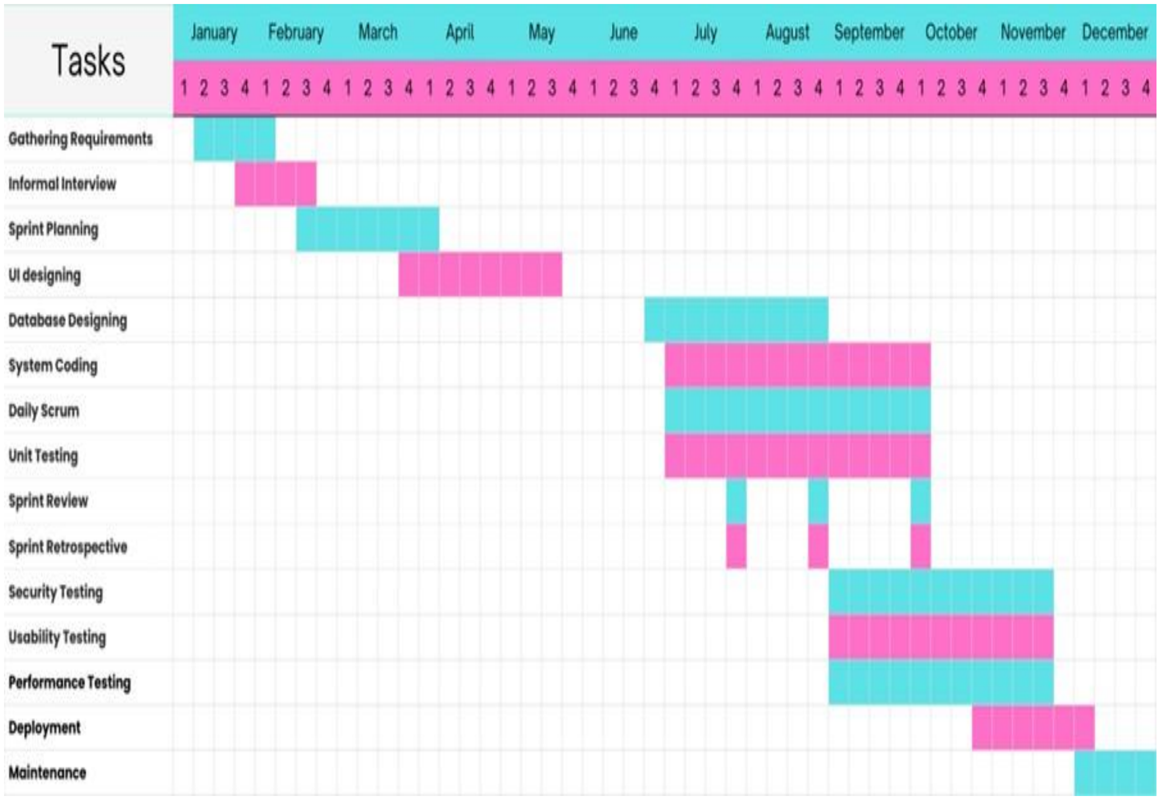

Figure 6. **Gantt Chart**

This Gantt chart illustrates the general project timeline, which indicates clearly the sequence, duration and overlap of the activities across the year presented in months and weekly basis. This project begins by collecting requirements and informal interviews as it gives a clear picture of the needs of the user and expectations of the system. Additionally, sprint planning and UI designing will be accomplished so that the system structure and user experience are properly laid out after which the development will commence. Database designing is partially implemented along the initial development preparation to conform the data architecture to system functionality. When the project is in the agile phase and the system coding is taking place, it leads to agile development, which is facilitated by the daily scrum and unit test to achieve the purpose of collaboration, early detection of issues and incremental improvement. Sprint reviews and retrospectives will be organized periodically to evaluate the development progress and optimize development practice. Security testing, usability testing and performance testing are subsequently carried out concurrently later in the cycle to ensure that the system is secure, user friendly, and efficient with the anticipated workloads. Lastly, deployment completes the project, the system is now operational and then comes the maintenance, continuous support, updates and enhancements which are conducted to ensure that the system is stable and functional over time.

4.1.1 Material and Statistical Tools

In this section, the materials, statistical tools, and development tools that were applied during the research and development process are described. Only the tools were chosen based on their appropriateness to collect data, analyze it and develop a system, which guaranteed adherence to the study goals, as well as the Agile Scrum methodology adopted in the project.

4.1.2 Research Design

This research is mixed-method research, which involves the integration of both qualitative and quantitative research to achieve the aims of creating and testing a multi-branch dental practice management system with point-of-sale and business analytics capability. The framework directs the refinement of the questions to be used in the research and the identification of proper development and testing methodologies and data collection processes to be used in assessing the effectiveness of the system. The design will guarantee an all-inclusive perspective of the user's needs as well as the performance of the system by using a variety of data sources and testing methods.

4.1.3 Data Gathering Procedures

This study employed two data collection methods to develop and evaluate the Multi-Branched Dental Practice Management System.

Semi-Structured Interviews. Interviews were conducted with dentists, dental staff, and patients to identify operational challenges. Participants revealed pain points including inefficient patient record management, appointment scheduling conflicts, billing errors, and inadequate business reporting. These insights directly informed the design of core system modules: patient management, appointment scheduling, point-of-sale integration, and business analytics.

Survey via Questionnaire. Quantitative feedback was quantified through administering structured questionnaires to test users after the development. Dentists and dental data input patient management characteristics and billing and scheduling capabilities, and patients assessed appointment booking instrumentation. The survey was used to determine the efficiency of the operations, the reduction of errors, system responsiveness and user satisfaction. Feedback was used to determine the strengths of the system and areas of weakness that were utilized in the refining of the system to be used practically in a clinic.

4.1.4 Sampling Techniques

To obtain credible outcomes, it is necessary to pay close attention to the study participants. This research adopted three different sampling methods depending on the nature and presence of each group of respondents, they include: Total Enumeration, Convenience Sampling and Purposive Sampling.

Total Enumeration was used for dentists and staff due to their limited population within the clinic. This technique involves including all members of these groups in the study, which eliminates sampling error and provides complete coverage of perspectives from key stakeholders who directly manage and operate the system daily.

Convenience Sampling was applied to select patients as respondents. Given the large patient population and practical constraints of data collection in a clinic setting, patients who were available and willing to participate during the study period were included. This approach ensures efficient data gathering while achieving a sufficient sample size to represent patient experiences and satisfaction with the system.

Purposive Sampling was applied to the selection of IT experts who would have the corresponding technical expertise to analyze the security testing of the system. This is to ensure the security assessment is done by qualified personnel.

Such mixed sampling methods assure holistic data gathering of all the interested groups of stakeholders that strike a balance between practical limitations and the demand to obtain representative and credible findings accurately describing the effectiveness of the system in terms of functionality, operational, and technical aspects.

4.1.5 Respondents

There was a variety of sampling methods used to all groups of respondents depending on their nature and role in the research. The dentist and staff were selected to use Total Enumeration as there were only a limited number of them in the clinic, which fully represents all the personnel involved in the management and maintenance of the system. To choose the most convenient patients, Convenience Sampling was used to choose 30 patients according to their availability and willingness to take part during the period of study and offer practical and efficient data collection and at the same time represent patient experience. Purposive Sampling was used in the selection of 5 IT experts possessing relevant technical expertise to carry out security testing so that the security test would be conducted by qualified professionals.

Table 5 presents the frequency and percentage distribution of the 40 respondents who participated in the study.

Table 5. Frequency Distribution of the Respondents

RESPONDENTS	FREQUENCY	PERCENTAGE
Dentist/Admin	1	2.38%
Staff	4	9.52%
Patient	30	71.43%
IT Experts	5	11.90%
TOTAL	**40**	**100%**

Table 5 reveals that there were four major groups which had a representation of the respondents Dentist/Admin, Staff, Patients, and IT Experts. Such groups have been sampled in a thoughtful manner with the right sampling methods to achieve value and comments regarding the system in various viewpoints so that the users, managers and technical evaluators are all involved in the study.

The highest percentage of respondents was 71.43 by patients since they were first end-users of the system. This high representation of patients will ensure real-world usability and user satisfaction in the evaluation. Total enumeration of the staff and dentists offers organizational and administrative anatomy of the entire operation, and the involvement of the IT experts will offer technical authority to the security examination. Such a composition of respondents allows a full assessment of the Dental Clinic Management System in functional, operational, and technical aspects.

4.1.6 Statistical Analysis Tools

The researchers employ the Arithmetic Mean as a statistical method to analyze the obtained data. This method will facilitate easier interpretation and understanding of results made by the researchers and is commonly applied to obtain the average value of a bunch of data points.

Pass/Fail Assessment. The researchers used binary pass/fail test in the evaluation to assess the user satisfaction and system effectiveness. The respondents rated a series of statements regarding the functionality, usability, and overall performance of the system as to whether it met acceptable levels (Pass), or not (Fail). This dichotomous measurement scale offers plain cut data, which can be interpreted into taking action and which areas in the system need to be improved and which ones are performing within the standards of the system.

Arithmetic Mean. The arithmetic mean is in other terms known as the average or the arithmetic mean. It is determined by subdividing the total number of elements into a data set with the total number. In this work, the arithmetic mean has been employed to test the outcomes of some of the system test scenarios. Besides the feedback provided by the users on how the system was functioning, this study also assisted in identifying areas that needed to be improved.

Frequency and Percentage. The demographic profile of the respondents will be analyzed by the researcher using Frequency and Percentage, which will highlight the percentage of the segments within the sample. It is calculated by dividing frequency with the total number of responses, and times 100.

$$P = \frac{f}{n} X\, 100$$

Where:

P = Percentage

f = Frequency

n = Total number of respondents

Table 6 presents the unit testing assessment framework based on ISO/IEC 25010 Functional Suitability [29], which evaluates functional completeness, correctness and appropriateness. Black box unit testing was conducted using PHPUnit [28], with results marked as Pass or Fail. Pass requires 100% test case success with zero failures and errors, while any failure results in a Fail. This standardized approach ensures consistent evaluation across all system modules before proceeding with integration testing.

Table 6. Unit Testing System Status using Black Box Testing

STATUS	DESCRIPTION
Pass	All test cases pass (100% success rate); handle valid/invalid inputs correctly; zero failures and zero errors; fully optimized and ready for integration.
Fail	One or more test cases fail (<100% success rate); contains errors or bugs; requires debugging and retesting before integration.

Unit testing results framework offers a procedural approach to test the individual components using ISO/IEC 25010 Functional Suitability prior to integration. Passing modules show complete functional completeness passing all PHPUnit test cases (100% pass rate) and failing modules show functionality deficiencies that need to be debugged and retested. Such a binary analysis will guarantee that only those components that are fully functional progress to integration testing that will minimize system-level defects and enhance software quality. The developed Pass/Fail norms offer an objective criterion of component readiness, which has uniformity in quality evaluation and creates a robust base of the later testing stages.

Table 7 presents the usability testing framework based on ISO/IEC 25010 [29], assessing the system's user experience and interface effectiveness. Evaluation was conducted through black-box testing with actual end-users and measured on a 1.00 – 0.00 scale, classifying results into five usability levels based on established usability assessment criteria.

Table 7. Level of Usability using Black Box Testing

RESULT	USABILITY LEVEL	DESCRIPTION
1.00	Completely Usable	100% – Completely Usable: Users complete all tasks effortlessly; interface and navigation are fully intuitive; exceeds usability standards.
0.67 – 0.99	Highly Usable	67–99% – Highly Usable: Users perform tasks easily; minor issues may exist; interface is clear; minor refinements recommended.
0.34 – 0.66	Moderately Usable	34–66% – Moderately Usable: Basic tasks completed with moderate difficulties; some navigation or interface inconsistencies; improvements needed.
0.01 – 0.33	Least Usable	1–33% – Least Usable: Users struggle with most tasks; interface and guidance are confusing; major usability improvements required.
0.00	Not Usable	0% – Not Usable: System unusable; tasks cannot be completed; interface and navigation severely flawed; complete redesign required.

The usability testing model will assess the system effectiveness in terms of the system-user interaction as per the ISO/IEC 25010 Usability standards [29]. Systems with a score of 1.00 exhibit a high level of user-centered design in all the sub-characteristics such as learnability, operability, protection against errors, interface aesthetics, and accessibility. Highly Usable (0.67-0.99) systems are highly usable with great satisfaction and productivity in completing tasks, Moderately Usable (0.34-0.66) systems need enhancements in protecting against errors, interface design or accessibility. A score lower than 0.34 signifies severe problems with usability and 0.00 systems do not meet any of the usability requirements. This framework is used to make sure that systems are functional as well as intuitive and satisfying and accessible so that they increase adoption, productivity, and user acceptance and offer a standardized means of measuring the quality of user experience.

Table 8 presents the performance testing framework based on ISO/IEC 25010 Performance Efficiency [29], evaluating system responsiveness, resource usage, and capacity under normal, peak, and stress conditions. Results are measured on a 1.00 to 0.00 scale and classified into five levels to indicate the system's operational efficiency based on established performance benchmarks.

Table 8. Level of Performance using Black Box Testing

RESULT	PERFORMANCE LEVEL	DESCRIPTION
1.00	Optimal Performance	100% – Optimal Performance: Response times are instantaneous; resources fully utilized; system handles maximum load flawlessly.
0.67 – 0.99	Highly Performance	67–99% – High Performance: Excellent response and resource use; minor optimizations possible; production-ready.
0.34 – 0.66	Moderately Performance	34–66% – Moderate Performance: Acceptable under normal load; performance degrades under stress; optimization needed.
0.01 – 0.33	Poor Performance	1–33% – Poor Performance: Slow responses; frequent timeouts or crashes; major improvements required.
0.00	No Performance	0% – No Performance: System fails to execute; unresponsive under any load; complete redesign required.

The performance testing system measures the systems to the ISO/IEC 25010 Performance Efficiency [29] to verify that they can be used in the real-world performance needs. Systems scoring

1.00 are at their best with instant response, efficient use of resources and complete capacity. High Performance (0.67-0.99) systems are already production ready with slight improvements possible. The moderate performance level (0.34-0.66) needs to be optimized to allow the maximum loads, whereas the scores under 0.34 imply serious gaps and thus such a system is not a viable option to implement. Those systems with a score of 0.00 have broken down and should be completely redesigned. This framework enables the stakeholders to evaluate deployment readiness, infrastructure requirements, and optimization priorities, to be able to have scalable, reliable and efficient system operations under different load requirements.

Table 9 presents the security testing framework based on ISO/IEC 25010 Security [29].

Table 9. Level of Security using Black Box Testing

RESULT	SECURITY LEVEL	DESCRIPTION
1.00	Completely Secure	100% – Completely Secure: All security sub-characteristics are fully implemented and validated. Advanced encryption, multi-factor authentication, audit logging, and threat detection function optimally, providing complete protection against all threats.
0.67 – 0.99	Highly Secure	67–99% – Highly Secure: Strong security mechanisms are effectively implemented, with minor enhancements possible. System meets professional standards and is deployment ready.
0.34 – 0.66	Moderately Secure	34–66% – Moderately Secure: Basic security measures are present, but notable vulnerabilities exist in authentication, authorization, or resistance. Targeted improvements required before deployment.
0.01 – 0.33	Least Secure	1–33% – Least Secure: Critical vulnerabilities compromise confidentiality, integrity, and access control. Immediate and extensive security remediation required.
0.00	Not Secure	0% – Not Secure: No effective security measures; system is fully vulnerable and unsuitable for deployment.

The structure assures systems compliance to ISO 25010 Security standards [29] and possible safeguard against cyber-attack sensitive data. There is no compromise in the implementation of the six sub-characteristics in Completely Secure systems (1.00), which exceeds the industry standards. Highly secure (0.67-0.99) systems are very strong and are suitable in the majority of production environment. Moderately Secure systems (0.34-0.66) are systems that need specific improvement to eliminate vulnerabilities, whereas low scores of 0.34 or lower show serious gaps, which need to be improved without delay. This tiered classification system enables organizations to objectively assess their security posture and prioritize remediation efforts based on quantifiable risk levels. Systems with a score of 0.00 do not work and need to be redesigned in terms of fundamental security.

Table 10 displays the overall system evaluation system in the light of ISO/IEC 25010:2011 Systems and software engineering - Systems and software Quality Requirements and Evaluation (SQuaRE) - System and software quality models [29] .

Table 10. Overall System Evaluation Result using Black Box Testing

RESULT	OVERALL SYSTEM RATING	DESCRIPTION
1.00	Completely Ready	100% – Completely Ready: All components demonstrate flawless functionality, usability, performance, and security; system exceeds standards and is fully deployment-ready.
0.67 – 0.99	Excellent	67–99% – Excellent: High functionality, usability, performance, and security; minor refinements possible; ready for deployment.
0.34 – 0.66	Good	34–66% – Good: Core requirements met, but deficiencies in one or more areas; further refinement and testing needed before deployment.
0.01 – 0.33	Poor	1–33% – Poor: Multiple critical issues across quality characteristics; major redevelopment required before operational use.
0.00	Unacceptable	0% – Unacceptable: Complete failure across all metrics; system unusable; requires full redesign and redevelopment.

The general assessment will package the ISO 25010 quality features [29] into a single rating, which will give a clear-cut reading on system preparedness. Systems with a score of 1.00 are excellent on all dimensions. Very good (0.67-0.99) systems are at production level but with some slight modification. Good (0.34-0.66) systems are those that are functional but cannot be improved adequately and need more tests. Any score below 0.34 suggests significant failures, which require redevelopment and significant investment. Systems with a score of 0.00 fail miserably and must be ground up to be redesigned.

4.1.7 Identified Problems and Proposed Solutions

In the initiation, the researchers found that in Dr. Wennie A. Samper Dental Clinic, a number of operational issues were present such as manual retirement of patients records which then led to loss of data and inefficiency, scheduling of appointments proved to be inefficient and thus led to overlapping schedules and patient dissatisfaction, the billing system was clumsy, and, therefore, created errors and lack of financial transparency, no systematic inventory management which resulted in stock-outs and expired materials, inefficiencies in communication, and, ultimately, no business insights were made. These interconnected problems created a cascade effect that negatively impacted both operational efficiency and patient care quality. The researchers suggested that these challenges should be overcome by developing an integrated, comprehensive Electronic Medical Record (EMR) and Electronic Dental Chart (EDC) to have a simplified patient record management system, creating an automated system to make appointments with real-time updates and to pay with many options and automated billing, an inventory management system with real-time tracking and automated notifications, communication and workflow system to facilitate staff cooperation, and a business analytics dashboard to provide data-driven insights to make informed decisions.

Table 11 presents the identified operational problems at Dr. Wennie A. Samper Dental Clinic along with their descriptions and proposed solutions.

Table 11. Identified Problems and Proposed Solution

PROCESS	IDENTIFIED PROBLEMS	PROPOSED SOLUTION
Patient Registration and Record Management	The clinic relied solely on paper-based patient records, making it difficult to retrieve patient information quickly and leading to potential loss or damage of important medical documents.	To enhance record management, the study proposes an Electronic Medical Records (EMR) system with Electronic Dental Chart functionality that digitizes patient information and provides secure, instant access to treatment histories.
Appointment Scheduling and Treatment History Tracking	Manual appointment scheduling is time-consuming, prone to double-booking conflicts, and lacks automated reminder systems, resulting in high no-show rates and inefficient time management.	The study developed an appointment scheduling module with real-time availability checking and automated email or SMS reminders to reduce no-show rates and improve clinic efficiency.
Billing, Invoicing, and Payment Processing	The manual billing and invoicing process consumed significant administrative time and was prone to errors, with limited payment method options and inadequate financial tracking capabilities.	The Point-of-Sale (POS) integration automates billing by linking treatment records directly to invoice generation, processes multiple payment methods, and provides accurate financial reporting with real-time transaction tracking.
Inventory and Supply Management	The clinic lacks a systematic inventory management system for dental supplies and equipment, resulting in stock-outs, overstocking, expired materials, and inability to track usage patterns or reorder levels efficiently.	The Inventory and Supply Management module provides real-time tracking of dental supplies with automated low-stock alerts, expiration monitoring, and usage analytics to ensure optimal stock levels and reduce waste.
Communication and Workflow Coordination	Communication and workflow coordination among dentists, assistants, and administrative staff is inefficient, leading to delays, miscommunication, and reduced operational productivity in daily clinic operations.	The Communication and Workflow Coordination module enables seamless collaboration through task management, internal messaging, staff notifications, and workflow automation for enhanced team coordination.
Monitoring Clinic Performance and Revenue Trends	The clinic lacks comprehensive business analytics and reporting capabilities, making it difficult to track revenue trends, patient demographics, and operational performance for strategic decision-making.	The system includes a Business Analytics and Reporting Module that collects and visualizes clinic data through interactive dashboards and reports, showing revenue trends, patient demographics, appointment statistics, and inventory performance for data-driven decision-making.

Table 11 summarizes the operational issues in the Dr. Wennie A. Samper Dental Clinic and the solutions to be used to solve these problems. The clinic is currently using conventional paper practices. The interview with staff members exposed the challenges of handling the physical patient files and especially where more than two members of the team are required to access the file at a given time. Inefficiency in scheduling and billing processes was also found by researchers. The medical practitioners observed that the scheduling method using a manual system often creates conflicts during booking, whereas the financial processing is time-consuming and prone to mistakes. The clinic has handwritten books of accounts that provide little strategic planning on growth. There is also the lack of proper supply tracking, team communication, and performance monitoring systems that result in operational bottlenecks.

The suggested solutions focus on the introduction of a unified digital management platform to improve the processes and increase their reliability. A transition to a web-based system would give up the dependency on paper, but would facilitate real time tracking of patient information, appointments, financial information, supplies, staff communications and performance indicators.

The first significant challenge was felt in Patient Registration and Record Management. The use of paper-based documentation was slow in terms of information retrieval and provided a risk of loss or damage of the records. The researchers established an Electronic Medical Records (EMR) capable of an Electronic Dental Chart system that allows digitalisation of all patient data that can be accessed securely and immediately to access all treatment history. During visits, healthcare providers are now able to immediately examine full medical backgrounds and enhance continuity in care, as well as decrease administrative load.

The second obstacle was in the form of Appointment Scheduling and Treatment History Tracking. The time spent on manual scheduling was too high and, due to the absence of reminders, no-show rates were too high, and there were cases of double booking. The researchers developed a scheduling system capable of real-time availability checks and email or SMS auto notifications.

The web platform has made it simple to manage appointments with both the patients and by the staff members and this has greatly minimized missed appointments.

The third hurdle was made by Billing, Invoicing, and Payment Processing. Traditional methods of financial processing consumed administrative resources and had recurrent inaccuracies, had limited payment choices, and had inadequate financial accounting. The Point-of-Sale (POS) integration will automate the billing process by connecting the records of the treatment to the creation of the invoices, several methods of payment will be accepted, and the detailed financial records will be maintained with real-time monitoring. This has minimized billing errors, improved speed of collecting payment and financial transparency.

The fourth issue was Inventory and Supply Management. Lack of proper tracking led to stock-outs, overstocking, expired materials and no means of tracking patterns of usage in the clinic. The Inventory Management module is designed to track supply in real-time, issue low-stock notifications automatically, track expiration, and usage analytics to operate at optimal levels and minimize waste to make informed procurement decisions.

The fifth challenge was Communication and Workflow Coordination. This was due to the lack of information flow between the dentists, assistants, and the administrative staff, which led to delays, miscommunication and reduced productivity. The Communication and Workflow module will helps to work together by managing the tasks, sending messages inside the organization, notifying staff, and automating workflows, making the work processes easier and communication among staff more efficient.

The last issue that was dealt with was the Monitoring Clinic Performance and Revenue trends. The clinic also did not have analytical tools on the reported revenue patterns, patient demographics, and performance of operations. The Business Analytics and Reporting Module would automatically gather and visualize clinic data using the interactive dashboards of revenue trends, demographics, appointment statistics, and inventory performance. This provides the management with the knowledge enabling them to take informed decisions on the allocation of resources, the services offered and the expansion of the business.

4.1.8 Operational Feasibility

The operational feasibility is assessing the viability of the offered Development of Multi-Branched Dental Practice Management System with Point-of-Sale and Business Analytics for DR. Wennie A. Samper Dental Clinic has the potential to solve the operational issues that were found in Dr. Wennie A. Samper Dental Clinic, and it would be successfully accepted by all stakeholders. The evaluation will look into technical competence and readiness of the clinic staff to move away with manual and paper-based systems to the digital platform in view of the issue of current computer literacy, training needs, and change resistance. It further examines the capability of the system to simplify the existing working processes, cut down on the administrative load and enhance patient delivery without interfering with daily operations. The analysis will be conducted regarding the ability of the suggested solution to practically address the specified issues of scheduling disputes, billing mistakes, lost records, and inability to use the system to its fullest capabilities by integrating the ideas of dentists and other administrative personnel and patients. The operational feasibility analysis will reveal the compatibility of the integrated system with the organizational structure and operational process of the clinic, which will prove that the technology solution is feasible and can be implemented in the existing healthcare setting. The information in this sub section includes a fishbone diagram as a way of establishing the root causes of the operational issues as well as a functional decomposition diagram that shows how the system components are responding to these issues in a systematic way.

Fishbone Diagram

A fishbone diagram is a pictorial representation that illustrates the connection of different factors that contribute to an issue, and it looks like a bone structure of a fish. It is also called an Ishikawa diagram (named after its creator), it is often employed together with the five-whys approach to finding root causes.

Figure 7 presents the fishbone diagram analyzing operational inefficiencies at Dr. Wennie A. Samper Dental Clinic. The analysis revealed critical issues including paper-based records, manual scheduling, training gaps, outdated systems, and communication breakdowns. This structured approach enabled the team to develop targeted solutions and establish a shared understanding among stakeholders, ensuring systematic problem-solving and improved development outcomes.

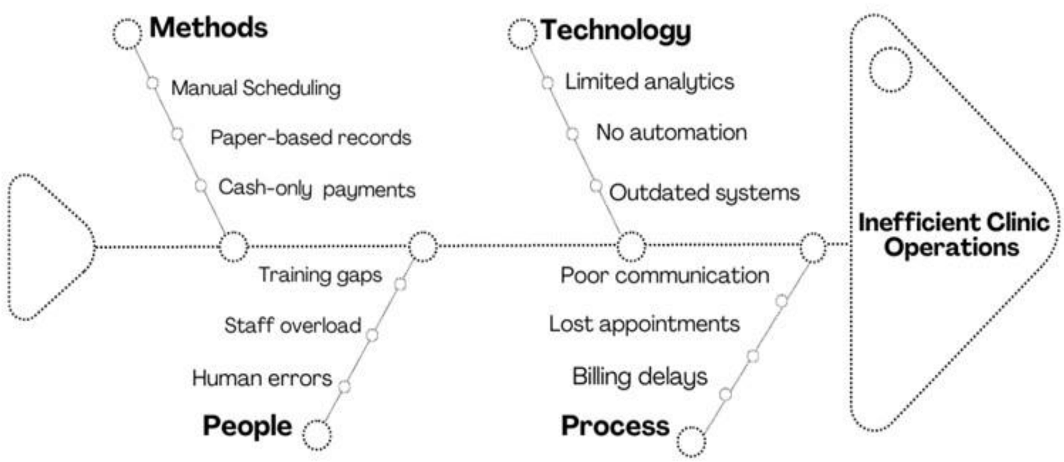

Figure 7. **Fishbone Diagram**

Figure 7 is a fishbone diagram that examines the issue of operational inefficiencies at Dr. Wennie. A. Samper Dental Clinic, divided root causes into four areas, namely Methods, Technology, People and Process.

The Methods category identifies reliance on paper-based records, manual scheduling, and cash-only transactions, indicating a need for digital transformation. Technology challenges include outdated systems, lack of automation, and limited analytics capabilities. People-related factors encompass data entry errors, staff administrative burden, and insufficient training. The Process category reveals inefficiencies in billing procedures, appointment management, and communication workflows.

This diagnostic analysis provides the foundation for developing the Multi-Branched Dental Practice Management System with Point-of-Sale and Business Analytics. The fishbone diagram serves as a strategic roadmap for implementing a comprehensive solution that modernizes operations, enhances data management, and improves service delivery at the clinic.

4.1.9 Risk Assessment Analysis

The Risk Assessment Analysis found out the possible risks and operation issues that would adversely impact the success of the system without addressing them. The biggest threat is that the current use of manual and paper-based operations generates errors in record-keeping, billing errors, and missed appointments causing losses in the administration and money. The risk associated with staff members involves the training issues, workload, and errors of human operators when changing to the new system; the new system should be easy to use, and role assignment should be done properly, and orientation should be carried out to the greatest extent to reduce resistance and abuse of the new system.

Technological risks include out-of-date hardware, bad internet connectivity, insufficient system upgrading, and inadequate robotization and scant analytics would decrease the capability of the system to facilitate data-driven decision-making. Security is a very serious issue because without the appropriate authentication and encryption, unauthorized access or loss of data may take place and, therefore, the security of patient and financial data is crucial to trust and regulatory compliance.

At the level of the process, there exist communication failures, time delays, and billing, which might linger, unless the system modules are integrated and tested properly. To overcome these threats, research is factored in business analytics, POS integration, role-based access control and good security measures as preventive measures.

This risk analysis was used to design the process so that such solutions as automation, analytic dashboards, secure authentication, and workflow integration were considered in the design, to provide technical safeguards and real-world concerns to make the dental practice management solution secure, efficient and reliable.

4.2 Execution Phase

Execution Phase is the heart of the development phase when the system is developed and constructed in the form of Agile Scrum iterative sprints. This stage converts the requirements of the Sprint Planning Phase into a working system using systematic design, development and constant refinement.

4.2.1 Designing

During this stage, the design of the system of the Integration and Business Analytics part of the Dr. Wennie A. Samper Dental Clinic system is planned in detail according to the rules of the Agile Scrum framework. The approach assists in the development of the system using iterative and adaptive cycles to ensure the development of responsive Integrated Dental Practice System. The design will include the core functional requirements which will include multi-branch patient record management, simplified appointment scheduling and the robust/complete Point-of-Sale (POS) integration, and the key non-functional requirements that ensures the reliability, usability and security of that system. These design elements are graphically modeled using Functional Decomposition Diagram (FDD), which splits features hierarchically by user roles and system flowcharts which describe user experiences when performing different tasks. This design is to bring together the various functions and make them cohesive and easy to use where the limitations inherent in the traditional paper-based administrative practice are defeated. With the integrated business analytics utilizing data-driven insights, the system will enable clinic administrators to make effective decisions concerning the allocation of resources, maximizing revenues, and operational enhancement at all levels of branches.

Functional Decomposition Diagram

Functional decomposition is an analysis method which decomposes complex systems into their bits. To the Multi-Branch Dental Practice Management System with Point-of-Sale and Business Analytics at Dr. Wennie A. Samper Dental Clinic, such technique determines and classifies the different activities and functions that comprise the web-based platform. Through breaking down of the system into smaller manageable components, the stakeholders can know how each of these functions affects the overall operation of the system and clinic workflow. The Functional Decomposition Diagram (FDD) is a graphical representation of the same process of decomposition and shows the hierarchical structure of the features and capabilities of the system. The functions are stratified based on the user privilege levels and access rights and hence give a clear picture of accessible capabilities to various user roles within the dental practice setting. Such a structured decomposition will help the development team members, clinic stakeholders and end-users communicate more effectively, since the system boundaries and functional relationships will be better understood. The diagram serves as a valuable tool for identifying potential gaps or redundancies in system functionality before implementation begins. Furthermore, it provides a foundation for creating comprehensive user documentation and training materials tailored to specific roles. Subsequent development stages also have the diagram as a reference point to be developed upon since the developers can easily determine which modules are dependent on each other and how to integrate them systematically. Also, the hierarchical structure facilitates the efficient testing strategies because quality assurance teams can ensure the correct operation of components and only afterwards make sure that they work well together in the whole structure.

Figure 8 The functional decomposition diagram of the Multi-Branch Dental Practice Management System is provided, and the system has been subdivided into three main user roles, which include: Dentist/Admin, Staff, and Patient. The diagram shows the way dentists deal with business analytics, booking, treatment history and point-of-sale services, the way staffing deal with bookings, patient records, inventory and billing transactions and the way the patient accesses appointment booking, personal details and billing.

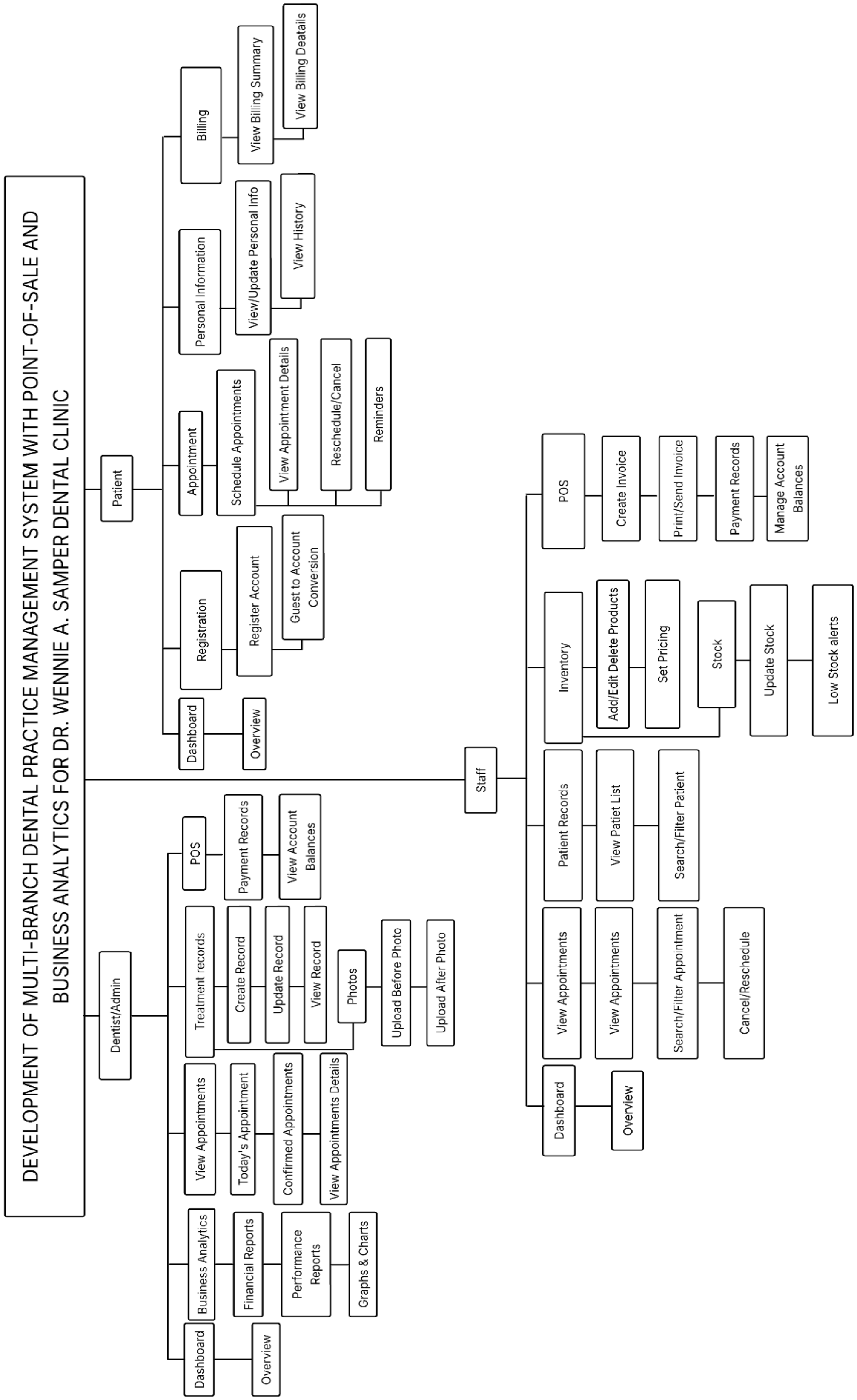

Figure 8. **Functional Decomposition Diagram**

The position of Dentist/Admin is related to strategic management, business intelligence, and clinical management. The Dashboard gives a view of the system, reports on finances including revenue and expenses, performance reports including efficiency of the clinic and graphics and charts. Business Analytics module allows making decisions based on data throughout all branches based on consolidated reporting and trend analysis. The Appointment Management enables dentists to see the scheduled appointments, the current appointments, the confirmed appointments as well as the detailed appointment information such as patient history and appointment plans. The Treatment Records module facilitates clinical recording and has the capability to enter, modify, and access treatment records of patients, as well as control the uploading of before-and-after photos that show the result of the treatment. The Point-of-Sale system helps the administrators to obtain financial records of payment and review account balances, which are used in financial supervision.

The operational core consists of the staff members who have access to five key modules that help with the daily clinic operations. The Dashboard will give access to system status and work description. View Appointments module has appointment management features such as seeing all appointments made, search and filter features and cancelling or rescheduling appointments. Patient Records module allows staff members to access the list of detailed patients and use search tools to find records effectively. Inventory Management module gives an opportunity to add new products, update information, establish prices, stocks and give regular low-stock messages. The Point-of-Sale module will empower the staff to generate invoices with detailed information, enable payments by a variety of methods, and even balance the account balances of patients.

The Patient role offers self-service/ administrative workload reduction capabilities of the role. The Registration module simplifies the process of onboarding where new patients can open accounts with their demographics and insurance details. The Appointment module allows the patient to make new appointments, see the remaining appointments, appointment history, and get reminders to minimize the rate of no-shows. The Personal Information module enables patients to access and modify demographic data, contact information and to check the entire history of treatment. Billing module will allow access to billing summaries, breakdown of charge detail and payment history. The functional decomposition diagram shows that there is a well-planned system architecture that offers multi-branch functions, and data consistency at all sites. The system provides an ideal system to handle contemporary dental practice through its effective structure of workflow by establishing a strict division of duties among the three key user roles and therefore negates redundancy and increases accountability, which makes it a perfect solution to handling dental practice on a broader scale.

Requirements Modeling

Requirements modeling is a methodical procedure in system and software development that entails the collection, assessment, documenting and graphical portrayal of the demands and anticipations of all the involved parties and users in a project. Critical activity that is undertaken prior to design and development undertaking is to enable the project team to have a common and clear knowledge of what the system is destined to achieve. It is through this process that both functional requirements (the actual operations, features, and tasks that the system would need to undertake) and non-functional requirements (quality attributes, such as usability, security, and performance) are established.

Graphical models and modeling notations, including Data Flow Diagrams (DFDs), Use Case Diagrams, and Entity-Relationship Diagrams (ERDs), are normally used in requirements modeling to represent requirements in a better structured and structured format. In the process of creating such models, development teams can minimize misunderstandings, detect problems at an early stage and provide a firm base in designing the system, hence making sure that the end product meets the needs of the stakeholders and the business objectives.

Input

Users securely log in with role-based access. Dentists and staff input patient information, appointments, dental charts, transactions, inventory (medicines and supplies), and service prices. Staff process payments via POS. The system automatically captures all data for reporting and analytics.

Process

The system validates and stores data in a centralized database. Patient records are linked and updated automatically. Dental charts reflect treatment history in real-time. POS calculates totals, generates receipts, and updates inventory automatically. Analytics tools aggregate data to produce insights on revenue, services, and supply usage with real-time synchronization across modules.

Output

The system generates updated dental charts, appointment schedules, transaction receipts, and inventory reports. The analytics dashboard displays visual reports on patient trends, sales performance, and supply consumption. Provides printable records, downloadable reports, and notifications for operations and decision-making.

Performance

Efficiently minimizes manual tasks and errors with fast data retrieval. Integrated modules maintain accuracy and quick access. Responsive interface with stable performance supporting multiple simultaneous users, enhancing clinic productivity.

Control

Secured through authentication, role-based permissions (dentist, staff, admin), and activity monitoring. Data validation ensures input accuracy. Regular backups, audit trails, and error handling maintain data integrity. Control measures ensure security, consistency, and compliance with clinical standards.

Context Diagram

Figure 9 presents the context diagram that illustrates the comprehensive interactions between external entities and the central DWAS Dental System. This high-level view demonstrates how the three primary stakeholders (Dentist/Admin, Staff, and Patient) interact with the system through various data flows and functional processes, providing a clear overview of the system's boundaries and information exchange.

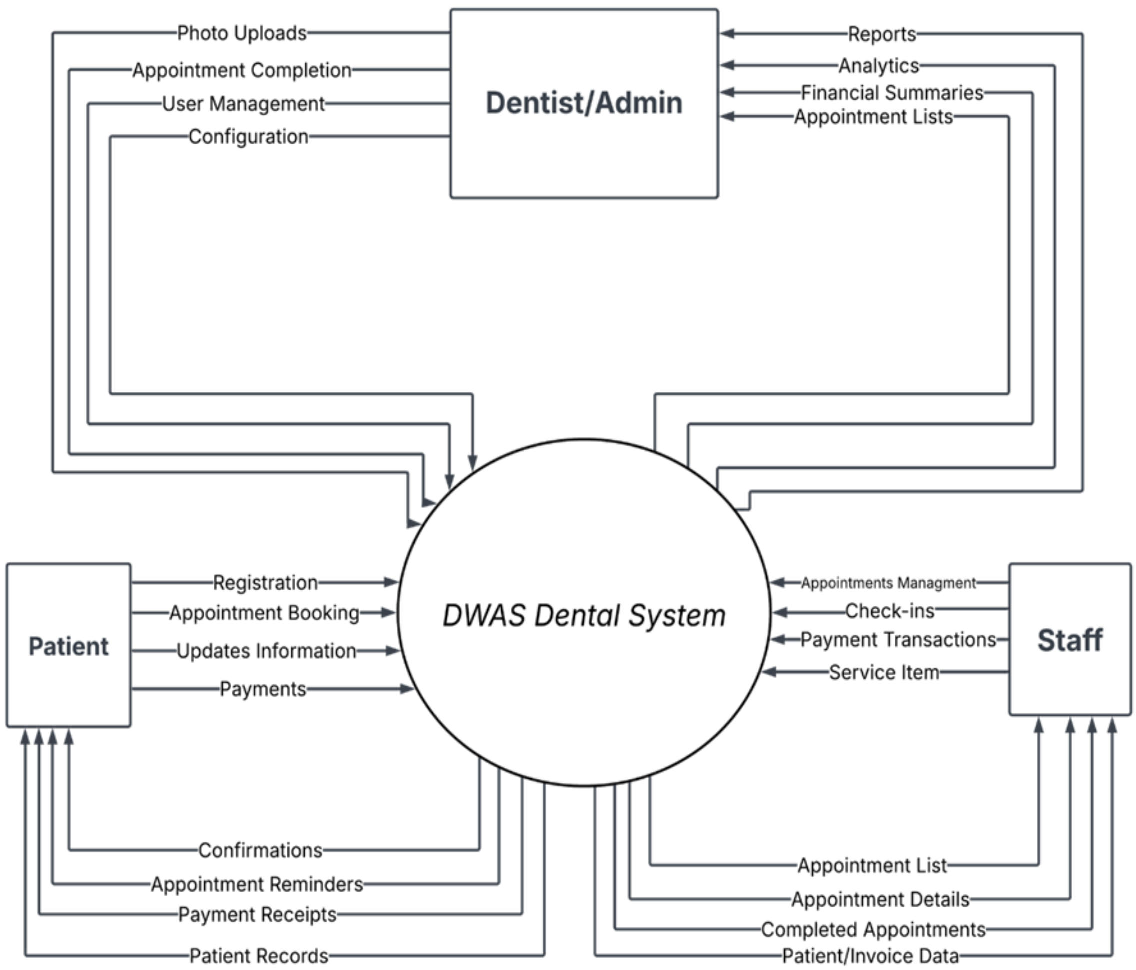

Figure 9. **Context Diagram**

Context diagram illustrates a centralized dental management system where there are three main external parties, namely Patient, Dentist/Admin, and Staff. Patients connect to the system by registering, making appointments, updating information, and payment options, receiving confirmations and appointment reminders, payment receipts, and access to patient records. The Dentist/Admin unit handles the most important administrative tasks such as photo uploads, tracking of appointment completion, user management and system configuration where the output includes reports, analytics, financial summaries and appointment lists. The operational duties of the staff are managed with the help of appointments management, check-ins, payment transactions, and service item management and access to appointment lists, appointment details, completed appointments, patient invoice information. The two-way data interchange between all parties and the central system illustrates an inclusive workflow that underlies clinical functions, administrative control and patient involvement which constitute the collaborative atmosphere in which all parties can effectively discharge their duties.

The context diagram demonstrates that the system is an architecture of a hub-and-spoke model whereby DWAS Dental System is the central processing core in all the stakeholder interactions. This design guarantees data consistency and integrity, as it directs all the transactions through one authoritative system instead of direct communication between the entities. This balanced flow of information is responsible to the symmetrical input and output flows where each stakeholder contributes data to the system and receives information processed by the system.

Data Flow Diagram

The Data Flow Diagram (DFD) of the system represents the data flow of the data on patients, medical and financial transactions in the system. In this DFD, standard notations are used, and the rounded boxes represent processing activities such as "Process Billing" or "Schedule Appointment" and the arrows indicate how the data flows, open boxes represent data storage such as in “patients”. EMR or inventory, and rectangle refer to external individuals such as "dentist, patient, admin." Hence, a comprehensive representation of the whole process is observed, which represents the information about a session with a patient being processed into business intelligence and a change in his/her medical file.

Figure 10 displays a data flow diagram of a dental practice management system, indicating interactions between three main actors (Patient, Dentist/Admin, and Staff).

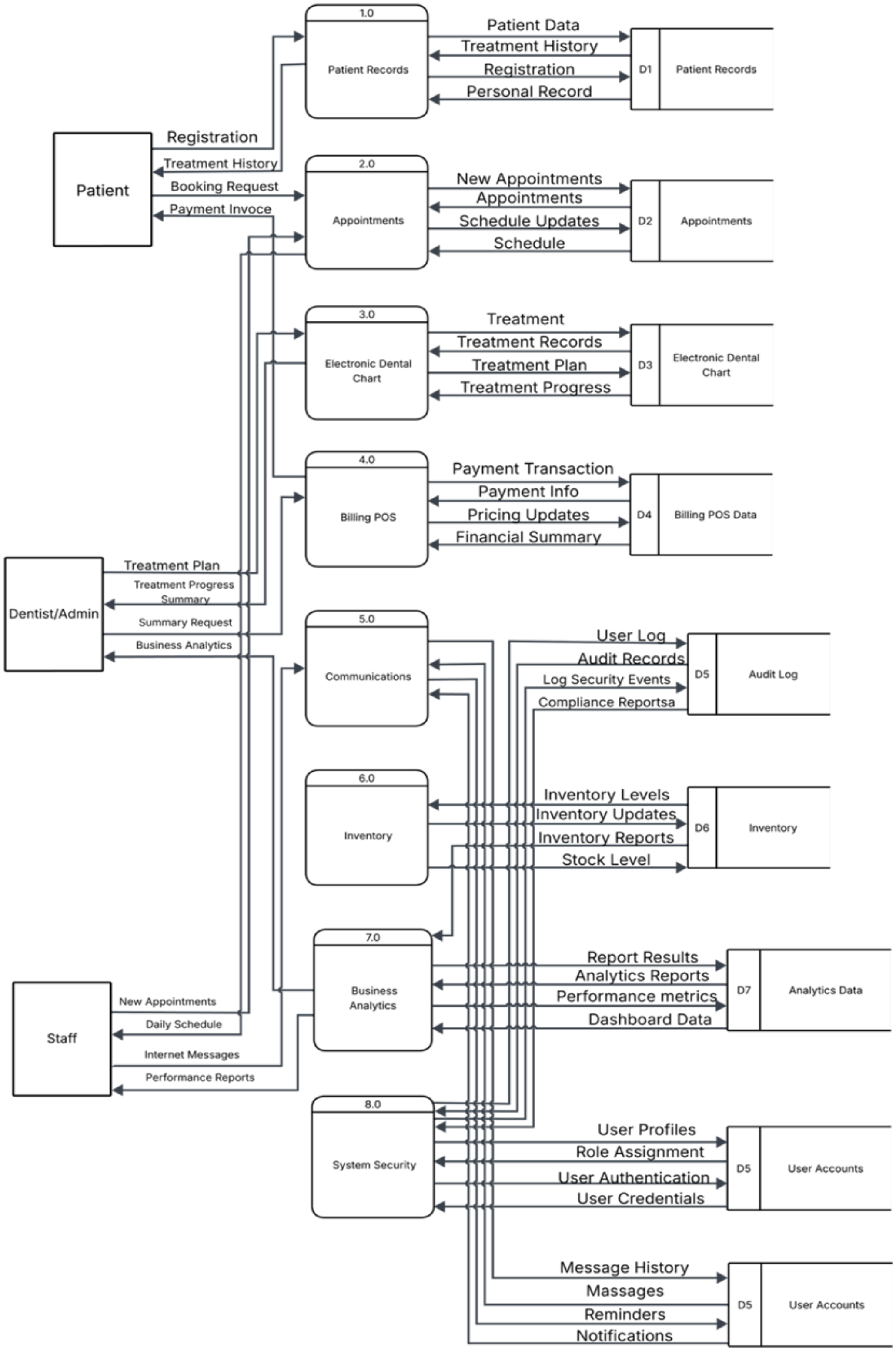

Figure 10. **Data Flow Diagram Level 0**

There is also separation of concerns between the different functional areas in the diagram. The process of Patient Records (1.0) is the backbone, which deals with registering and keeping the history of treatment that is passed downstream. The Appointments system (2.0) is a directional scheduling system that moves workflows with patients and provides the staff with visibility of the schedules. The Electronic Dental Chart (3.0) facilitates clinical processes and the creation of treatment plans and progress, and the Billing POS (4.0) supports all the financial transactions. Administrative operations are spread over Communications (5.0), Inventory Management (6.0), and Business Analytics (7.0), were each keeping their own data stores (D5 to D7). It is important to note that System Security (8.0) is a cross-cutting concern that means that it interacts with all the other processes to enforce authentication and authorization using the User Accounts data store (D5).

The diagram does a good job of showing the flow of information out of the patient-facing processes through the clinical systems to administrative analytics with the role of Dentist/Admin being the central organizer requesting the treatment plan, progress summaries, and business intelligence to inform a decision. This architecture can be used to fulfill the requirements of HIPAA compliance by using the audit logging features and role-based access controls of the security layer. The nature of such interconnection also assures data consistency throughout the system since modifications in one module would automatically spread to the related components and referential integrity would be preserved throughout the database. The multiple data stores (D1 to D7) offer logical separation of concern and allow the efficient retrieval and reporting of data and the stores are specialized to serve a particular functional domain. This modular design enables easier maintenance of the system, testing, and future upgrades since the changes made in the system can be done on the individual processes with little effects on other system components.

Use Case Diagram

Figure 11 shows a use case diagram, with the three main actors (Dentist/Admin, Patient, and Staff) interacting functionally, with a role-based access control to the system functionality.

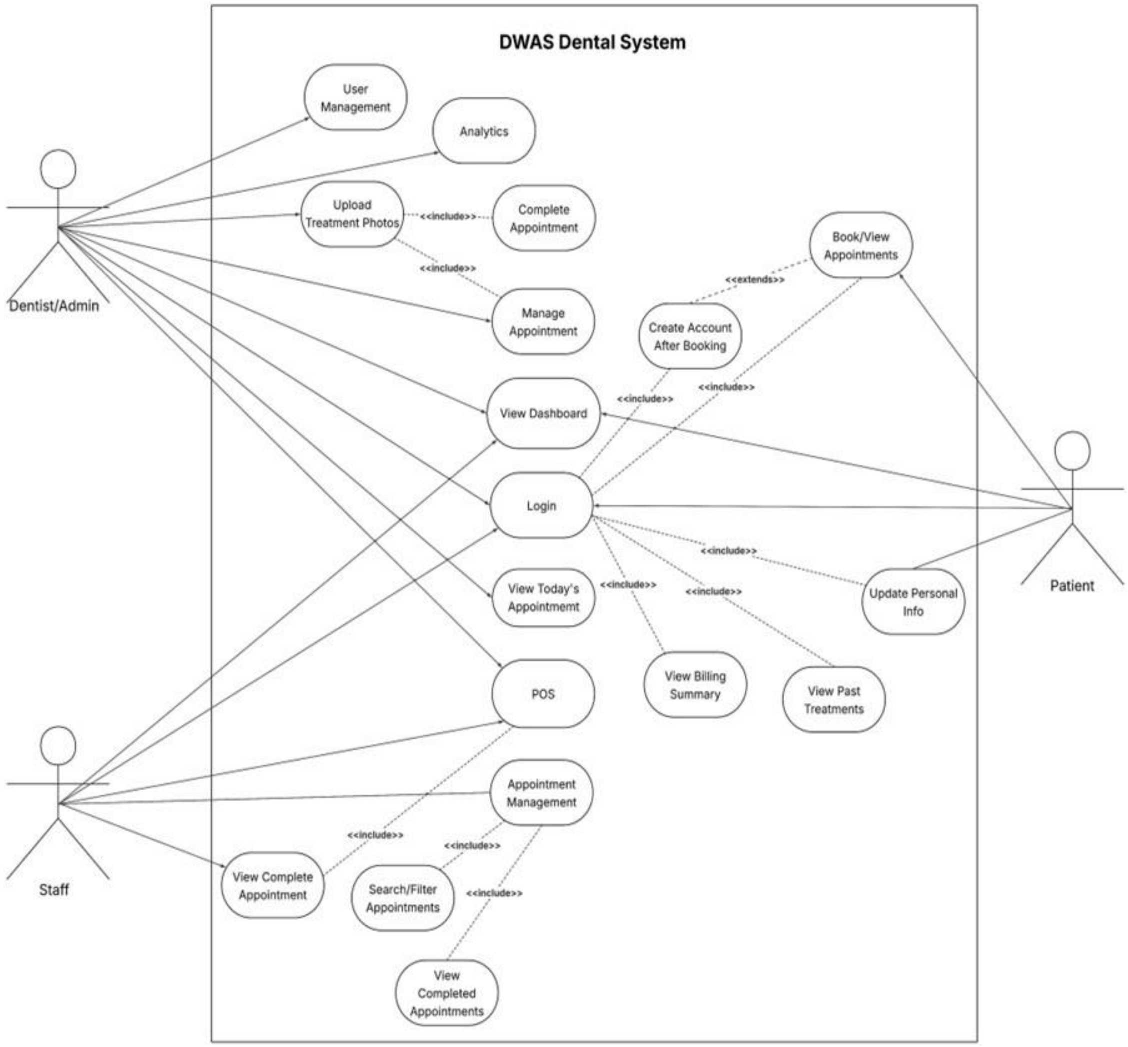

Figure 11. **Use Case Diagram**

The use case diagram demonstrates the hierarchical structure of access in which the Dentist/Admin actor has the most comprehensive system privileges, such as user management, analytics, photo treatment uploads, appointment completion and management, viewing the dashboard, user login access, and POS (Point of Sale) functions. The system mainly communicates with patient actors using self-service features to book and see appointment, create an account upon booking, log in, update personal details, see a summary of bills, check previous treatment, and see the appointment of the day. The staff members will receive operational access that is oriented on appointment management, viewing full appointments, appointment management with the search and filtering options, and viewing the already completed appointments. The diagram illustrates several include and extend relationships, especially, the booking process had extensions to account creation and number of use cases that integrate the use of the login functionality reflecting the modular architecture of the system. This type of design guarantees a good separation of concerns between the various types of users and has workflow effectiveness and data protection.

Object Modeling

Object modeling is a system analysis and design methodology that models important concepts as objects having data (attributes) and behavior (methods). It finds a wide application in object-oriented programming (OOP) and software engineering as a way of visualizing the interaction and communication between various elements of the system. The approach of modeling is aimed at determining real objects of the system, defining their features, and showing their interrelations. Object modeling tools like sequence diagrams are common UML diagrams that allow the developers and other stakeholders to gain an insight into system structure and behavior. Object modeling creates simple complex designs through system structure, facilitates reuse, and allows development through modules, thus simplifying the maintenance and subsequent expansion of software systems.

Sequence Diagram

A Sequence Diagram is a UML interaction diagram that visually presents the sequence of messages or interaction between objects in a system as they occur with time. It brings into focus the series of occurrences that take place in each situation where objects do interact with one another to perform a certain function or process. In a sequence diagram, the objects or entities are associated at the top and time flows downwards along vertical lifelines.

Horizontal arrows depict message or method call transmission between the objects and thus what is occurring when and in which order is easy to observe. The sequence diagrams are much used in the software analysis and design to establish the system behavior, check logic, and allow smooth communication between components. They assist developers and other concerned parties in picturing dynamic processes and are therefore useful in system documentation besides the implementation of plans.

Figure 12 illustrates the interaction between the dentist/admin, system, and database during login, patient management, treatment updates, and report generation.

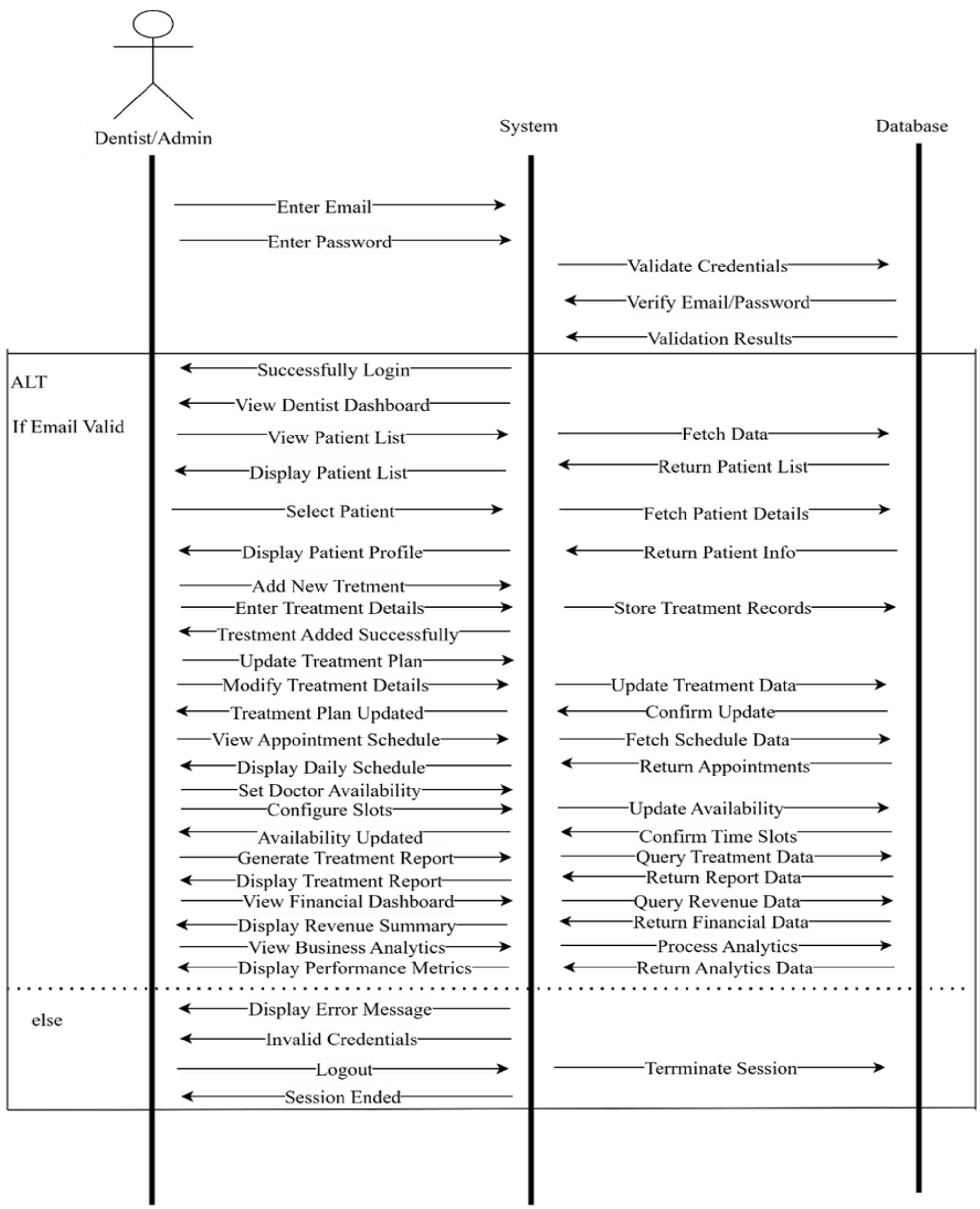

Figure 12. **<u>Sequence Diagram (Dentist/Admin)</u>**

Figure 12 This sequence diagram shows how the dentist/admin, system, and database in a dental management system interact to describe how the dentist or the admin may log in, handle patient records, update treatments, and schedule and record keeping. The system works by the system receiving the email and password of the dentist or the system user, through which the system will verify the credentials entering the system; when the email is confirmed as correct then the user is successfully logged in and he or she gains access to the dentist dashboard. The dentist, sitting at the dashboard, can view and select a patient in the list of patients and this makes the system to draw and display the information of the patient in the database. The dentist is then able to add or change the types of treatment or alter the treatment plans and the changes are logged and validated in the database. The system additionally allows the dentist to view and manipulate the appointments schedules, which involve the appointment availability and doctors time slot adjustment, whereas the appointment validation involves the query of the relevant data and its updating in the database. Moreover, the system will be able to generate various kinds of reports such as treatment reports, financial summaries, revenue analysis as well as performance measurements where the system automatically gathers and processes the necessary data in the database to display the current analytical and financial dashboards. The system will include an error message and end the session in case the invalid login credentials are entered. The flow of structured messages shows the distinct separation of concerns between the presentation, business logic and data persistence layers, which guarantee maintainable and scalable system design. Such an interaction pattern is complete, and the data integrity is assured by the correct validation and confirmation processes in every point of critical operations. Overall, this figure illustrates the whole workflow and chain of communication between the dentist/admin, system, and database to perform administrative, treatment, and analytical procedures within the dental management setting.

Figure 13 illustrates the interaction between the staff, system, and database during staff login, patient management, and appointment handling processes in a healthcare management system.

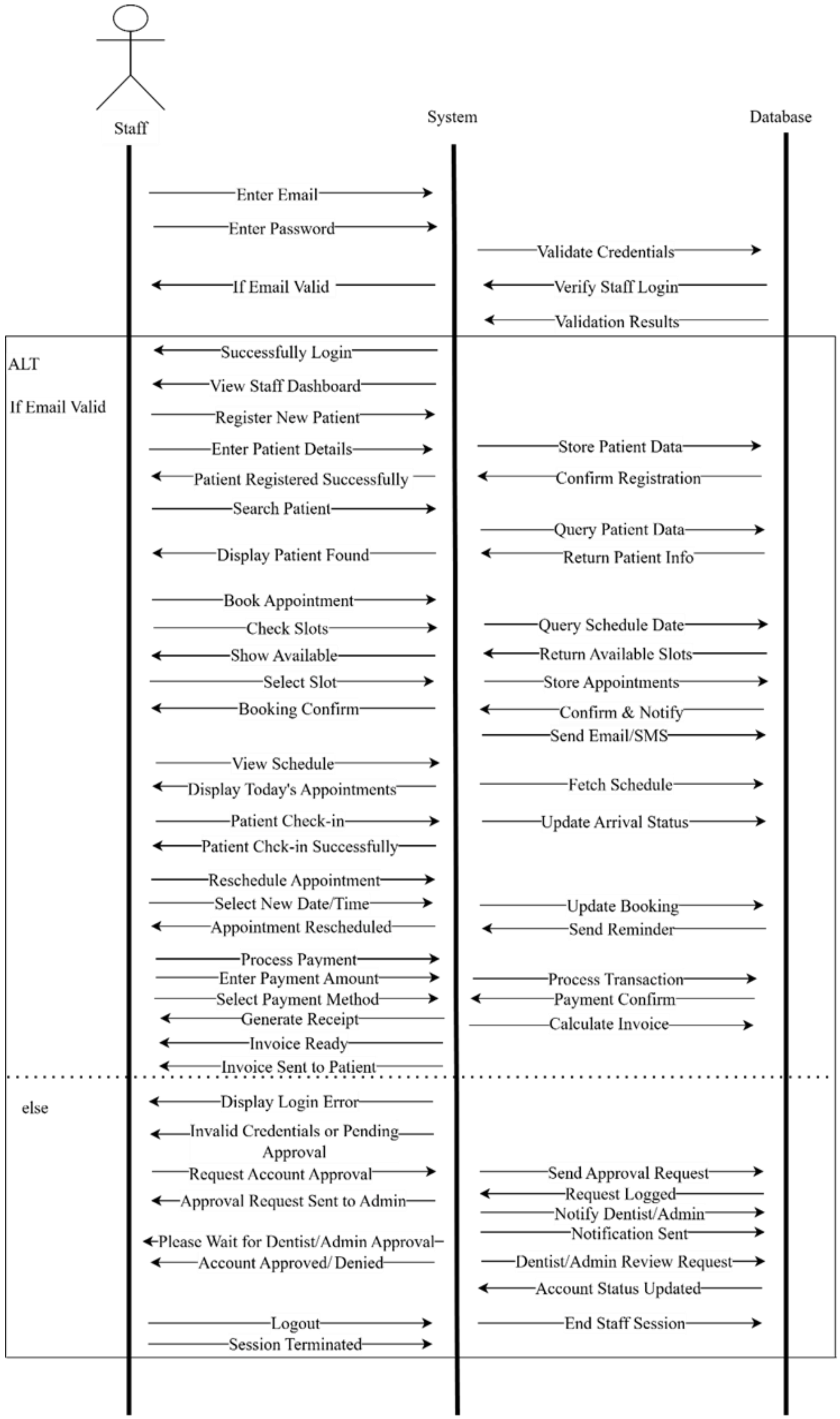

Figure 13. **Sequence Diagram (Staff)**

Figure 13 This sequence chart shows how the interaction between a staff user, the system and the database in a healthcare or an appointment management system take place with the processes of a staff user to log in, work with the patient records and book an appointment. It begins with the input of the email and password by the staff which is sent to the database where the credential is determined as either valid or invalid, in case of the former, then the staff is successfully logged in and can access their dashboard. The personnel over there are able to enroll a new patient by completing the patient demographics which are stored in the database, and they are verified upon successful enrolment. An existing patient can then be identified by the staff whereby the system will retrieve and present the specific patient in the database. The staff will then be able to make an appointment by searching through vacant slots and selecting one and confirming the appointment, and the system will update the schedule to the database and send an email or SMS notification to the patient. The staff can also display the schedule and mark the check-in of a patient, and the database will record the arrival status when the patient arrives at the time of appointment. On a case where an appointment is to be rescheduled, the staff chooses a different time, and the system does the update in the database. When processing the payment, the system will calculate the invoice, make the payment, send a confirmation to the database and subsequently an invoice will be generated and channeled to the patient. In case of email validation failure during the logging in, the system will not allow the user to proceed and instead the invalid credentials message will be displayed, and an account approval or review request will be sent to the database and administrator to review and then the staff can be informed about the account status. This workflow focuses on role-based access control and administration control where only the members of staff who have the right to perform the sensitive operations can do so after a due process of authentication and approval. The diagram also shows automated notification systems that make patients aware of the appointment cycle, improve the efficiency of communication and decrease the number of no-shows. Finally, as the employees log out, the system ends the session to end a transaction.

Figure 14 illustrates how the patient, system, and database interact step by step, demonstrating the way in which a patient log in, handles personal and appointment information, pays, and logs out of the dental management system.

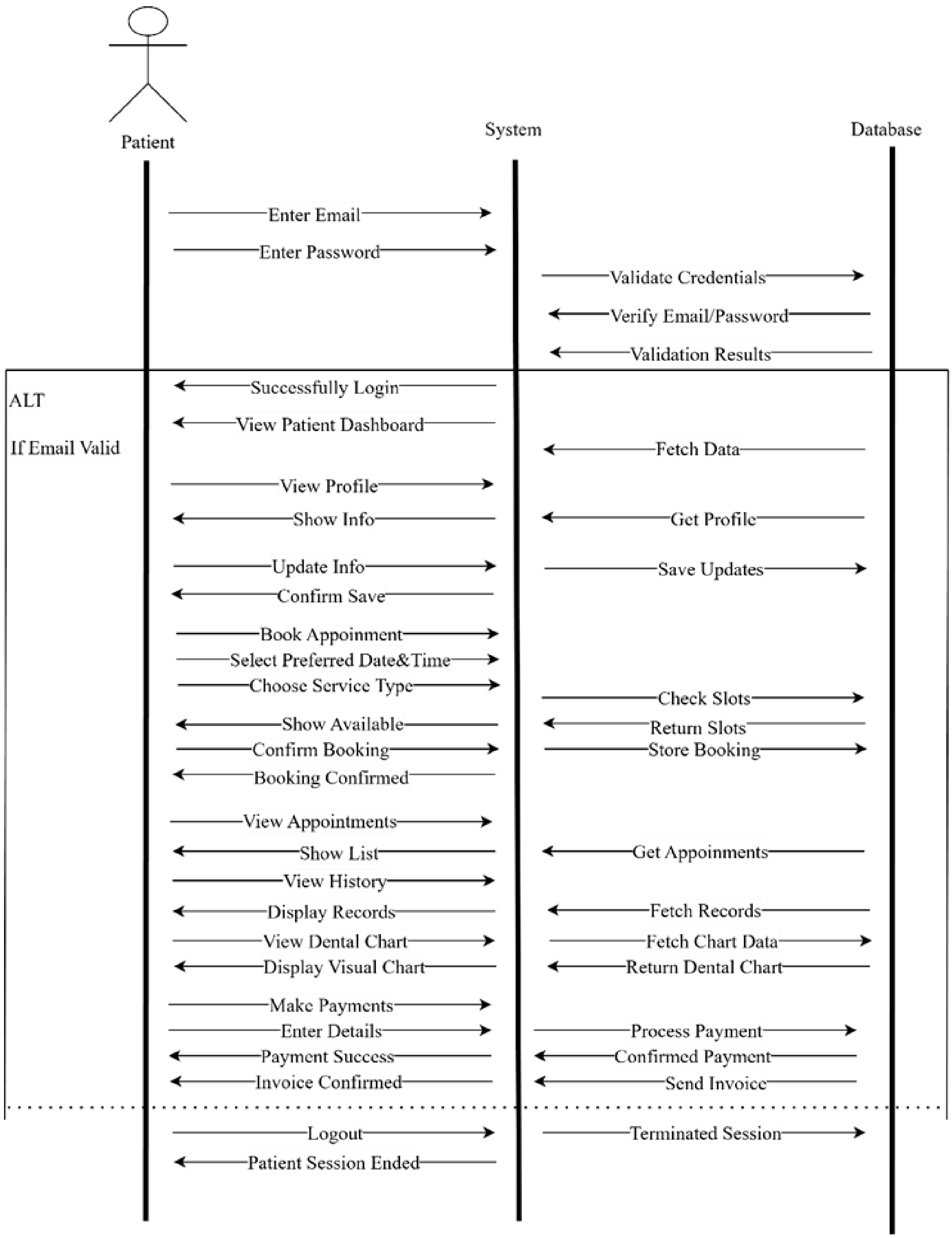

Figure 14. **Sequence Diagram (Patient)**

Figure 14 shows a sequence diagram of an online dental management system interaction process between the Patient, the System, and the Database that illustrates how a patient logs in, updates their personal and appointment details, and performs transactions. It starts when a patient enters their email and password, and the system verifies the data with the database through credential checks, allowing the patient to log in and view or update their profile. When the patient enters the desired date, time, and type of service, the system verifies available slots in the database, confirms the booking, and stores the appointment data. The diagram demonstrates the sequential flow of operations and the communication between different system components throughout the patient journey. Additionally, the diagram shows how patients can view their treatment history, access dental charts, and make payments, with each action triggering corresponding database operations.

Data Design

The Data Design step is aimed at establishing an appropriate and safe background of managing, storing, and organizing all the clinical, financial, and operational data of Development of Multi-Branched Dental Practice Management System with Point-of-Sale and Business Analytics to DR. By transforming the requirements into specific data models, relational designs, and database specifications that enable the achievement of accuracy, efficiency and scalability, in addition to supporting patient records, appointment scheduling, billing automation, inventory tracking and business analytics, in turn, enables the dental clinic to run its operations smoothly, ensure sensitive patient information is kept and reliable insights made on the basis of which informed decisions are made.

Entity Relationship Diagram

Entity- Relationship Diagram (ERD) is a data modeling diagram which aids in showing the arrangement of a database. The entities (concepts or objects), attributes (properties), and relationships between them are presented in ERDs.

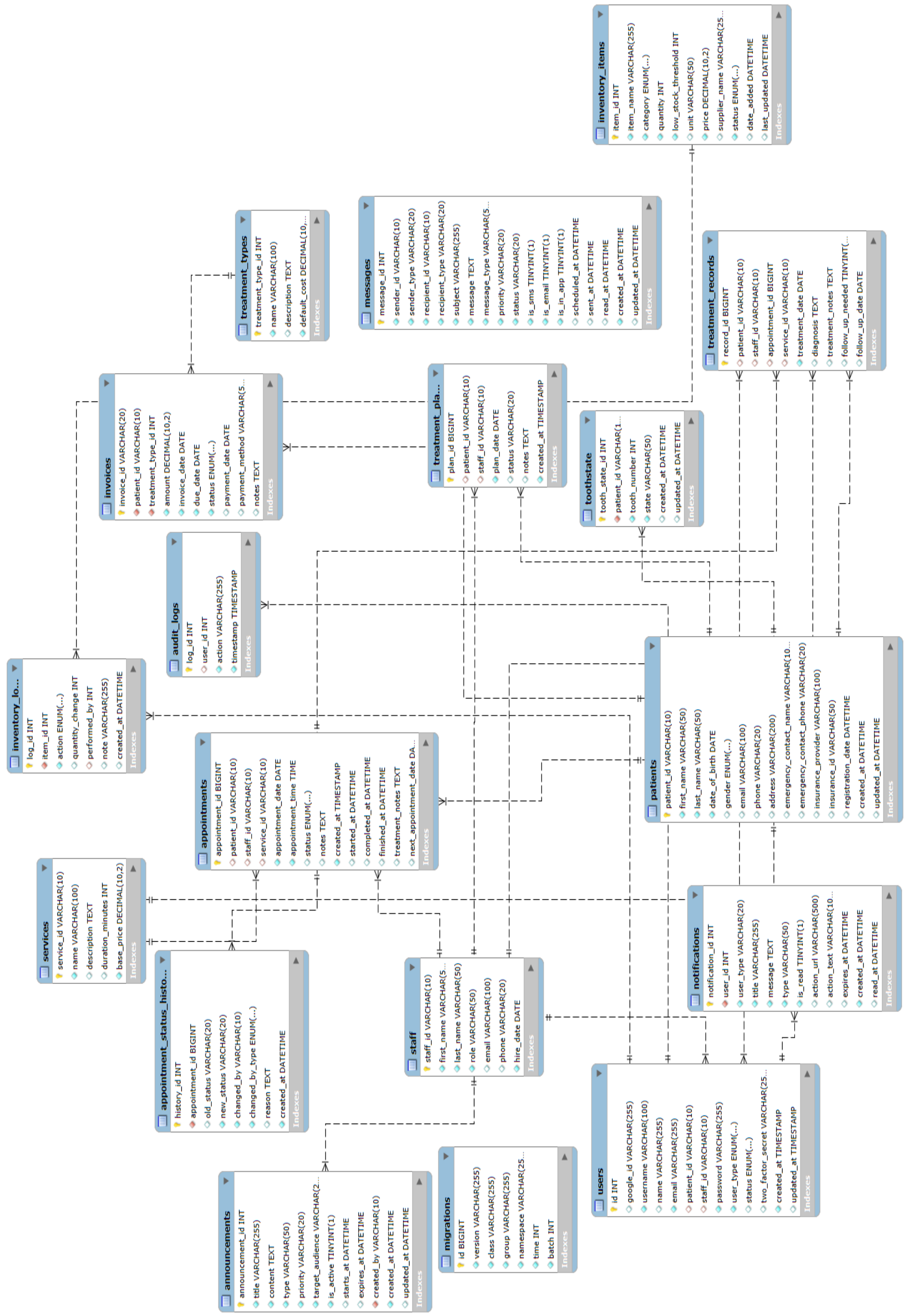

Figure 15. **Entity Relationship Diagram**

In this database, practice was organized around 17 tables that are connected. Patient management (demographics, records), appointment scheduling (status tracking and notifications), clinical operations (treatment records, plans, and types), billing (invoices and services), staff coordination, inventory tracking (medical supplies), and communication (messages, announcements, notifications) are the core functionality. It serves as a very all-encompassing system in the management of patient care, administrative work, and operational processes in a healthcare environment because the schema has audit logs on compliance, user authentication, and tooth state tracking specific to dental procedures.

4.2.2 Development

Development proceeded in a formal but iterative manner, with a few elements of Agile methodology to merge agility with comprehensive documentation and testing. The team then entered the development phase following the planning phase where issues of lack of operational effectiveness in areas like manual record keeping, ineffective scheduling and billing errors were realized. All the modules were tailored to the needs and project goals of the clinic.

The development team designed the system in iterations starting with mock-up designs of user interface and database schema, and continuous coding and integrating modules. With the iteration, it was accompanied by a regular review where functionality was tested based on usability, performance, and security tests with the output of these tests used to refine the next iteration. It was necessary to have collaboration between developers, clinic staff and stakeholders in order to make sure that the input of dentists, administrative staff and patients are included to maximize the usability and real-world applicability.

The core functionalities such as the Electronic Medical Records (EMR), Electronic, etc. Dental Chart (EDC), Appointment Scheduling, billing with POS, Business Analytics Dashboard, Inventory Management, Communication Workflow and Role-Based Access Control were developed and integrated, extensive testing was initiated. To ensure correct and consistent user input produced accurate results, black box testing was used to test functionality without analyzing internal code. We would complement testing with usability questionnaires and performance indicators to determine the efficiency, the error rates, and the user satisfaction.

The deployment followed the satisfaction of performance, functionality, and security requirements. The system was implemented in a real clinical setting at Dr. Wennie A. Samper Dental Clinic and automated record management, reduced cases of double booking, automated billing using real time POS integration, and offered business insights using business analytics. The successful implementation showed that the system was not only operational in the technically expected way but that it provided significant changes in operational efficiency, communication, and patient satisfaction. Under this development project, the Integrated Dental Practice System was born as an overall and easy to use platform that was successful in modernizing the clinic processes, making informed decisions as well as providing a future scalable foundation.

Output and User-Interface Design

This part explains how the system provides well-formatted, correct, and concise outputs like appointment schedule, patient records, receipts, reports and analytics dashboard that assist in the daily clinical and administrative activities. The user interface is set in such a way that it is easy to use and intuitive with consistent design, layout and basic navigation being given to ease of use by the various user roles. UI interfaces are structured and delivered in terms of the system core modules where each module discussed a particular area of functionality of the dental practice management system. Accessibility and responsive design are also integrated with a view to providing an efficient interaction system with different devices. The annotated prototypes of the main elements, such as the point-of-sale (POS) interface and dashboards, are shown to show how the information is presented in an organized way and the actions are fulfilled with minimum steps. Comprehensively, the output and user-interface design enhance efficiency, minimize errors, facilitate data-based decision making, and enhance easier and more consistent user experience to the dental practice management system.

Figure 16 displays the patient management interface from the dentist's side.

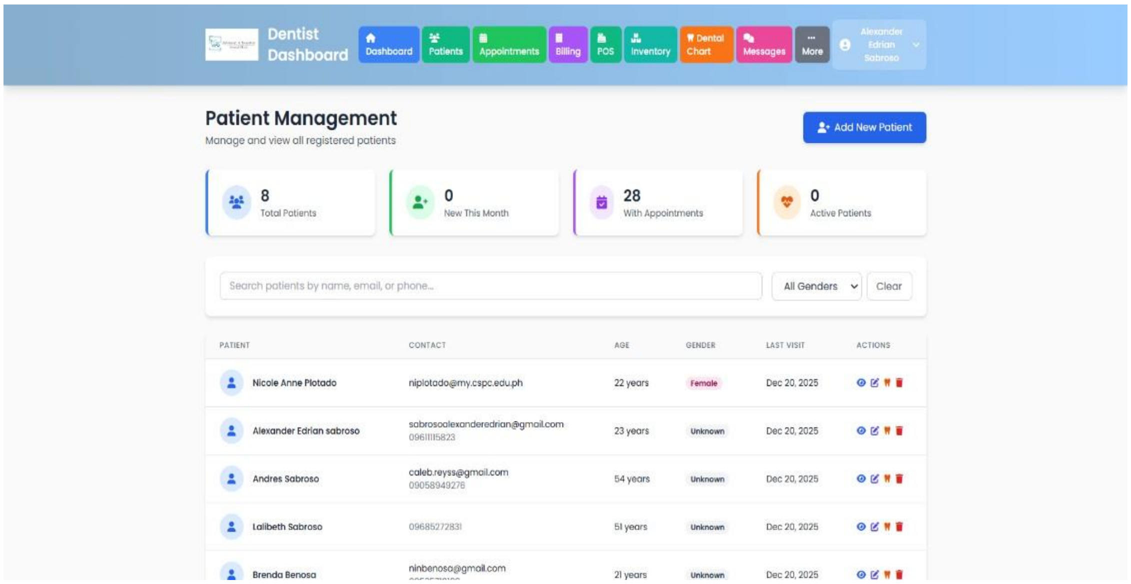

Figure 16. **Patient Management (Dentist Side)**

The patient management interface is the centralized point in which the patient information can be organized and accessed in the dental clinic system. The dashboard has a statistical overview section which displays real time patient metrics to ensure that dentists can track the growth of their patient base, appointment schedules, and the status of their patients without the need to view any of the sections. The interface also includes powerful search and filter systems, and healthcare providers will be able to find patient records with ease by using various filters such as the name, email, phone number, and demographic filters. The tabular format of the layout contains the necessary information on the patient in a systematized form where the columns in the table are clearly defined, and the indices of patient identification, contact information, age, gender typing, and visit history are displayed. This organized presentation ensures that staff can quickly scan through multiple records without experiencing information overload. The action controls have been placed strategically on every patient record to allow access to detailed profiles immediately, update the information available or delete the records in the system.

Figure 17 displays the patient records interface from the staff side, providing comprehensive tools for searching and managing patient information efficiently.

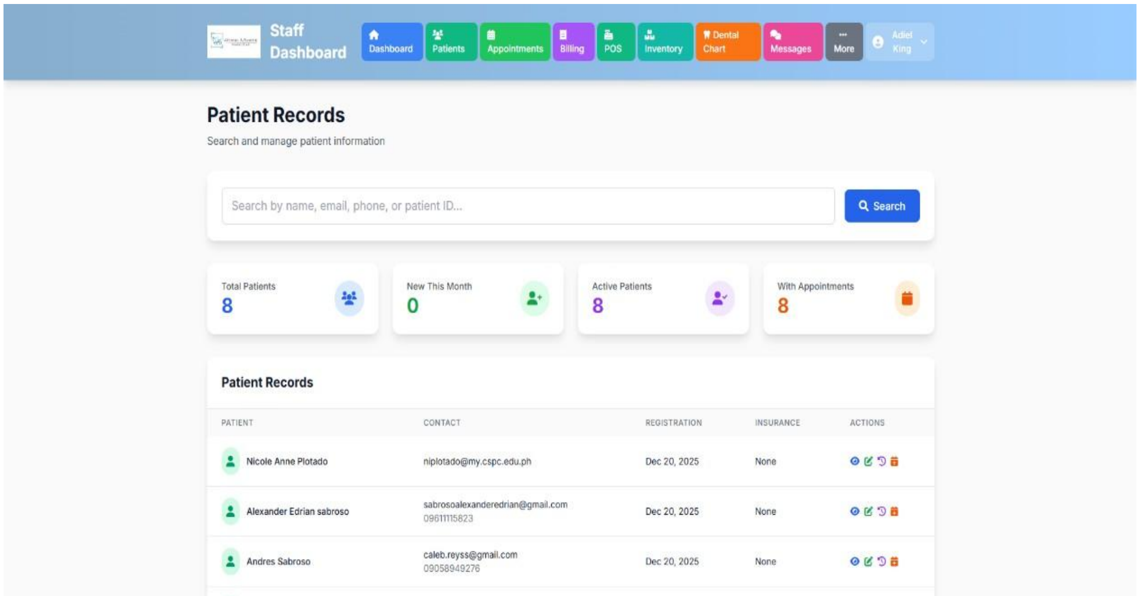

Figure 17. **Patient Management (Staff Side)**

The interface of the patient records allows the management of the work with the patients and the retrieval of the information about the patient with the help of the search bar which takes as parameters different parameters: name, email, phone number, unique patient identification codes. The dashboard also includes four essential performance indicators in the form of colored cards, which allow the staff to see the real-time access to the total amount of patients, number of monthly registrations, status of the active patients, and bookings to enable the staff to track the movement of patients and the use of the clinic. The patient records table is organized in a clean and systemic format and the columns used in the table have consisted of patient identification, contact details, registration dates, insurance coverage, and action control that allows instant access to functions such as viewing full profiles, updating details, down-loading records, and deleting entries. The interface is designed in a way that meets the needs of the staff and workflow efficiently, enabling them to take care of routine patient management functions with a few steps without losing the accuracy of their data and having a balanced administrative and clinical integration.

Figure 18 displays the patient profile interface from the dentist side, providing comprehensive access to patient information and treatment history.

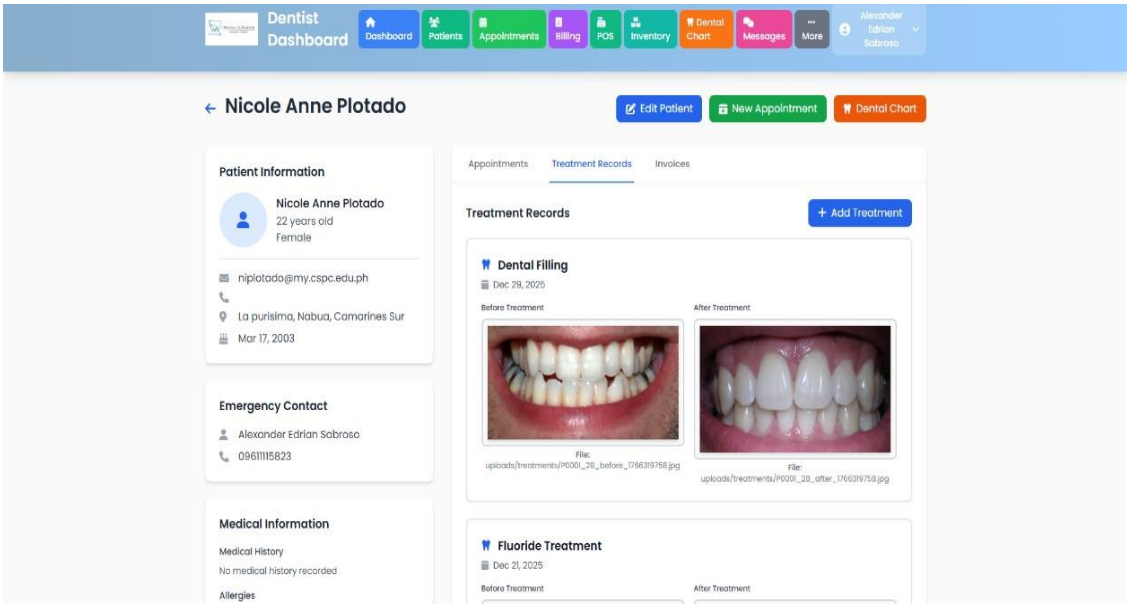

Figure 18. **Patient Profile Interface**

The patient profile interface consolidates all patient information and clinical history in a unified format for easy care provision and treatment tracking. The left sidebar displays key patient details including age, gender, contact information, physical address, date of birth, and emergency contacts in organized sections for quick access during consultations or emergencies. The main section features tabbed navigation separating patient information into appointments, treatment records, and invoices for easy navigation. The treatment records section includes a visual documentation system with before and after clinical photos, enabling dentists to track treatment progress and maintain visual records of outcomes. Each treatment entry shows procedure type, date, and clinical images with file path references for efficient recordkeeping. This comprehensive documentation approach supports both clinical decision-making and patient education by providing clear visual evidence of treatment effectiveness. The system also maintains a complete audit trail of all patient interactions, ensuring compliance with healthcare recordkeeping standards and facilitating continuity of care across multiple visits. Top action buttons provide quick access to essential functions like editing patient details, scheduling appointments, and accessing dental charts for seamless patient care management.

Figure 19 displays the dental chart system interface, providing access to patient dental charts for viewing and editing.

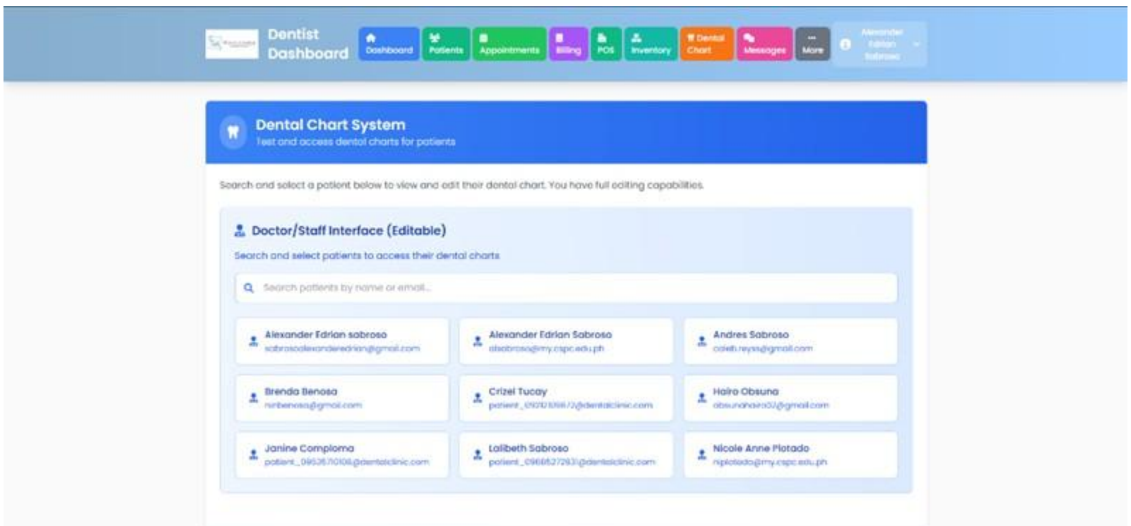

Figure 19. **<u>Electronic Dental Chart Module</u>**

The interface of dental chart system is the main entry point and access point to patient dental record management and search engine-based navigation system. The interface has a title bar prominently positioned that indicates well what the module is all about and contains teaching information on how the users can fully edit it. The doctor and staff interface part also has a search feature that allows patients to be searched by name or email quickly which simplifies the process of finding a particular record of the patient. The search bar is followed by the list of patient records arranged in a grid format and each card has the name of a patient and contact details to help the user scan and select them easily. The patient cards serve as a clickable entry point to the respective dental chart making navigation through the specific clinical documentation quick. The interface design focuses on accessibility and effectiveness by presenting patient data in clear and arranged format in that a user can find and access patient dental charts within a very short time when performing their clinical duties.

Figure 20 displays the interactive dental chart interface, enabling clinical documentation of tooth conditions and treatments through a visual representation of patient dentition with color-coded status indicators.

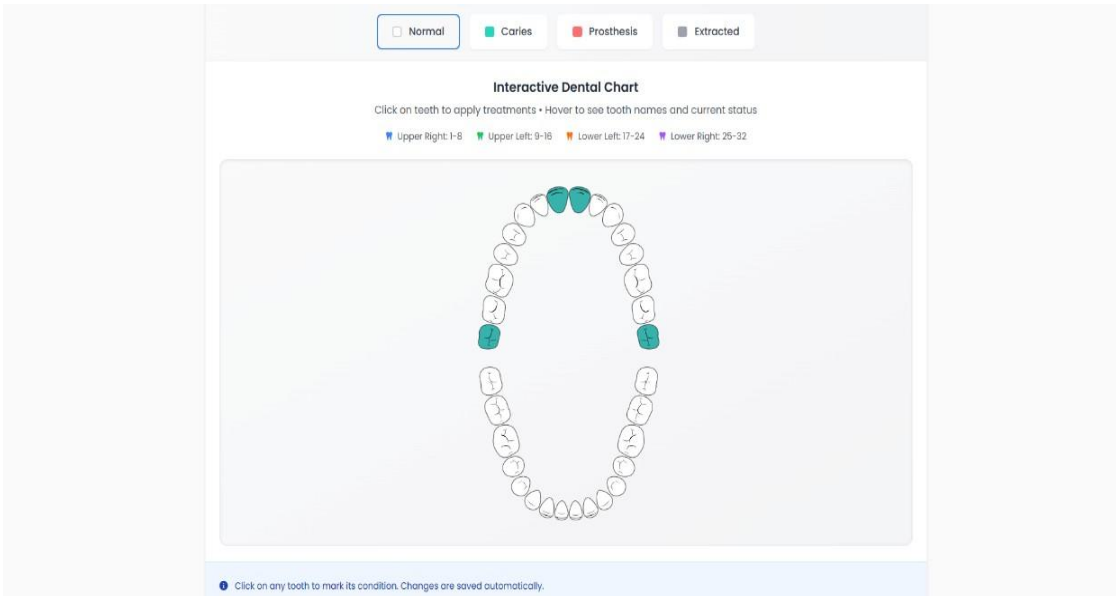

Figure 20. **Interactive Dental Chart Interface**

The interactive dental chart interface presents a visual display of patient dentition allowing real-time updates of tooth conditions and treatments. A color-coded legend at the top classifies tooth status as normal, caries, prosthesis, and extracted for easy identification of treatment states. The main dental chart shows complete adult dentition arranged in four quadrants (upper right, upper left, lower left, lower right) following conventional dental nomenclature. All teeth are clickable with visual feedback through color changes indicating the selected treatment category. The hover feature displays tooth names and status to assist proper documentation and reduce identification errors. This intuitive design enables dentists to quickly update multiple tooth conditions during patient examinations without interrupting the consultation flow. The system maintains a complete history of all chart modifications, allowing practitioners to review changes over time and track treatment progression across multiple visits. Additionally, the interface supports standardized dental coding systems, ensuring consistency in documentation and facilitating seamless communication between dental professionals. Instructional text guides users through the interface, with automatic saving of all changes to preserve data integrity and minimize disruptions to clinical workflow.

Figure 21 displays the appointment management interface for patients, allowing them to view, manage, and book their dental appointments with status tracking and scheduling options.

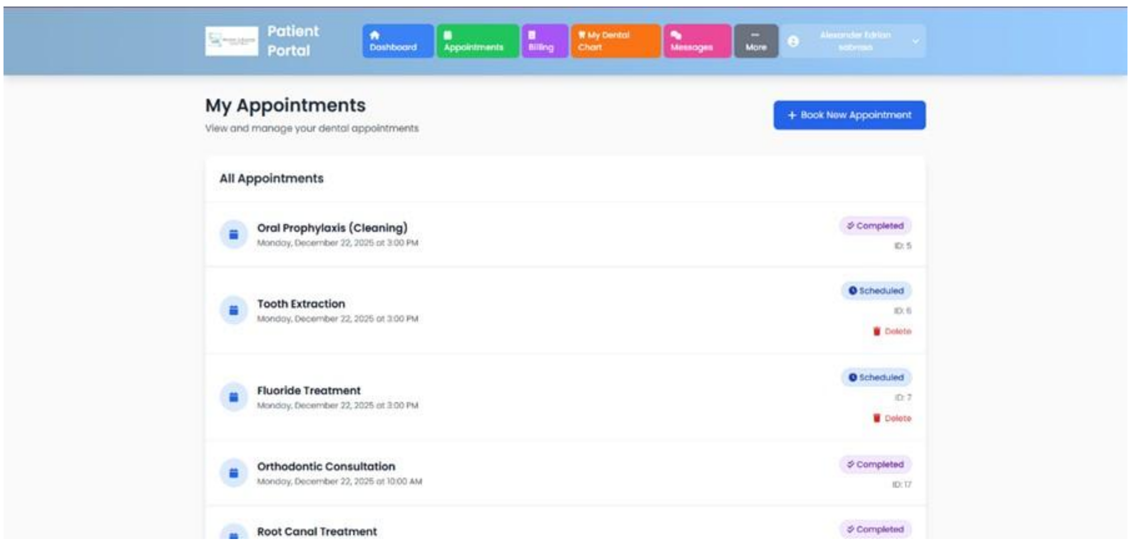

Figure 21. **Patient Appointment Interface**

The appointment management interface allows the patients to access their future and previous dental experiences at a centralized point where all the scheduled and completed dental procedures are shown. The interface presents the appointments in a chronological order with significant information about the procedure name, date, time, and identification numbers which can be used as reference. Status indicators will be presented as color code badge to determine the completed and scheduled appointments and help the patient have easy access to assess previous and future assignments. In the case of scheduled appointments, the system has a cancellation feature that will remove the necessity of making phone calls or e-mails to the clinic. This self-service capability enhances patient convenience while reducing administrative burden on clinic staff who would otherwise handle cancellation requests manually. The interface also provides appointment reminders and notifications to help patients stay informed about their upcoming visits and reduce no-show rates. The scheduling workflow is easily accessible, with the Book New Appointment button in the upper-right corner that promotes the involvement of patients and self-management of their oral health.

Figure 22 displays the appointment booking interface for patients, providing a structured form for scheduling new dental appointments with branch selection, service options, and emergency contact information fields.

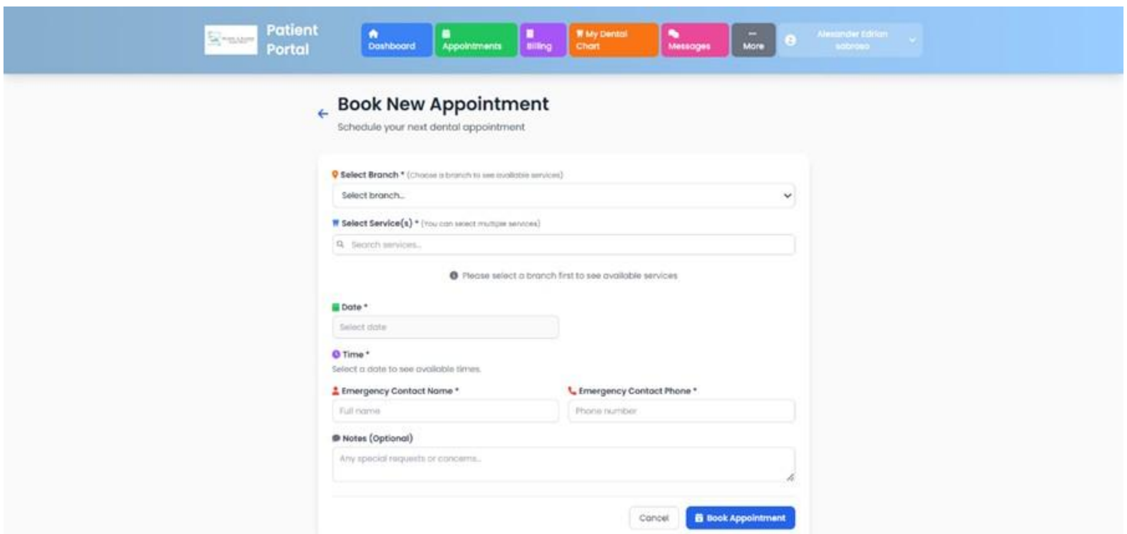

Figure 22. **Patient Appointment Booking Interface**

The booking appointment interface eases the process of booking an appointment by having a step-by-step form that directs the patients on the required information. It starts with the selection of branches where patients are given an opportunity to select a desired clinic location and see what services are available. In the service selection area, there are several choices activated by a searchable dropdown menu, allowing a patient to reserve various procedures and do them at the same time. The date and time fields become conditional on earlier selections and time field shows only available slots on selecting a date to avoid scheduling conflicts. Emergency contact fields will provide the clinic with an opportunity to contact other contacts in case of necessity. The option of notes section enables the patients to provide special requests, concerns or any other medical related information. There are easily accessible action Buttons at the bottom that offer easy navigation with a cancel and confirm option to enable patients to be in full command of the booking procedure.

Figure 23 displays the point-of-sale interface for processing patient transactions, featuring product selection, cart management, payment processing, and real-time billing functionalities for dental services and supplies.

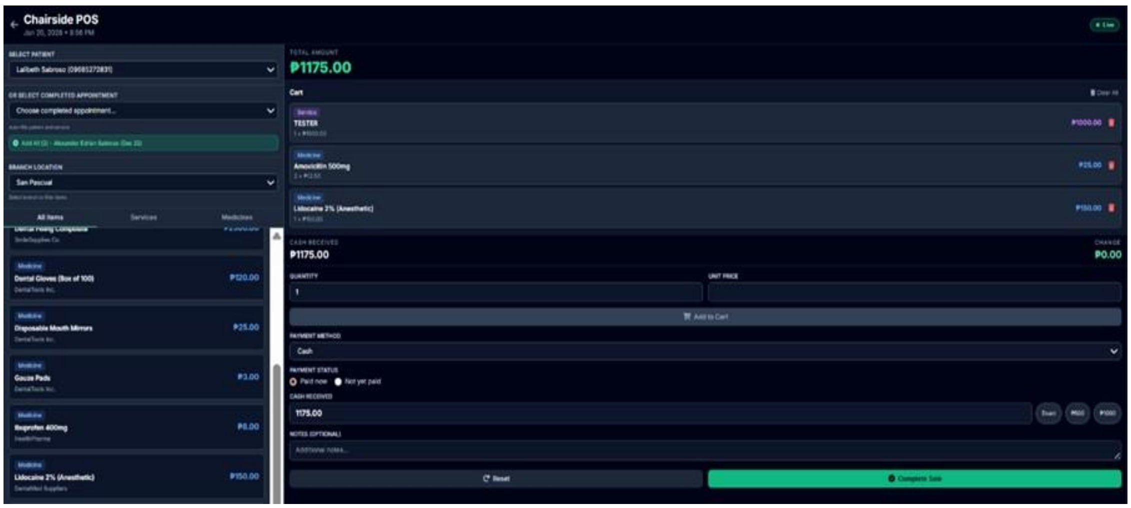

Figure 23. **<u>Point of Sale (POS) Interface</u>**

The point-of-sale interface offers a comprehensive transaction management system integrating patient selection, appointment linking, and multi-item billing into one workspace. The left side displays a categorized list of available items organized by tabs (all items, services, and medicines), enabling staff to quickly find and add products to the transaction cart. Patient selection and appointment dropdowns at the top allow staff to associate transactions with specific patients and appointments for accurate billing records. The right panel shows the working cart with selected items, prices, quantities, and removal options for transaction adjustments. Payment processing controls include payment method selection, cash received input with automatic change calculation, and payment status indicators. Denomination buttons expedite cash processing, while an optional notes field captures special transaction information. The Complete Sale button finalizes payment and generates receipts for clinic archives and patient records.

Figure 24 displays the inventory management interface for tracking medicines, supplies, and dental services across multiple clinic branches.

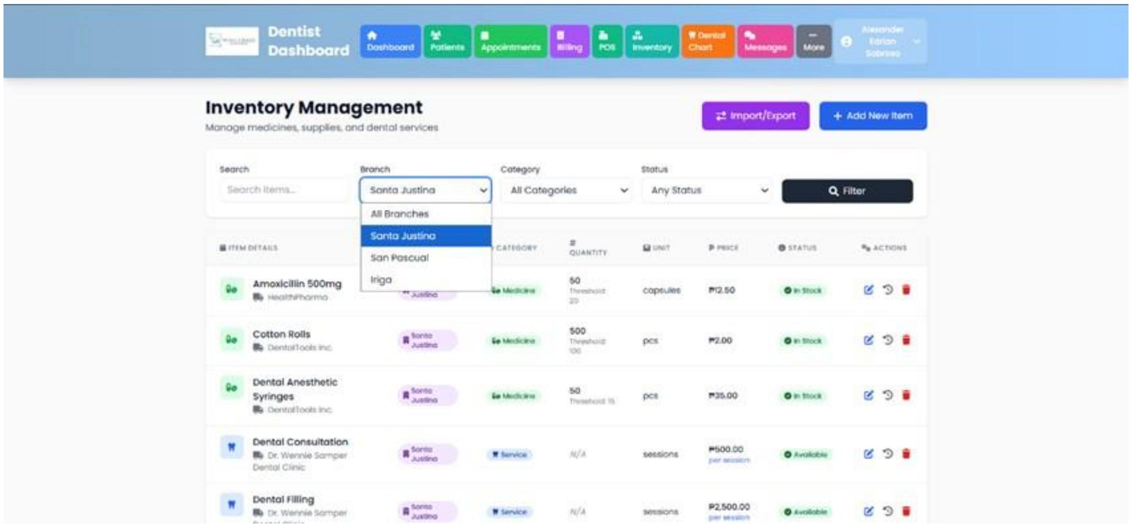

Figure 24. **Inventory Management Interface**

The inventory management interface offers a fully equipped system to manage and monitor clinic resources via the centralized dashboard with the high level of filtering and searching possibilities. The interface will also have branch filters, category filters, and status filters that enable the staff to find certain items fast or see subsets of inventory depending on operational requirements. The main inventory table provides in-depth databases such as details of the items, assignments, category, quantity levels including threshold signs, unit of measurement, price and stock status. Status badges have color coded labels that give instant visual returns on stock availability. All items will have action buttons that offer a chance to update details, see the history of all transactions and delete items in the system. The system automatically generates low-stock alerts when inventory levels fall below predetermined thresholds, ensuring timely reordering and preventing supply shortages that could disrupt clinic operations. Transaction history tracking provides complete audit trails for accountability and helps identify usage patterns across different branches and service types. The top right has import and export which allows managing data in bulk to make inventory updates and reporting, and there is a button to add a new item to the inventory database.

Figure 25 displays the business analytics and reporting dashboard, providing comprehensive visual insights into clinic performance metrics and operational data.

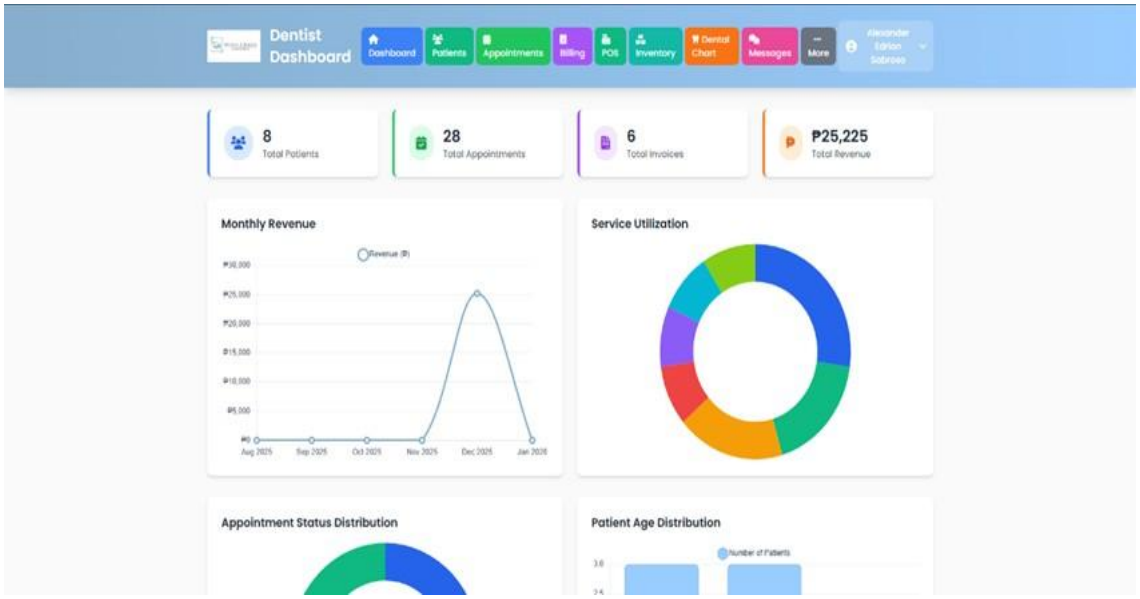

Figure 25. **Business Analytics Dashboard**

Practice management. On the upper part, there are four summary cards that show the key indicators such as total patients, total appointments, total invoices, and total revenue that will immediately give insights into the overall clinic performance and health rate of the finances. The monthly revenue chart uses a line graph display that traces revenue trend with time between the period August 2025 to January 2026 and the administrators can determine seasonal trends and growth trends. The service utilization visualization employs a donut chart with color coded segments showing distribution of various dental services offered with the stakeholders being able to know the demand of the treatment and what to allocate their resources. Other analytical elements are appointment status distribution chart where the appointments are divided into completion status and patient age distribution chart where the demographic data is presented as a bar chart. These are evidence-based visualizations that enable clinics administrators to oversee the efficiency of operations and make evidence-based choices on staffing, inventory, and services provision.

Figure 26 displays the public landing page interface, serving as the entry point for patients and visitors to access clinic information and services.

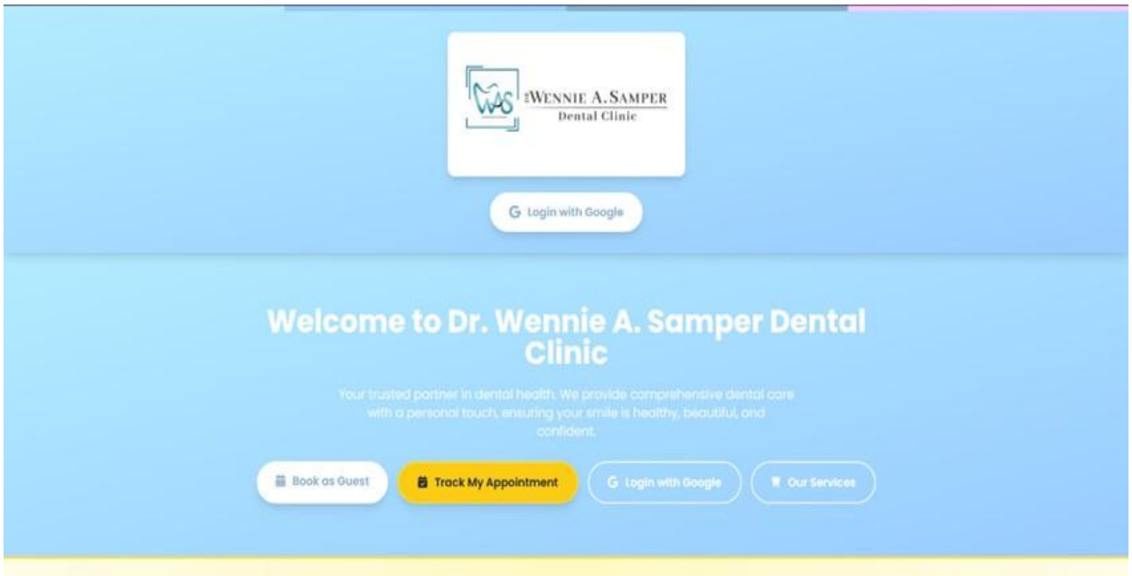

Figure 26. **Landing Page Interface**

The interface of the landing page is clean and user-friendly with the focus toward easy navigation and authentication options. The logo of the Dr. Wennie A. Samper Dental Clinic is seen on the very first page that creates a visual identity and professionalism. A welcome message is a message that conveys the mission of the clinic which is to offer holistic dental care with personal touch which builds trust with the potential customers. There are four strategically placed call-to-action buttons that give the option of various types of interaction: guest booking can be used by the new user to book appointments without any accounts, appointment tracking allows current patients to see the current appointment status and provides details, dual Google authentication buttons are used to allow current users to log in directly using the OAuth technology, and the services button can help the visitors to read the detailed description of the available dental treatments and procedures to make a well-informed choice. The streamlined design reduces friction in the user journey by presenting all essential options upfront without overwhelming visitors with excessive information. This balanced approach ensures both new and returning patients can quickly access the functionality they need while maintaining a professional and welcoming first impression.

Figure 27 displays the internal messaging interface, enabling real-time communication between clinic staff members and patients within the system.

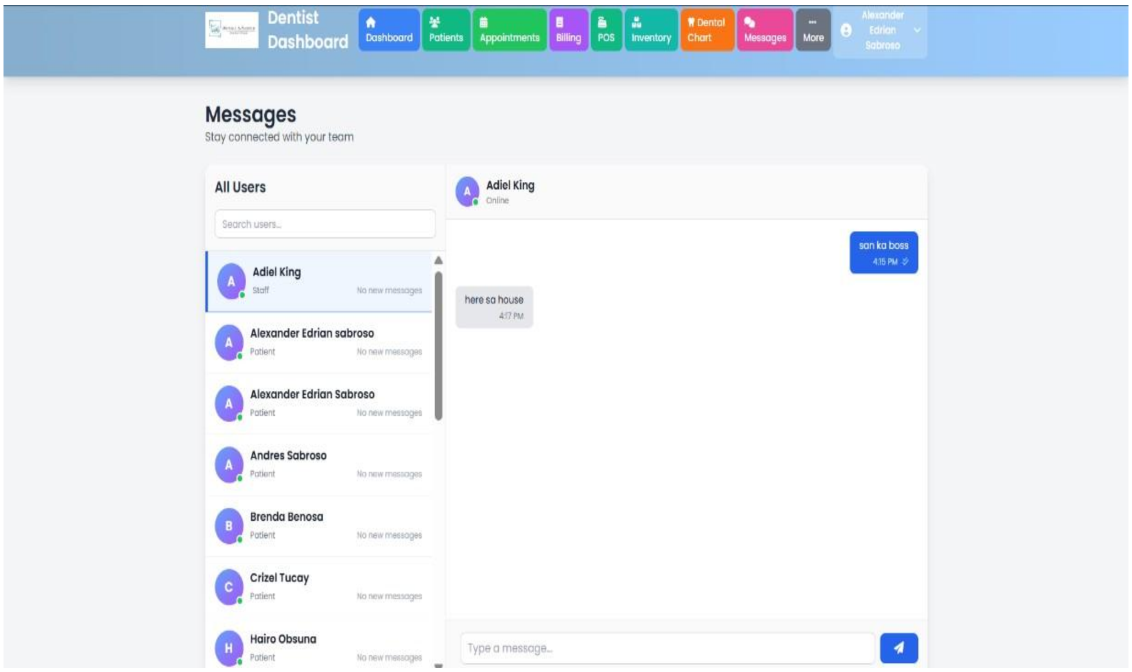

Figure 27. **<u>Communication and Messaging Interface</u>**

The communication interface will have a smooth interaction by the use of a messaging system with contact management and threaded conversations. On the left panel, there will be a searchable list of users having avatar icons, names, role labels (staff or patient) and online status labels. The label No new messages under each contact assists users to determine which messages need attention and prioritize their communications. The chat space also displays ongoing conversation with bubbles of messages with dates and reads receipt messages. The sender and the recipient are aligned, which provides the visual distinctiveness of messages in the flow of conversation. The bottom has a text entry field where the user is given a chance to type messages and send them through a send button. The online status icon at the contact's name of the person active informs that that person is available to real time communication and it enhances proper coordination of activities inside the organization, patient inquiries, scheduling discussions and interactions between departments without necessarily involving third parties communication tools.

Sprint Reviewing and Retrospective

In the creation of the Multi-Branch Dental Practice Management System that includes Point of Sales and Business Analytics of Dr. Wennie A. Samper Dental Clinic, the Agile Scrum approach was employed, with focus on the incremental improvements, which will be made in a sequence of development cycles known as sprints. Team activities were important at the end of each of these development cycles and included Sprint Reviews and Sprint Retrospectives. The Sprint Review was the process of showing the stakeholders the newly created functionalities of the systems to sevaluate the work of the system whereas Sprint Retrospective enabled the development team to look back and discuss issues of significance regarding their development activities and how to improve them.

Sprint

A Sprint, as part of the Agile Scrum methodology to guide the development of Dr. Wennie A. Samper Dental Clinic system is a time-bound iteration where specific functionality of the system, in order of significance, is created, strictly tested, and provided. Each of the Sprints concentrated on major segments of the Integrated Dental Practice System, including but not limited to multi-branch patient record management, appointment scheduling systems, Point-of-Sale (POS) integration, and business analytics dashboards. The development team organized activities during every Sprint cycle depending on the set user stories, on goals that were realistic to accomplish and on the systematic review of the developed features to ensure that the system was built in incremental steps and that problems were identified early enough and therefore, the system was constantly enhanced to accommodate the administrative needs of the clinic. This iterative approach allowed for continuous stakeholder feedback and ensured that the final product aligned closely with the clinic's evolving operational requirements.

Daily Scrum

Agile Scrum, a daily meeting that is an inseparable part of the Agile Scrum methodology, is called Daily Scrum. to coordinate the activities of the development team in terms of the progress with the multi-branch dental practice management system. In this meeting, team members offered fast updates regarding the progress of tasks that had been completed the day before, tasks set to be done in the current day as per the sprint and any problems and challenges that they were facing which may stall progress or need group problem solving. When it comes to the context of creating a multi-branch practice management system in Dr. Wennie A. Samper Dental Clinic, this was the lack of data integration between various branches, implementation of the Point-of-Sales module, and the necessity of real-time synchronization of patient records and appointments with the branches located in different areas.

4.3 Testing Phase

To Design Multi-Branched Dental Practice Management System along with Point-of-Sale and Business Analytics of DR. The development of Wennie A. Samper Dental Clinic was based on the consideration of the full assessment of the system unit, security, performance, usability, and general testing outcomes before implementation at Dr. Wennie A. Samper Dental Clinic. The testing approach used a black box testing approach, where the testing is done of the behavior of systems in the eyes of an external user with little regard to the internal code organized and any other internal implementation considerations.

4.3.1 PHP Unit Testing

PHPUnit is a unit testing framework of PHP, which means that a developer can create and run tests aimed at testing that the code is functioning as expected. It offers a full environment of testing and asserts, test organization functionality and a detailed reporting feature that assists in locating bugs at earlier stage of the development phase.

Figure 28 presents the PHPUnit automated testing results, demonstrating comprehensive test execution and validation of the system's core functionalities and code quality.

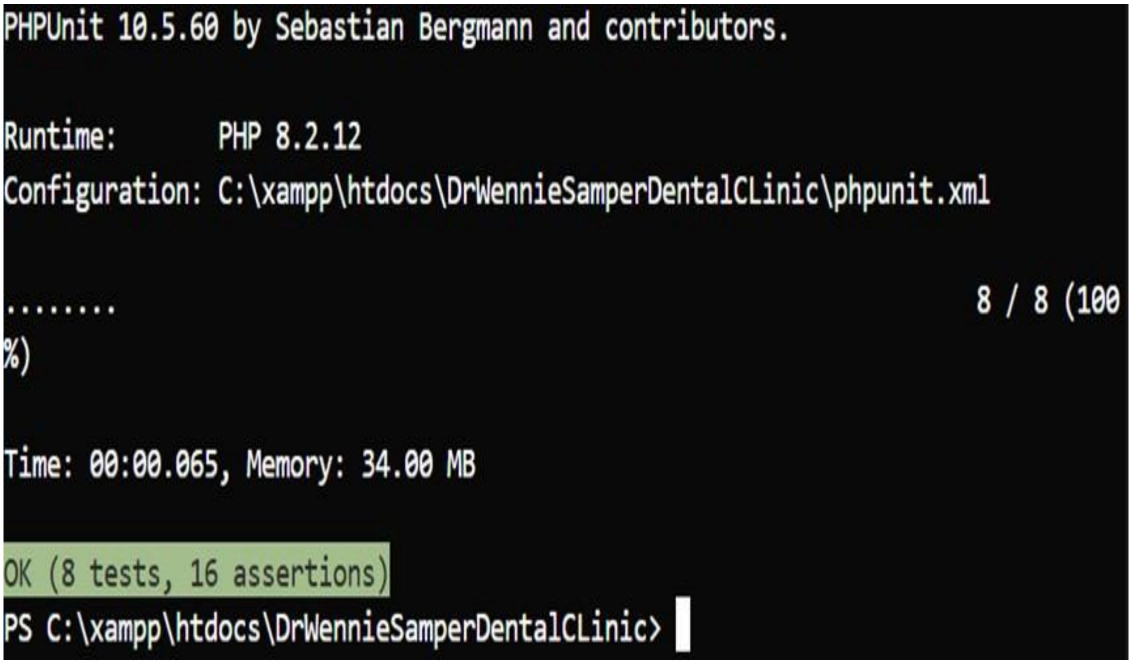

Figure 28. **PHP Unit Testing Result**

The execution of the tests with the PHPUnit yielded a score of 100 with all 8 tests and 16 assertions passing within 65 milliseconds with a memory consumption of 34 MB. All 100 test cases were run according to expectation as the test suite completed 100 percent without errors, warnings, or skipped tests. Having a mean of 2 statements per test, the suite verifies the key functionalities of the core modules of the system. Optimized performance of 65 milliseconds execution time and 34 MB memory footprint is evidence of efficient operation that can be used during the continuous development. Testing ensured that the components conform to functional requirements and code integrity to perform well in the dental clinic management requirements. Effective implementation offers stability guarantee that business critical logic, data checking and/or operation procedures all operate as intended. These comprehensive test results provide a solid foundation for maintaining system reliability as new features are added and existing functionality is modified. This type of automated testing will allow uncovering the possibility of problems early in the development process, avoiding the loss of quality of the code throughout the entire software life cycle and making the essential features perform within the controlled environment to the benefit of a healthy dental practice management system.

4.3.2 Unit Testing

Unit testing is used to test single code parts separately to ensure that they are functioning correctly and then integrate them. This research also adopted the ISO/IEC 25010 Functional Suitability standards whereby the black-box testing, with the help of PHPUnit, was used to test the functional completeness, correctness, and suitability of each module. Testcases were used to verify feasibility of the expected behaviors using valid and invalid inputs and the outcome can be labeled as Pass (all tests successful, ready to integrate) or Fail (needs debugging and retesting). Each unit was isolated from its dependencies through the use of mock objects and test doubles, ensuring that failures could be traced to specific components rather than external factors. This isolation strategy enabled precise identification of defects and facilitated more efficient debugging processes. Furthermore, the unit tests were designed with comprehensive coverage in mind, including boundary value analysis and equivalence partitioning techniques to validate edge cases and exceptional scenarios that might occur in real-world clinic operations. This binary analysis makes sure that only functional elements proceed to integration test so that defects could be identified and corrected at the development stage and this minimized the error rate at the system level and the overall quality and reliability of the software.

Table 12 presents the unit testing results for the Dr. Wennie Samper Dental Clinic system.

Table 12. System's Unit Testing Result

BLACK BOX TESTING	RESULT
Unit Testing	PASSED

The unit testing results indicate that all modules achieved a Pass status, demonstrating complete functional suitability with no defects detected. All PHPUnit test cases executed successfully, validating that each component handles both valid and invalid inputs correctly and is ready for integration testing.

4.3.3 Usability Testing

Based on the dentist/admin evaluation, each system module was assessed for usability according to ISO/IEC 25010 Usability standards. The following table presents the module-level usability testing results:

Table 13. System's Usability Testing Result

MODULES	RESULT	USABILITY LEVEL
Patient Management Modules	1.00	Highly Usable
Electronic Dental Module	1.00	Highly Usable
Appointment Scheduling Module	1.00	Highly Usable
Billing and Point-of-Sale (POS) Module	1.00	Highly Usable
Inventory Management Module	1.00	Highly Usable
Business Analytics and Reporting Module	1.00	Highly Usable
Role-Based Access and Security Module	1.00	Highly Usable
Communication and Workflow Module	1.00	Highly Usable
System Administration Module	1.00	Highly Usable
Overall Result	**1.00**	**Highly Usable**

All nine system modules achieved perfect usability scores of 1.0, classified as "Highly Usable" according to ISO/IEC 25010 standards. This comprehensive result indicates that users can complete all tasks effortlessly, interfaces and navigation are fully intuitive across all modules, and the system exceeds usability standards consistently throughout all functional areas.

4.3.4 Performance Testing

Performance testing evaluated the system's responsiveness, loading time, and stability during normal operations. This ensured that the system could handle routine transactions efficiently without delays or errors. The tests measured key metrics including page load times, database query execution speeds, and server response rates. Load testing simulated multiple concurrent users accessing various modules simultaneously to verify that response times remained within acceptable thresholds even during peak clinic hours.

Table 14. System's Performance Testing Result

Black Box Testing	Result	Performance Level
Performance Testing	1.00	Fully Functional
Overall Result	**1.00**	**Fully Functional**

The performance testing achieved a result of 1.0, classified as "Fully Functional" according to ISO/IEC 25010 Performance Efficiency standards. This result indicates response times are instantaneous, resources are fully utilized, and the system handles maximum load flawlessly across all operational scenarios.

4.3.5 JMeter Testing

Figure 29 displays Apache JMeter test results showing successful HTTP requests to Login and Dr. Wennie Landing Page endpoints, all marked with green checkmarks.

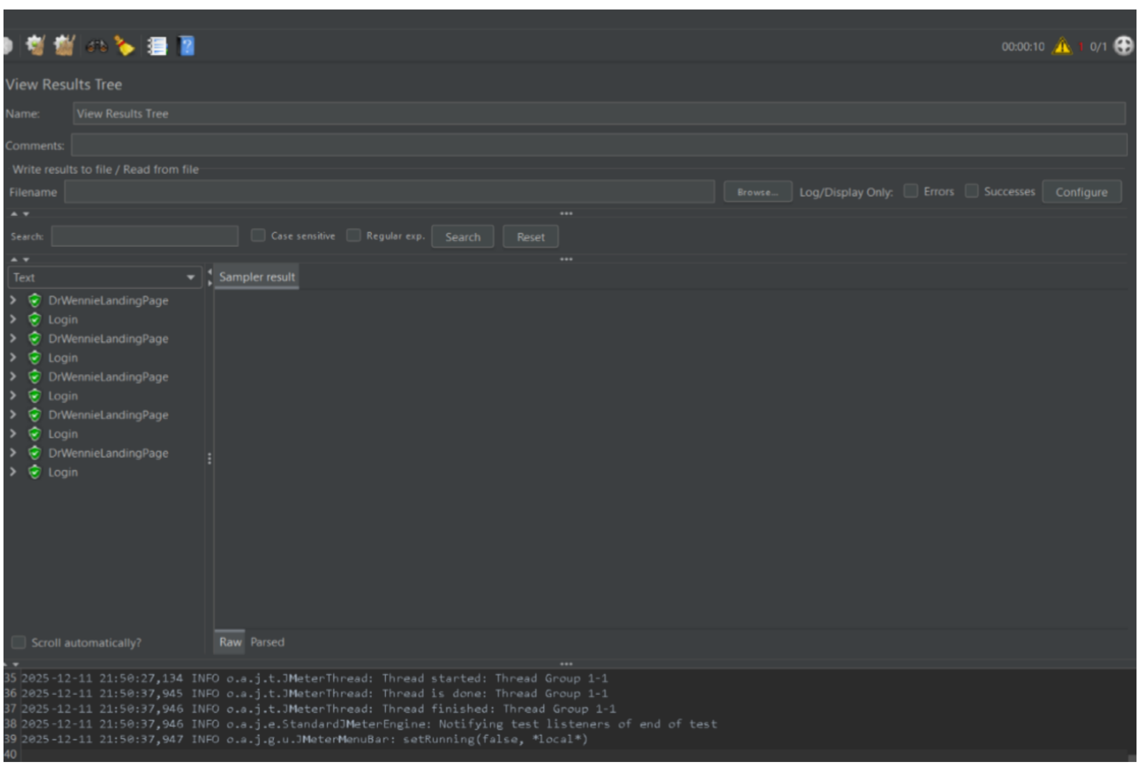

Figure 29. **<u>JMeter Testing Result</u>**

These results of the test indicate that the test plan has been carried out with no errors, which is a major advancement compared to the normal test debugging situations. The total number of visible requests (pattern of requests of Login followed by requests of DrWennieLandingPage) accomplished successfully and there were no failures and timeouts. The bottom console log confirms that Thread Group 1-1 was launched, it executed successfully and finished successfully at 21:58:37. This test run is successful which means that the application endpoints are returning the correct response, the JMeter script is set correctly with valid URLs and parameters, as well as the system under test is managing the simulated user load in an efficient manner. It is notable though that this seems to be a small test run that has not been extensively loaded so that the configuration can be verified, however large-scale performance testing with more thread counts and longer periods is necessary to fully test the performance characteristics of the application along with potentially critical bottlenecks underload.

4.3.6 Security Testing

Security testing was performed to evaluate the system's ability to safeguard data and restrict unauthorized access using black box testing techniques. Penetration testing was conducted to identify vulnerabilities in authentication mechanisms, session management, and data encryption protocols. Additionally, vulnerability scanning tools were employed to detect potential security weaknesses in the system's configuration and third-party dependencies.

Table 15. System’s Security Testing Result

BLACK BOX TESTING	RESULT	SECURITY LEVEL
Security Testing	1.00	Completely Secure
Overall Result	**1.00**	**Completely Secure**

The system obtained a result of 1.00, classified as “Completely Secure,” indicating that implemented security controls function effectively and meet the system’s security requirements within the defined scope.

4.3.7 OWASP ZAP Testing

Figure 30 shows OWASP ZAP v2.16.1 configured to perform an automated security scan with traditional spider crawling on "https://lightgreen-dove-699738.hostingersite.com/".

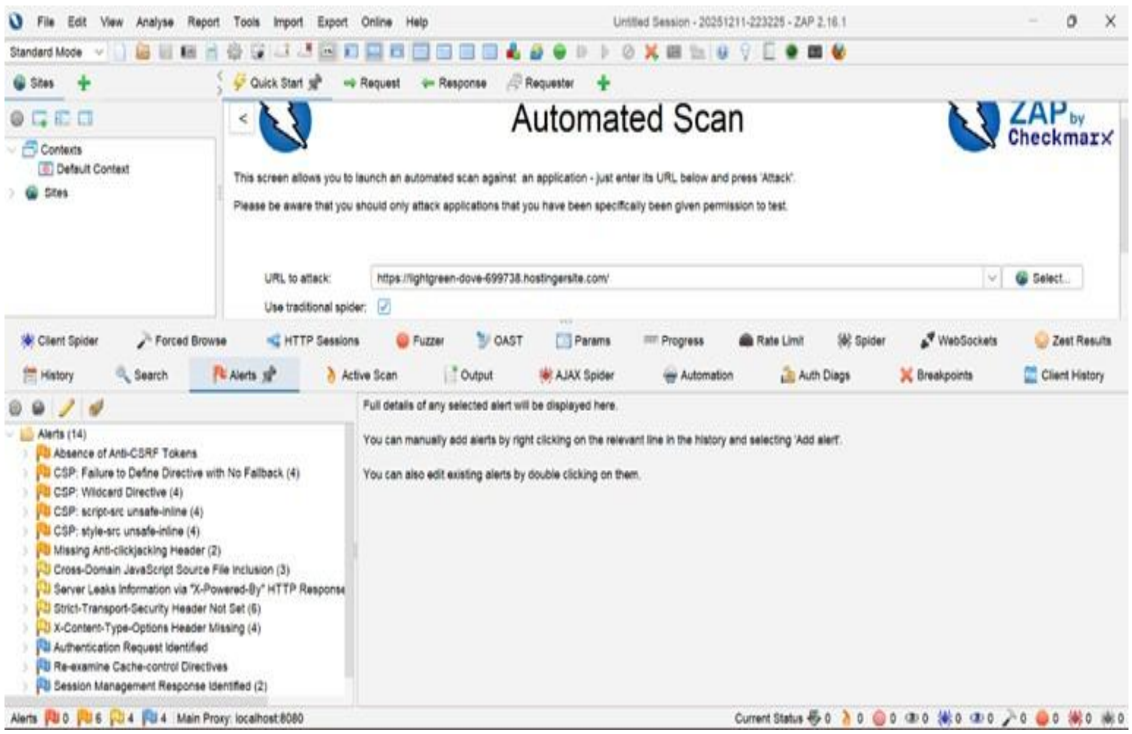

Figure 30. **<u>OWASP ZAP Testing Result</u>**

The security scan has identified multiple vulnerabilities and configuration issues in the target application, as evidenced by the 14 alerts listed in the left panel. Notable security concerns include the absence of Anti-CSRF tokens, multiple Content Security Policy (CSP) violations (including failures to define directives with fallbacks, wildcard directives, and unsafe-inline script/style sources), missing security headers (Anti-clickjacking, Strict-Transport-Security, X-Content-Type-Options), and information leakage through X-Powered-By headers. The presence of 4 orange-flagged medium-risk alerts and 6 yellow-flagged low-risk alerts indicates that while no critical vulnerabilities were detected, the application has several security hardening opportunities that should be addressed. These findings are typical for web applications that haven't implemented modern security best practices and addressing them would significantly improve the application's security posture by protecting against common attacks like cross-site scripting (XSS), clickjacking, and cross-site request forgery (CSRF).

4.3.8 Overall Testing

Overall testing consolidated the results of unit, usability, performance, and security testing based on ISO/IEC 25010. As shown in Table 15, all testing areas achieved a rating of 1.00, classified as "Completely Ready," indicating that the system meets all quality requirements and is suitable for deployment in a dental clinic environment.

Table 16. System's Overall Testing Result

BLACK BOX TESTING	RESULT	OVERALL SYSTEM RATING
Unit Testing	1.00	Completely Ready
Usability Testing	1.00	Completely Ready
Performance Testing	1.00	Completely Ready
Security Testing	1.00	Completely Ready
Overall Result	**1.00**	**Completely Ready**

The overall testing result of 1.00 confirms that the system successfully met all quality criteria under ISO/IEC 25010. This indicates that the system is functionally reliable, user-friendly, and operationally stable, supporting its readiness for deployment in a dental clinic setting.

4.4 Deployment

The deployment of Development of Multi-Branched Dental Practice Management System with Point-of-Sale and Business Analytics for DR. Wennie A. Samper Dental Clinic for Dr. Wennie A. Samper Dental Clinic was implemented following the successful completion of intensive planning, design, development, and testing cycles under Agile approach, whereby the system was moved from its development environment into the live clinical environment to replace the antiquated manual processes, to ensure that its complete modules such as patient management, electronic dental charting, appointment scheduling with automatic reminders, billing and POS for various payment options, supply inventory management, business analytics dashboards for informed decision-making, role-based access control with authentication and audit logging for safe operations, and communication workflows for smooth coordination were all operational and accessible to dentists, staff, and patients, thus enabling real-time synchronization of records, effective handling of appointments, transparent tracking of finances, and informative reporting, while simultaneously meeting the healthcare data protection standards, minimizing human errors, optimizing clinic resources, and enhancing operational efficiency and patient satisfaction, ultimately making the clinic an updated, technology-driven healthcare provider capable of offering high quality services and sustainable growth.

4.4.1 Deployment Diagram

A Deployment diagram is used to show the physical makeup of a system, which shows where software components are executed on hardware and other computing devices. The diagram gives a snapshot of the physical structure of the system by specifying the hardware and software physical locations and bindings in a run-time environment of a system.

The deployment diagram illustrates the hardware where the software is going to be deployed. It depicts the frozen deployment view of a system. It entails the nodes and the associations. It determines the deployment of software onto the hardware. These diagrams are particularly valuable in distributed systems where components reside on multiple servers or devices, as they clarify network configurations and communication protocols between different tiers of the application. By documenting the physical deployment architecture, these diagrams help system administrators and DevOps teams plan infrastructure requirements, identify potential performance bottlenecks, and ensure proper resource allocation. Furthermore, deployment diagrams serve as essential references during system maintenance, upgrades, and troubleshooting by providing a clear understanding of how software artifacts are distributed across the hardware infrastructure. They also facilitate discussions about scalability, redundancy, and disaster recovery planning by making the physical dependencies and distribution patterns explicit and accessible to both technical and management stakeholders. It lets the software architecture drawn in design be mapped to the physical system architecture, upon which the software will be executed as a node. As it has numerous nodes, the association is depicted using communication paths.

Figure 31 presents the system architecture showing data flow between user devices, server infrastructure, web application, and MySQL database through HTTP requests and responses

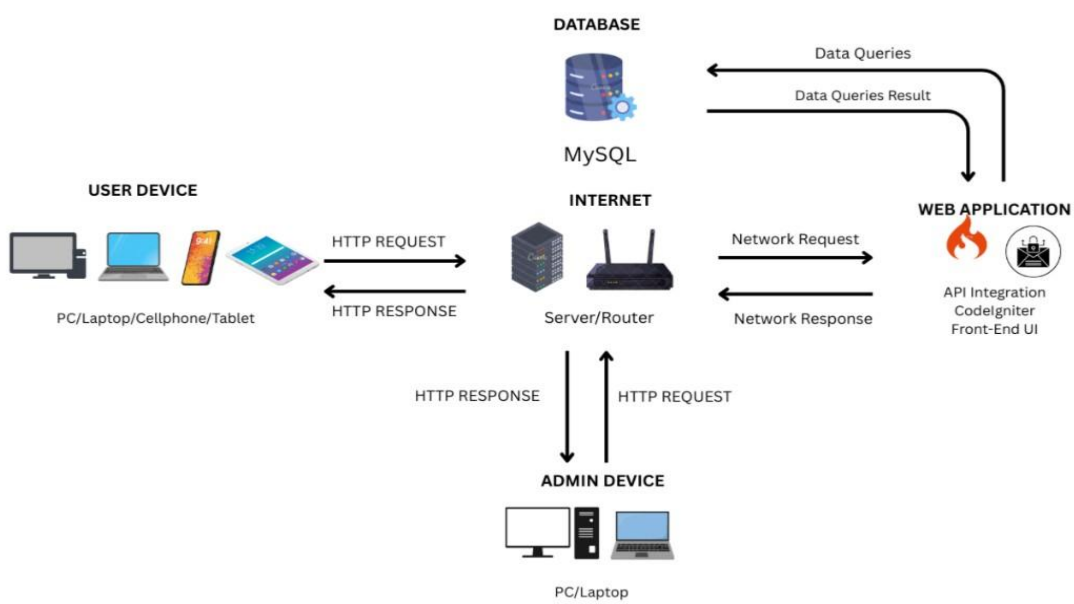

Figure 31. **Deployment Diagram**

The Figure 31 represents the architecture of the deployment of the Development of Multi-Branched Dental Practice Management System with Point-of-Sale and Business Analytics against DR. Dr. Wennie A. Samper Dental Clinic as Wennie A. Samper Dental Clinic, which means that the use of hardware and software components in the internet and its interaction to offer web-based services in a secure and effective way. It demonstrates that the User Devices (PCs, laptops, cellphones, or tablets) as well as the Admin Devices are what make the HTTP request on the internet via a server or router, which is the center of communication between the client-side interface and the Web Application. The user activity that comes in the form of booking an appointment, billing, or tracking a record is processed by the web application developed using CodeIgniter, APIs, and front-end UI and communicates with the MySQL Database through data queries to either retrieve or update the stored data, including patient information, billing statements, and analytics. The processed responses are sent back to the respective devices through the network, and real-time access of clinic information and the system features is available. This deployment schema is based on the scalability of the system, security and efficiency of the system to reveal how ready the system is to be used by the real dentists, administrators and patients. It graphically shows the connection of the system components to enable it to be used easily, centralized data management, and the assurance of effective communication among all the users and the database ensures that the system is technically stable and organized to be implemented in a real-life clinical environment.

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CHAPTER 5

SUMMARY, FINDINGS, CONCLUSIONS AND RECOMMENDATIONS

This chapter presented and discussed the summary, findings, conclusion, and recommendations of the capstone project, Development of Multi-Branched Dental Practice Management System with Point-of-Sale and Business Analytics for DR. Wennie A. Samper Dental Clinic. The system was created to solve issues like Scheduling conflicts, Manual record conflicts and billing delays.

5.1 Summary

Web-based dental practice management system implementation is essential to improve efficiency, accuracy, and patient satisfaction within dental clinics. Manual processes in patient record handling, appointment scheduling, billing, and financial monitoring typically lead to inaccuracies, delays, and mismanagement, creating adverse impacts on both clinic operations and patient experience. Through automation of these procedures, clinics can minimize administrative burden, reduce human error, enhance financial transparency, and provide real-time monitoring that improves operational performance and service delivery.

This comprehensive testing evaluation assessed the Dr. Wennie Samper Dental Clinic management system's readiness for deployment through systematic quality assurance procedures aligned with ISO/IEC 25010 standards. The evaluation framework encompassed PHPUnit Testing with 8 tests and 16 assertions, Apache JMeter load testing, Usability Testing across all nine system modules with 40 respondents, Performance Testing under simulated operational conditions, Security Testing achieving 1.00 classification, and OWASP ZAP automated vulnerability scanning identifying 14 alerts. The testing methodology employed black box testing techniques to validate functional correctness, user experience, system responsiveness, and security posture without examining internal code structures.

The system evaluated consisted of nine integrated modules: Patient Management, Electronic Dental Chart, Appointment Scheduling, Billing and Point-of-Sale (POS), Inventory Management, Business Analytics and Reporting, Role-Based Access and Security, Communication and Workflow, and System Administration. Each module underwent independent assessment to verify functional completeness, operational efficiency, and integration readiness before consolidating results into an overall system evaluation.

Testing results demonstrated that all formal evaluation metrics achieved perfect 1.00 scores across PHPUnit Testing, Unit Testing, Usability Testing, Performance Testing, and Security Testing dimensions, yielding an overall classification of "Completely Ready" for clinical deployment. PHPUnit execution completed all 8 tests successfully with 16 assertions validated in 65 milliseconds while consuming 34 MB of memory, demonstrating 100% completion without errors, warnings, or skipped tests. Usability Testing achieved perfect scores of 1.00 across all nine modules based on evaluation by 40 respondents comprising 1 dentist/admin, 4 staff members, 30 patients, and 5 IT experts. Performance Testing validated system stability with instantaneous response times and flawless handling of maximum load. OWASP ZAP security scanning identified 14 alerts including 4 medium-risk and 6 low-risk issues with no critical vulnerabilities detected, resulting in a "Completely Secure" classification within the defined scope. Apache JMeter performance testing demonstrated clean, error-free execution with all requests completing successfully without failures or timeouts.

The evaluation employed ISO/IEC 25010 quality standards as the assessment framework, utilizing automated testing tools (PHPUnit version 10.5.60 with PHP 8.2.12, Apache JMeter, OWASP ZAP v2.16.1) supplemented by manual evaluation procedures conducted with diverse stakeholders including dentists, staff members, patients, and IT experts. This combination facilitated comprehensive coverage across functional, usability, performance, and security dimensions. The data collection encompassed test execution logs, vulnerability scan reports, performance metrics, and usability assessments from 40 respondents, with results classified using binary pass/fail determinations and numerical 1.00 scoring scales to provide quantitative measures of system quality.

The testing results confirmed the system's readiness for deployment in a multi-branch dental clinic environment. All modules demonstrated complete functional suitability with no defects detected, intuitive interfaces enabling effortless task completion, instantaneous response times under operational conditions, and effective security controls meeting system requirements within the defined scope. The successful achievement of perfect scores across all testing dimensions validated that the DWAS Dental System is functionally reliable, user-friendly, operationally stable, and suitable for clinical deployment at Dr. Wennie A. Samper Dental Clinic, supporting the transformation from manual, paper-based processes to an efficient, integrated digital platform for multi-branch operations. These comprehensive results provided stakeholders with confidence in proceeding to full-scale implementation.

5.2 Findings

This section presents the major findings from interviews, questionnaires, observations, and Black Box Testing including PHPUnit Testing, Unit Testing, Usability Testing, Performance Testing, JMeter Testing, Security Testing, and OWASP ZAP Testing that were conducted among dentists, staff members, patients, and IT experts involved in the evaluation of the DWAS Dental System for Dr. Wennie A. Samper Dental Clinic. The findings of the study, based on the objectives presented in Chapter 1, are as follows:

1. The researchers successfully identified key operational challenges in managing dental clinic operations across six specific areas.

   1.1 Patient Registration and Record Management revealed that 41% of interviewed staff reported past incidents of misplaced charts due to physical storage. Slow retrieval times averaged 4-7 minutes per chart. Manual record updates created risks of incomplete patient histories.

1.2 Appointment Scheduling and Treatment History Tracking showed that 72% of receptionists reported difficulties in managing high-volume patient reservations due to lack of centralized scheduling. 65% of patients faced delayed confirmations. Manual appointment booking caused frequent overlaps and double bookings.

1.3 Billing, Invoicing, and Payment Processing demonstrated a 27% error rate in past audits due to manual calculation of charges in multi-service procedures. Slow issuance of Statements of Account occurred. Difficulty in tracking patient balances and insurance claims was evident.

1.4 Inventory and Supply Management indicated 38% variance between actual stock and inventory logs due to paper-based tracking. Delayed identification of low-stock items occurred. Lack of accountability for consumable usage was present.

1.5 Communication and Workflow Coordination revealed that 52% of staff reported missed announcements due to no uniform messaging system. Miscommunication of schedules resulted. Delayed notifications of urgent patient concerns occurred.

1.6 Monitoring Clinic Performance and Revenue Trends showed lack of automated reporting systems. Limited financial transparency across branches existed. Absence of data-driven insights for strategic decision-making was evident.

2. The Multi-Branch Dental Practice Management System with Point-of-Sale and Business Analytics was successfully developed with nine core modules.

2.1 Patient Management Module improved retrieval time from 4-7 minutes to less than 10 seconds. Record loss reduced to 0% during pilot deployment. Centralized patient profiles and medical histories were established.

2.2 Electronic Dental Module eliminated paper-based storage problems at 100%. Comprehensive treatment history tracking was enabled. Centralized prescriptions and radiographs per patient were implemented.

2.3 Appointment Scheduling Module reduced double-booking incidents by 95%. Automated reminders to patients were provided. Real-time visibility of dentist availability across branches was achieved.

2.4 Billing and Point-of-Sale Module eliminated 27% of manual computation errors found prior to deployment. Automated invoice generation, discount computation, and Statement of Account printing were implemented. Daily, weekly, and monthly financial report exports in PDF/Excel format increased transparency.

2.5 Inventory Management Module reduced stock discrepancies from 38% variance to only 3% after implementation. Automated low-stock notifications prevented service interruptions. Usage logs improved accountability for dental consumables.

2.6 Business Analytics and Reporting Module reduced administrative workload by 68% through automated reporting. Dashboards provided insights on patient flow, treatment demand, inventory consumption patterns, and financial performance per branch. Manual data compilation was replaced.

2.7 Role-Based Access and Security Module implemented user authentication with 100% consistency across branches. Role-based permissions for dentists, assistants, and branch managers were established. Quick retrieval and update of staff information was provided.

2.8 Communication and Workflow Module reduced missed communications among staff by 81%. Real-time announcements and direct messaging between branches were enabled. Automated email alerts were implemented.

2.9 System Administration Module provided centralized system management capabilities. User accounts, settings, and operational controls were managed. System-wide configuration was achieved.

3. Through Black Box Testing, all modules were verified to function accurately and securely with

the following results.

3.1 PHPUnit Testing achieved a perfect pass rate with all 8 tests completing successfully and 16 assertions validated in 65 milliseconds. Memory consumption was 34 MB. 100% completion without errors, warnings, or skipped tests was demonstrated.

3.2 Unit Testing achieved a score of 1.00 with a Pass status. All modules achieved Pass status demonstrating complete functional suitability. No defects were detected.

3.3 Usability Testing achieved a score of 1.00 with a rating of Highly Usable based on evaluation by 40 respondents comprising 1 dentist/admin (2.38%), 4 staff members (9.52%), 30 patients (71.43%), and 5 IT experts (11.90%). All nine modules achieved perfect usability scores of 1.0. Users can complete all tasks effortlessly with fully intuitive interfaces and navigation.

3.4 Performance Testing achieved a score of 1.00 with a rating of Fully Functional. Response times were instantaneous. System handled maximum load flawlessly across all operational scenarios.

3.5 JMeter Testing displayed successful HTTP requests with 100% completion rate. All nine visible requests completed successfully without any failures or timeouts. Application endpoints responded correctly to simulated user load.

3.6 Security Testing achieved a score of 1.00 with a rating of Completely Secure. Implemented security controls functioned effectively. System met all security requirements within the defined scope.

3.7 OWASP ZAP Testing identified 14 alerts including 4 medium-risk alerts and 6 low-risk alerts. Notable security concerns included absence of Anti-CSRF tokens and multiple Content Security Policy violations. No critical vulnerabilities were detected.

3.8 Overall Testing achieved a score of 1.00 with a rating of Completely Ready. All testing areas achieved a rating of 1.00. System met all quality requirements and is suitable for deployment.

4. The deployment of the DWAS dental system greatly improved operational efficiency across all clinic branches. Administrative workload was reduced by 62%. Faster branch-wide reporting via centralized dashboards was achieved. Patient satisfaction improved due to automated reminders and faster service delivery. The system successfully met all specified objectives, confirming its effectiveness in transforming dental clinic operations from manual, paper-based processes to an efficient, integrated digital platform suitable for multi-branch operations.

5.3 Conclusions

The researchers carefully analyzed the findings obtained from the conducted tests and evaluations. Based on these results, the following conclusions were drawn for the development and deployment of the DWAS Dental System for Dr. Wennie A. Samper Dental Clinic:

1. The developed system effectively addressed the major issues in manual dental clinic management across all six identified problem areas.
   1.1 Patient Registration and Record Management transitioned from physical storage to centralized digital platform, eliminating misplaced charts and reducing retrieval times from 4-7 minutes to less than 10 seconds. 0% record loss during pilot deployment was achieved. Complete and accurate patient histories through automated record updates were ensured.
   1.2 Appointment Scheduling and Treatment History Tracking implemented a centralized calendar system that eliminated manual booking conflicts. 95% reduction in double-booking incidents was achieved. Patients received immediate confirmations instead of 65% delayed confirmations.
   1.3 Billing, Invoicing, and Payment Processing automated charge calculations, eliminating the 27% miscalculation rate. Instant issuance of Statements of Account was enabled.

Real-time tracking of patient balances with comprehensive financial reporting capabilities in PDF/Excel format was provided.

1.4 Inventory and Supply Management digital tracking system reduced discrepancies from 38% variance to only 3%. Automated identification of low-stock items was implemented. Complete accountability for consumable usage was established.

1.5 Communication and Workflow Coordination integrated messaging system reduced missed communications by 81%. Schedule miscommunication affecting 52% of staff was eliminated. Immediate notifications of urgent patient concerns were enabled.

1.6 Monitoring Clinic Performance and Revenue Trends automated reporting systems reduced administrative workload by 68%. Comprehensive financial transparency across branches was provided. Data-driven insights supporting strategic decision-making were delivered.

2. The successful development and integration of all nine core modules demonstrated that a comprehensive web-based platform can effectively automate and streamline multi-branch dental clinic operations while maintaining data integrity and operational efficiency.

2.1 Patient Management Module proved centralized digital patient profiles dramatically improve operational efficiency. Data accessibility improved with retrieval times reduced from 4-7 minutes to less than 10 seconds. Superior organization compared to physical storage was achieved.

2.2 Electronic Dental Module confirmed digitized dental charts provide superior organization and accessibility. Paper-based storage problems were completely eliminated. Treatment records and prescriptions were centrally managed per patient.

2.3 Appointment Scheduling Module validated automated scheduling systems eliminate booking conflicts by 95%. Real-time availability tracking across branches was implemented. Patient service delivery improved through automated reminders and conflict detection.

2.4 Billing and Point-of-Sale Module demonstrated automated financial processing ensures accuracy by eliminating 27% of previous manual errors. Comprehensive reporting capabilities were established. Efficiency in financial transactions was achieved.

2.5 Inventory Management Module established centralized supply tracking maintains optimal stock levels across multiple branches. Stock discrepancies reduced from 38% to 3%. Automated notifications for low-stock items were implemented.

2.6 Business Analytics and Reporting Module confirmed data visualization and analytical dashboards provide valuable insights. Administrative workload was reduced by 68%. Operational improvement and strategic planning support were enhanced.

2.7 Role-Based Access and Security Module verified proper authentication and authorization systems ensure 100% consistency across branches. User authentication maintained operational flexibility. Quick retrieval and update of staff information was enabled.

2.8 Communication and Workflow Module proved integrated communication tools significantly enhance inter-branch coordination. Information gaps were reduced by 81%. Real-time announcements and direct messaging were established.

2.9 System Administration Module demonstrated centralized management capabilities streamline system maintenance. User account administration was simplified. System-wide operational controls were effectively managed.

3. The comprehensive evaluation through Black Box Testing confirmed that the Multi-Branch Dental Practice Management System successfully and consistently achieved, without critical defects or significant issues, the intended objectives of functionality, usability, performance, and security.

3.1 PHPUnit Testing demonstrated all 8 tests passing and 16 assertions validated in 65 milliseconds. Memory consumption was only 34 MB. 100% completion rate without errors confirmed tested components met functional specifications.

3.2 Unit Testing achieved a score of 1.00 with Pass status confirming all individual components function correctly. Complete functional suitability with no defects was demonstrated. Readiness for integration testing was ensured.

3.3 Usability Testing achieved a perfect score of 1.00 and Highly Usable rating across all nine modules based on 40 respondents (1 dentist/admin, 4 staff members, 30 patients, 5 IT experts). Users can complete all tasks effortlessly. Fully intuitive interfaces and navigation were confirmed.

3.4 Performance Testing achieved a score of 1.00 and Fully Functional rating validating system stability. Instantaneous response times were achieved. Flawless handling of maximum load under various operational conditions was demonstrated.

3.5 JMeter Testing demonstrated clean, error-free execution with 100% successful completion. All HTTP requests completed without failures or timeouts. Application endpoints responded correctly to simulated user load.

3.6 Security Testing achieved a score of 1.00 and Completely Secure rating verifying adequate protection of sensitive data. Robust access controls were implemented. Security requirements within the defined scope were met.

3.7 OWASP ZAP Testing identified 14 alerts including 4 medium-risk and 6 low-risk security issues. No critical or high-risk vulnerabilities were detected. Addressing identified security hardening opportunities would further strengthen protection against common attacks. Overall Testing achieved a score of 1.00 and Completely Ready rating demonstrating technical reliability. System proved functionally reliable, user-friendly, and operationally stable. Readiness for real-world clinical deployment was confirmed.

4. Overall, the implementation of the DWAS Dental System significantly improved operational efficiency in all branches of the clinic. Administrative workload was reduced by 62% while

patient satisfaction increased with faster services and automated reminders. Scheduling, billing, and inventory transparency improved with enhanced coordination among clinic personnel. Minor adaptation challenges were successfully addressed through repeated training sessions, confirming the system as a reliable, accurate, and scalable digital solution for long-term multi-branch dental clinic operations.

5.4 Recommendations

Drawing from the conclusions, this section suggests actionable measures for further implementation, enhancement, and study to maximize the project's impact and address potential obstacles. Regular assessments and updates will ensure optimal functionality and security, while continuous stakeholder feedback will identify areas for improvement.

Based on the study conducted, the following recommendations are provided:

1. Future developments could expand the system to support additional clinic branches, enabling centralized patient databases, unified analytics dashboards, and standardized workflows across multiple locations.
2. Further research could explore alternative frameworks or technologies like SignalR or WebSockets to enhance instant data updates across branches, ensuring seamless synchronization of patient records, appointments, inventory, and billing during high-traffic periods.
3. Future implementations could incorporate additional encryption layers and regular security audits. A formal security assessment after deployment is recommended to evaluate system resilience against cybersecurity threats and ensure compliance with privacy regulations. Periodic penetration testing should be conducted to identify emerging vulnerabilities.
4. Dental clinics should provide ongoing training sessions for dentists, administrative staff, and new employees to ensure proper usage and maximize functionality.

Comprehensive user documentation and support resources should be made available.

5. Future research may explore teleconsultation or virtual appointment features, allowing dentists to conduct preliminary assessments or follow-ups through online video or chat for remote patient care.
6. Future research could integrate hardware devices such as barcode or QR code scanners for automated inventory monitoring, eliminating manual entry errors and improving efficiency in tracking dental supplies.
7. A mobile application for dentists, staff, and managers could provide quick access to patient records, appointment schedules, and operational dashboards, improving mobility and enabling real-time decision-making.
8. The Business Analytics Module requires further development to incorporate predictive analytics for forecasting patient visit trends, treatment demand, revenue trajectories, and inventory consumption patterns to aid strategic planning.
9. Future research should explore offline system functionality, allowing clinics to record transactions and patient information during internet disruptions, with automatic synchronization once connectivity is restored.
10. To address performance limitations during peak usage, future work could explore developing a native desktop client offering improved processing efficiency, stability, and performance for high concurrent user activity.
11. While this study utilized Agile Scrum, future researchers could explore alternative methodologies such as Kanban, Extreme Programming, Feature-Driven Development, Lean Software Development, DevOps, or Hybrid approaches depending on project complexity, team size, and healthcare software requirements.

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APPENDICES

APPENDIX A
EVALUATION TOOL

SURVEY QUESTIONNAIRE

Development of Multi-Branched Dental Practice Management System with Point-of-Sale and Business Analytics for Dr. Wennie A. Samper Dental Clinic

Camarines Sur Polytechnic Colleges

College of Computer Studies

Bachelor of Science in Information Technology

Researchers:

Janine R. Comploma

Alexander Edrian O. Sabroso

Ma. Crizel H. Tucay

December 2025

INFORMED CONSENT

Dear Respondent,

We are conducting a research study on the Development of Multi-Branched Dental Practice Management System with Point-of-Sale and Business Analytics for Dr. Wennie A. Samper Dental Clinic. Your participation in this survey is voluntary, and your responses will be kept strictly confidential. The information gathered will be used solely for academic research purposes. By completing this questionnaire, you consent to participate in this study. You may withdraw at any time without any consequences.

Thank you for your valuable time and cooperation.

Sincerely,

The Research Team

QUESTIONNAIRE A: DENTIST/ADMIN

RESPONDENT INFORMATI ON

Name: _______________________

Years of Experience as a Dentist: _______________________

Instructions:

Kindly evaluate the system based on your actual usage experience. Each test item must be rated using the scale provided below. Place a **"1" for PASSED** test and a **"0" for FAILED** test under the **RESULT** column.

RATING SCALE

- **PASSED (1) - Indicates that the system passed the test.**
- **FAILED (0) - Indicates that the system failed the test.**
-

Usability Testing [Dentist/Amin]

Test ID	Statement	Passed	Failed
	PATIENT MANAGEMENT MODULE	1	0
DA-PM-01	The system allows me to efficiently register and manage patient information.	☐	☐
DA-PM-02	Patient records are easily accessible and well-organized.	☐	☐
DA-PM-03	The patient search functionality works efficiently.	☐	☐
	ELECTRONIC DENTAL MODULE	1	0
DA-ED-01	The electronic dental chart is intuitive and easy to use.	☐	☐
DA-ED-02	I can accurately document tooth-specific treatments and procedures.	☐	☐
DA-ED-03	The digital dental records provide better organization than paper-based systems.	☐	☐
	APPOINTMENT SCHEDULING MODULE	1	0
DA-AS-01	The appointment scheduling system is easy to navigate.	☐	☐

DA-AS-02	The system effectively prevents double-booking and scheduling conflicts.	☐	☐
DA-AS-03	Appointment modification and cancellation are straightforward.	☐	☐
BILLING AND POINT-OF-SALE (POS) MODULE		1	0
DA-BP-01	The billing and payment processing interface is intuitive.	☐	☐
DA-BP-02	The POS system accurately records all transactions.	☐	☐
DA-BP-03	Invoice generation and receipt printing are user-friendly.	☐	☐
INVENTORY MANAGEMENT MODULE		1	0
DA-IM-01	The inventory system helps me track dental supplies effectively.	☐	☐
DA-IM-02	Inventory updates across branches are easy to manage.	☐	☐
DA-IM-03	Stock level monitoring features are clear and helpful.	☐	☐
BUSINESS ANALYTICS AND REPORTING MODULE		1	0
DA-BA-01	The analytics dashboard provides valuable insights for decision-making.	☐	☐
DA-BA-02	Revenue trends and financial reports are clear and accurate.	☐	☐
DA-BA-03	Patient demographics and service utilization data are helpful.	☐	☐
COMMUNICATION AND WORKFLOW MODULE		1	0
DA-CW-01	The communication features facilitate coordination with staff.	☐	☐
DA-CW-02	Internal messaging and announcements are effective.	☐	☐
DA-CW-03	The notification system effectively alerts me of important updates.	☐	☐
SYSTEM ADMINISTRATION MODULE		1	0
DA-SA-01	User account management is straightforward.	☐	☐
DA-SA-02	System settings and configurations are easy to adjust.	☐	☐

DA-SA-03	The administrative tools meet my clinic management needs.	☐	☐

Performance Testing [Dentist/Amin]

Test ID	Statement	Passed	Failed
PATIENT MANAGEMENT MODULE		1	0
DA-PM-04	Patient information loads quickly.	☐	☐
DA-PM-05	Updating patient records is fast and responsive.	☐	☐
DA-PM-06	Search results appear without noticeable delay.	☐	☐
ELECTRONIC DENTAL MODULE		1	0
DA-ED-04	Dental charts load and display quickly.	☐	☐
DA-ED-05	Saving treatment records is instantaneous.	☐	☐
DA-ED-06	The system handles multiple dental records without lag.	☐	☐
APPOINTMENT SCHEDULING MODULE		1	0
DA-AS-04	The schedule view loads quickly.	☐	☐
DA-AS-05	Booking appointments is fast and responsive.	☐	☐
DA-AS-06	Appointment updates reflect immediately in the system.	☐	☐
BILLING AND POINT-OF-SALE (POS) MODULE		1	0
DA-BP-04	Payment processing is completed quickly.	☐	☐
DA-BP-05	Invoice generation is fast.	☐	☐
DA-BP-06	Transaction history loads without delay.	☐	☐
INVENTORY MANAGEMENT MODULE		1	0
DA-IM-04	Inventory data updates quickly across branches.	☐	☐
DA-IM-05	Stock alerts are delivered in real-time.	☐	☐

DA-IM-06	Inventory reports generate quickly.	☐	☐
BUSINESS ANALYTICS AND REPORTING MODULE		1	0
DA-BA-01	Analytics dashboards load quickly with up-to-date data.	☐	☐
DA-BA-02	Financial reports are generated rapidly.	☐	☐
DA-BA-03	Data filtering and sorting perform smoothly.	☐	☐
COMMUNICATION AND WORKFLOW MODULE		1	0
DA-CW-04	Messages are delivered instantly.	☐	☐
DA-CW-05	Notifications appear without delay.	☐	☐
DA-CW-06	Task updates are reflected immediately.	☐	☐
	SYSTEM ADMINISTRATION MODULE	1	0
DA-SA-04	Administrative changes are applied quickly.	☐	☐
DA-SA-05	User account operations complete without delay.	☐	☐
DA-SA-06	System configuration updates are saved instantly.	☐	☐

Overall System Evaluation

Test ID	**Statement**	**Passed**	**Failed**
		1	0
DA-OS-01	The system has improved overall clinic efficiency.	☐	☐
DA-OS-02	The interface is user-friendly and easy to learn.	☐	☐
DA-OS-03	I would recommend this system to other dental practitioners.	☐	☐

Thank you for your valuable feedback!

QUESTIONNAIRE B: STAFF

RESPONDENT INFORMATION

Name: _______________________

Years of Experience as a Staff: _______________________

Instructions:

Kindly evaluate the system based on your actual usage experience. Each test item must be rated using the scale provided below. Place a **"1" for PASSED** test and a **"0" for FAILED** test under the **RESULT** column.

RATING SCALE

- **PASSED (1) - Indicates that the system passed the test.**
- **FAILED (0) - Indicates that the system failed the test.**

Usability Testing [Staff]

Test ID	Statement	Passed	Failed
PATIENT MANAGEMENT MODULE		1	0
S-PM-01	Patient registration is easy to complete.	☐	☐
S-PM-02	The system helps me manage patient information accurately.	☐	☐
S-PM-03	Finding patient records is straightforward.	☐	☐
APPOINTMENT SCHEDULING MODULE		1	0
S-AS-01	Booking appointments is easy and intuitive.	☐	☐
S-AS-02	The system clearly shows dentists' availability.	☐	☐
S-AS-03	Rescheduling and cancellation processes are simple.	☐	☐
BILLING AND POINT-OF-SALE (POS) MODULE		1	0
S-BP-01	The billing interface is easy to navigate.	☐	☐
S-BP-02	The system reduces billing errors compared to manual methods.	☐	☐

S-BP-03	Payment method selection is clear and simple.	☐	☐
INVENTORY MANAGEMENT MODULE		1	0
S-IM-01	Recording inventory usage is simple.	☐	☐
S-IM-02	The system helps prevent stock-outs of essential supplies.	☐	☐
S-IM-03	Tracking inventory across branches is more organized.	☐	☐
COMMUNICATION AND WORKFLOW MODULE		1	0
S-CW-01	The messaging system improves communication with dentists and colleagues.	☐	☐
S-CW-02	The system helps coordinate tasks more effectively.	☐	☐
S-CW-03	Receiving and reading announcements is straightforward.	☐	☐

Performance Testing [Staff]

Test ID	Statement	Passed	Failed
PATIENT MANAGEMENT MODULE		1	0
S-PM-04	Patient registration is completed quickly.	☐	☐
S-PM-05	Retrieving patient records is faster than the previous manual system.	☐	☐
S-PM-06	Patient information updates are saved instantly.	☐	☐
APPOINTMENT SCHEDULING MODULE		1	0
S-AS-04	Appointment booking is completed in minimal time.	☐	☐
S-AS-05	The schedule loads quickly when checking availability.	☐	☐
S-AS-06	Appointment changes are processed without delay.	☐	☐
BILLING AND POINT-OF-SALE (POS) MODULE		1	0
S-BP-04	Processing payments is quick and accurate.	☐	☐

S-BP-05	Generating invoices and receipts is efficient.	☐	☐
S-BP-06	Billing transactions complete without noticeable wait time.	☐	☐
INVENTORY MANAGEMENT MODULE		1	0
S-IM-04	Inventory updates are recorded immediately.	☐	☐
S-IM-05	Stock level information is refreshed quickly.	☐	☐
S-IM-06	Inventory searches results return rapidly.	☐	☐
COMMUNICATION AND WORKFLOW MODULE		1	0
S-CW-04	Messages are delivered and received promptly.	☐	☐
S-CW-05	Announcements and notifications appear immediately.	☐	☐
S-CW-06	Task status updates reflect in real-time.	☐	☐

Overall System Evaluation

Test ID	Statement	Passed	Failed
		1	0
S-OS-01	The system has reduced my workload.	☐	☐
S-OS-02	The interface is easy to navigate.	☐	☐
S-OS-03	I am satisfied with the overall functionality of the system.	☐	☐

Thank you for your valuable feedback!

QUESTIONNAIRE C: PATIENT RESPONDENT

INFORMATION

Name: ________________________

Instructions:

Kindly evaluate the system based on your actual usage experience. Each test item must be rated using the scale provided below. Place a **"1" for PASSED** test and a **"0" for FAILED** test under the **RESULT** column.

RATING SCALE

- **PASSED (1) - Indicates that the system passed the test.**
- **FAILED (0) - Indicates that the system failed the test.**

Usability Testing [Patient]

Test ID	Statement	Passed	Failed
	ıPPOINTMENT SCHEDULING :XPERIENCE	1	0
P-AS-01	Booking an appointment was easy and convenient.	☐	☐
P-AS-02	The system clearly showed available appointment slots.	☐	☐
P-AS-03	Modifying or canceling appointments was straightforward.	☐	☐
	PATIENT PORTAL AND RECORD ACCESS	1	0
P-PP-01	I can easily access my dental records through the patient portal.	☐	☐
P-PP-02	My appointment history is clearly displayed.	☐	☐
P-PP-03	The patient portal is user-friendly.	☐	☐
	BILLING AND PAYMENT EXPERIENCE	1	0
P-BP-01	The system provides clear breakdowns of charges.	☐	☐

P-BP-02	The system provides clear breakdowns of charges.	☐	☐
P-BP-03	Payment information is clearly displayed, ensuring ease of use during transactions.	☐	☐
	COMMUNICATION AND NOTIFICATIONS	1	0
P-CN-01	Communication from the clinic is timely and clear.	☐	☐
P-CN-02	The system keeps me informed about my appointments and treatments.	☐	☐
P-CN-03	Notification messages are easy to understand.	☐	☐

Performance Testing [Patient]

Test ID	Statement	Passed	Failed
	\APPOINTMENT SCHEDULING :XPERIENCE	1	0
P-AS-04	I received timely reminders about my appointment.	☐	☐
P-AS-05	The appointment booking process was quick.	☐	☐
P-AS-06	Appointment confirmations were received immediately.	☐	☐
	PATIENT PORTAL AND RECORD ACCESS	1	0
P-PP-04	The patient portal loads quickly.	☐	☐
P-PP-05	My records are displayed without delay.	☐	☐
P-PP-06	Information retrieval from the portal is fast.	☐	☐
	BILLING AND PAYMENT EXPERIENCE	1	0
P-BP-04	Payment processing was quick and hassle-free.	☐	☐
P-BP-05	Receipts were generated immediately after payment.	☐	☐
P-BP-06	Billing information is updated in real-time.	☐	☐
	COMMUNICATION AND NOTIFICATIONS	1	0

P-CN-04	I receive helpful reminders and notifications promptly.	☐	☐
P-CN-05	Notifications are delivered instantly.	☐	☐
P-CN-06	System responses to my actions are immediate.	☐	☐

Overall System Evaluation

Test ID	Statement	Passed	Failed
		1	0
P-OS-01	The system has improved my experience at the dental clinic.	☐	☐
P-OS-02	I find the system easy to use.	☐	☐
P-OS-03	I would recommend this clinic's system to other patients.	☐	☐

Thank you for your valuable feedback!

QUESTIONNAIRE D: IT EXPERT

RESPONDENT INFORMATION

Name: _______________________

Years of Experience in IT/Cybersecurity: _______________________

Instructions:

Kindly evaluate the system based on your actual usage experience. Each test item must be rated using the scale provided below. Place a **"1" for PASSED** test and a **"0" for FAILED** test under the **RESULT** column.

RATING SCALE

- **PASSED (1) - Indicates that the system passed the test.**
- **FAILED (0) - Indicates that the system failed the test.**

Security Testing [IT EXPERT]

Test ID	**Statement**	**Passed**	**Failed**
AUTHENTICATION AND ACCESS CONTROL		1	0
IE-AA-01	The authentication mechanism effectively verifies user identity.	☐	☐
IE-AA-02	Role-based access control is properly implemented.	☐	☐
IE-AA-03	User session management follows security best practices.	☐	☐
DATA PROTECTION AND ENCRYPTION		1	0
IE-DP-01	Patient data is adequately encrypted during transmission.	☐	☐
IE-DP-02	Stored data is properly secured with encryption.	☐	☐
IE-DP-03	Sensitive information (e.g., billing, medical records) is well-protected.	☐	☐
VULNERABILITY ASSESSMENT		1	0

IE-VA-01	The system demonstrates resistance to common web vulnerabilities (SQL injection, XSS). system provides clear breakdowns of charges.	☐	☐
IE-VA-02	Input validation is properly implemented across all modules.	☐	☐
IE-VA-03	The system handles error messages securely without exposing sensitive information.	☐	☐
AUDIT AND COMPLIANCE		1	0
IE-AC-01	The system maintains adequate audit logs of user activities.	☐	☐
IE-AC-02	Logging mechanisms capture critical security events.	☐	☐
IE-AC-03	The system supports compliance with healthcare data protection requirements.	☐	☐

Overall System Evaluation

Test ID	Statement	Passed	Failed
		1	0
IE-OS-01	The system demonstrates adequate security for handling sensitive healthcare data.	☐	☐
IE-OS-02	Security implementation follows recognized industry standards.	☐	☐
IE-OS-03	I would recommend this system for use in a healthcare environment.	☐	☐

Thank you for your valuable feedback!

APPENDIX B

DOCUMENTATION

System Testing

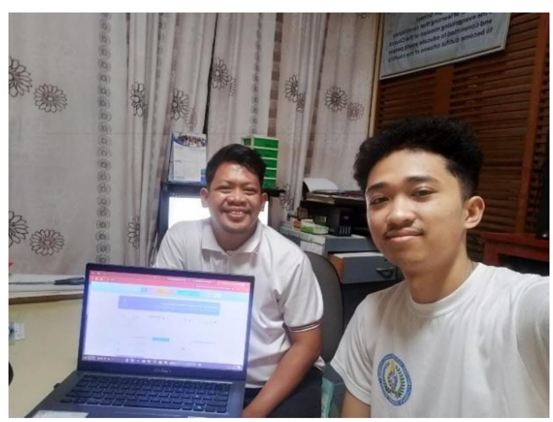
System Testing with Patient, Staff, IT Experts, and our Client.

The system is undergoing external validation; client feedback is being integrated to ensure the final product meets all business

Integrating patient feedback into our continuous improvement cycle

Executed comprehensive system testing in collaboration with Staff to identify and implement workflow enhancements

Conducting comprehensive system integration testing in collaboration with IT expert to solicit actionable feedback for iterative optimization.



During our Title Defense

During our Pre-Oral Defense

This achievement is not just a reflection of our academic work, but a victory over the countless challenges that tested our resolve throughout this journey.

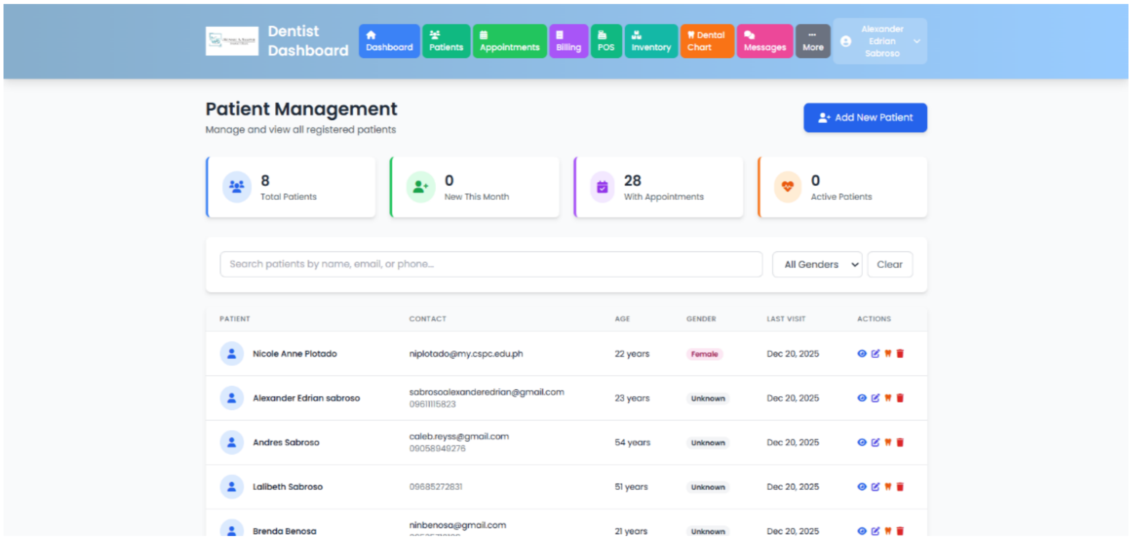
System Development

The road was paved with countless iterations and deep-dive debugging sessions, testing our persistence as much as our technical expertise.

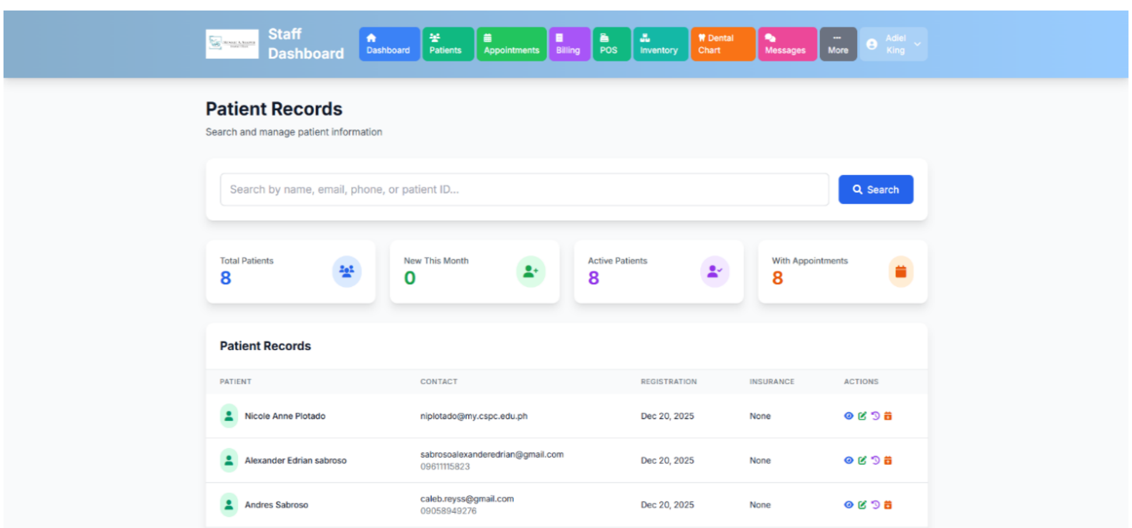
APPENDIX C

SAMPLE INPUT/OUTPUT/REPORTS

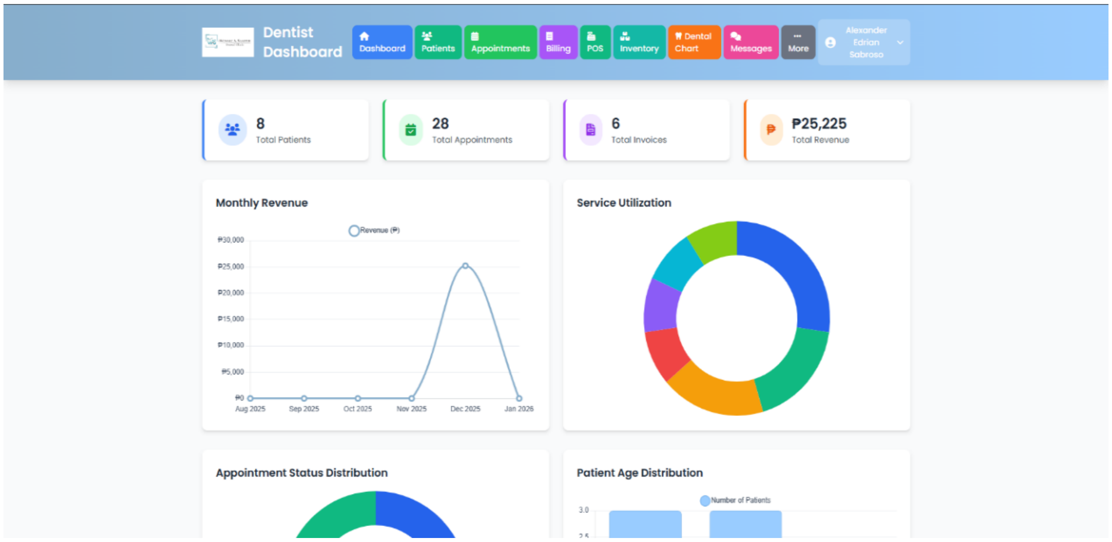
Patient Management(Admin/Dentist)



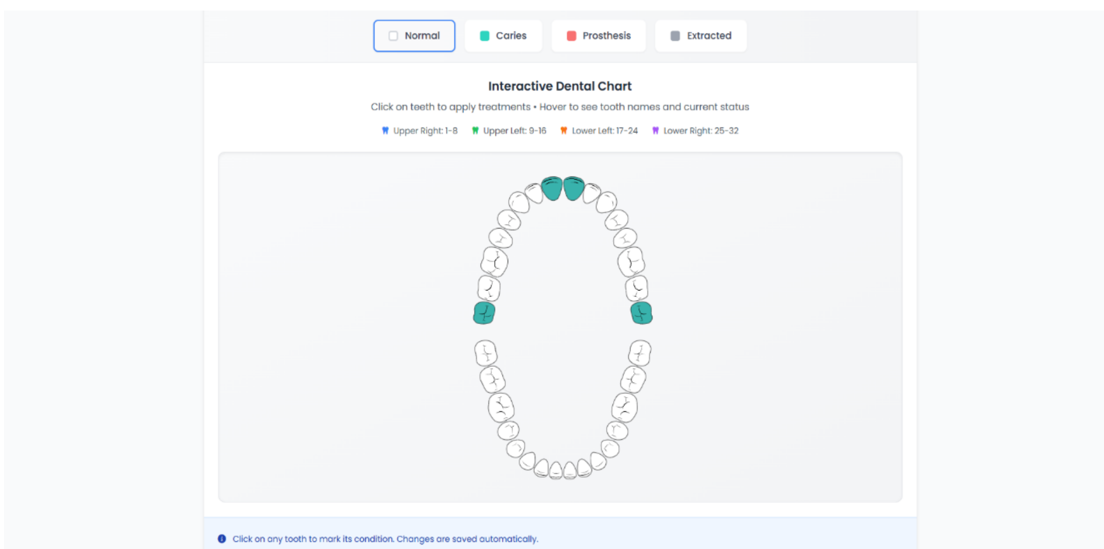
Patient Management(Staff)



Business Analytics(Admin)



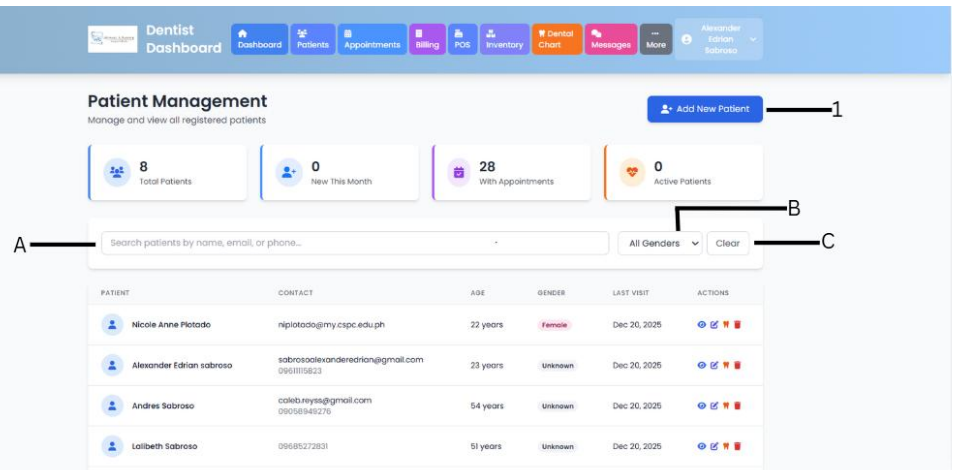
Dental Chart



APPENDIX D

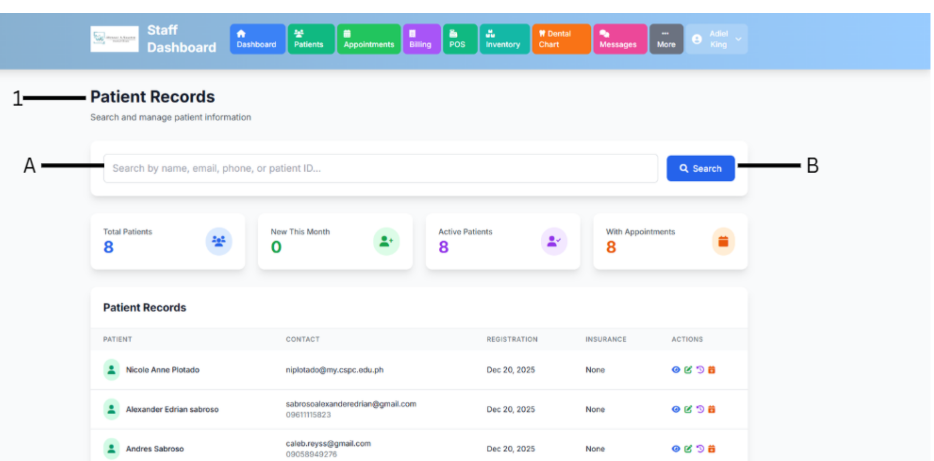
USER'S GUIDE

Patient Management(Admin/Dentist)



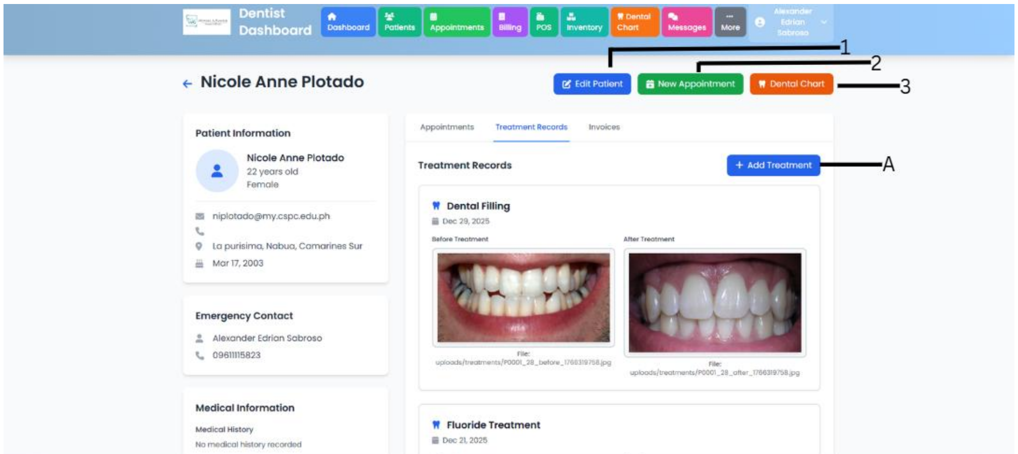
1. Click the blue "Add New Patient" button to register a new patient.
A. Type name, email, or phone in the search box to find a patient quickly.
B. Use the "All Genders" dropdown to filter by gender.
Click "Clear" to reset.

Patient Records(Staff)



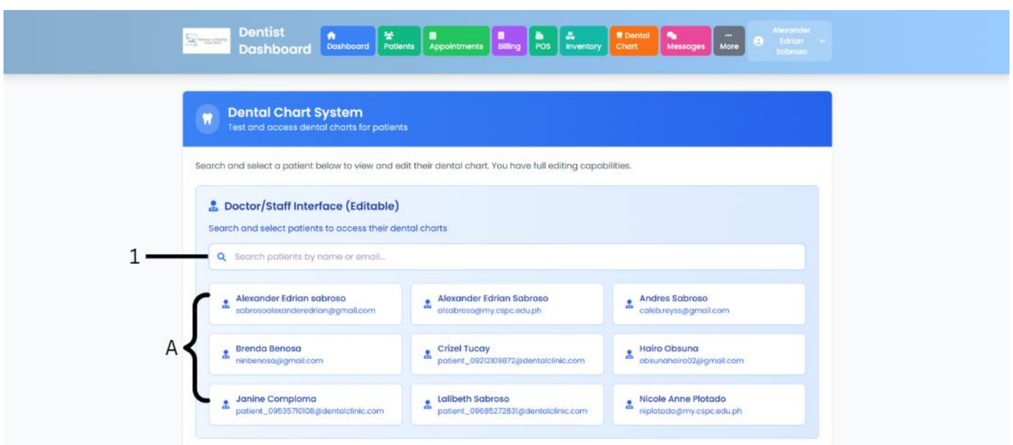
A. Type in the search box to find patients by name, email, phone, or patient ID.
B. Click the blue "Export" button to download patient records.

Patient Profile Interface(Patient)



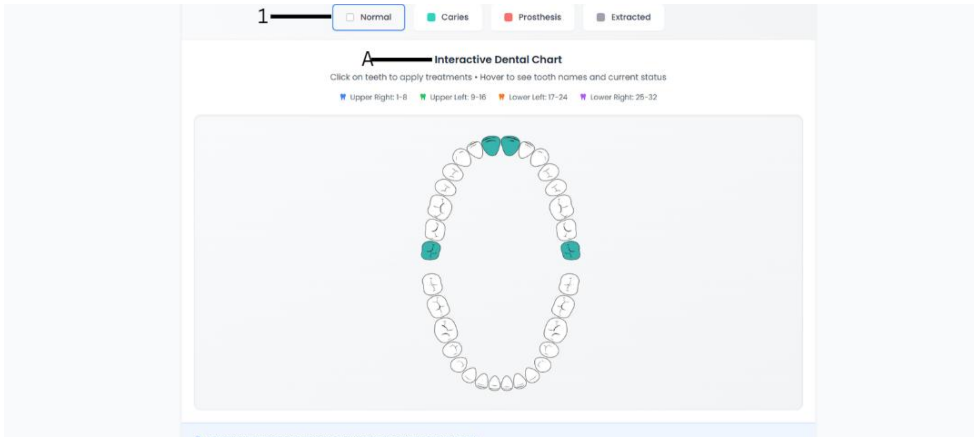
1.Click the blue **"View"** button to see full patient details.
2. Click the green **"Edit Patient"** button to update information.
3. Click the orange **"Actions"** button for additional options.
A. Click the blue "Add Treatment" button to record a new treatment or procedure.

Electronic Dental Chart



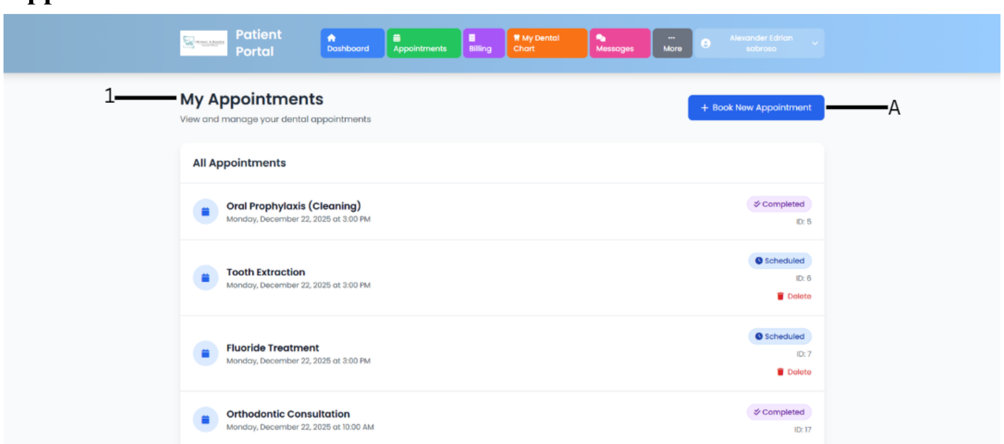
A. 1. Type patient name or email in the search box to find their dental chart.
B. A. Click on any patient card to open and edit their dental chart.

Interactive Dental Chart Interface



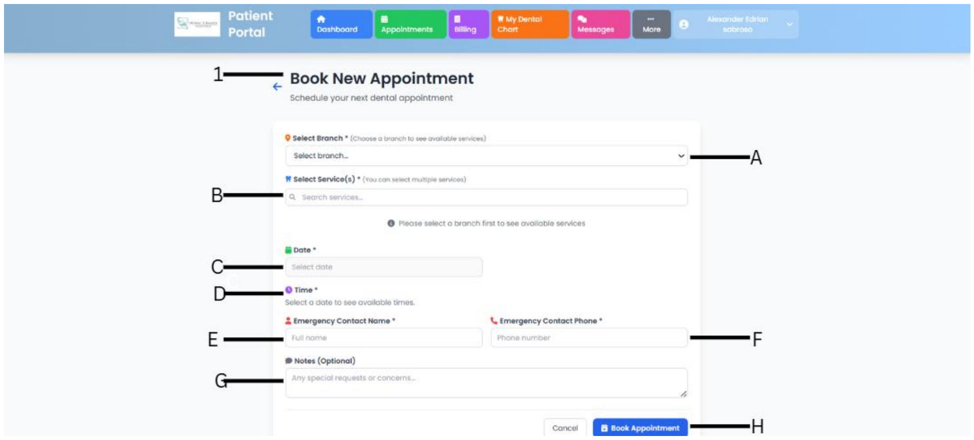
1. Select Tooth Status - Choose the condition (Normal, Caries, Prosthesis, or Extracted)
A. Click on Teeth - Apply the status by clicking teeth on the chart

Patient appointment Interface



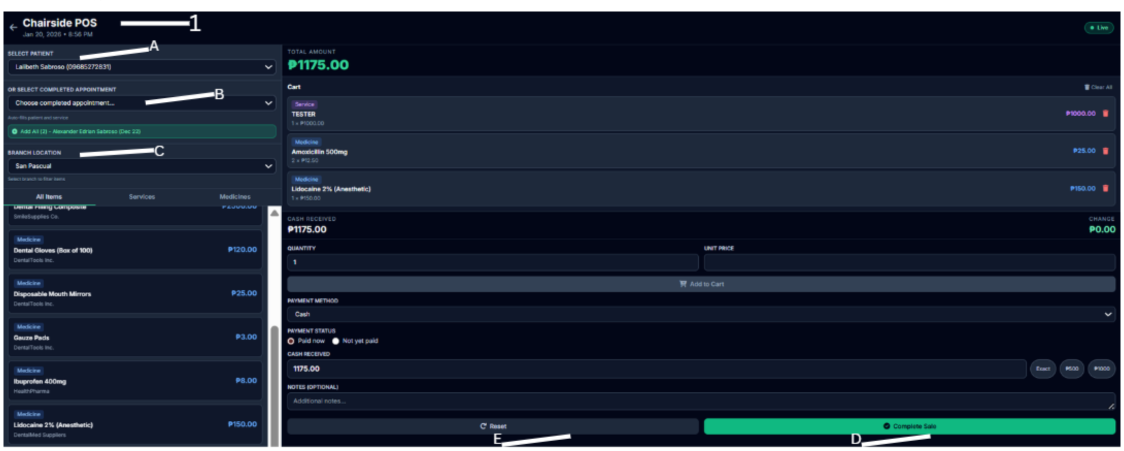
1. View Appointments - See all your scheduled and completed appointments
A. Book New Appointment - Schedule a new dental visit

Patient Appointment Booking Interface



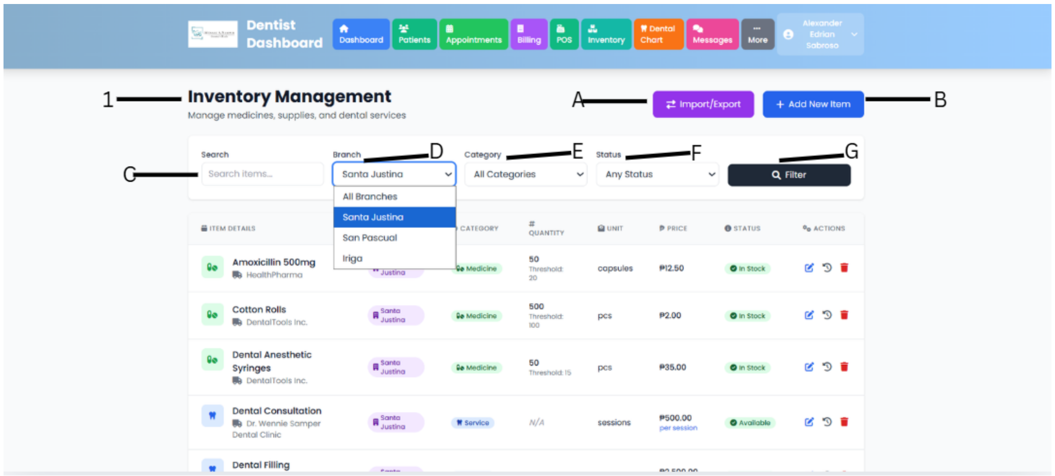
1.Click the back arrow to return to appointments list.
A. Choose a dental clinic branch from the dropdown to see available services.
B. Search and select the dental service(s) you need (you can choose multiple).
C. Pick your preferred appointment date from the calendar.
D. Choose an available time slot for your appointment.
E. Enter the full name of your emergency contact person.
F. Emergency Contact Phone
F. Enter the phone number of your emergency contact.
G. Add any special requests or health concerns for the dentist.
H. Click the blue **"Book Appointment"** button to confirm your booking.

Point of Sale Interface



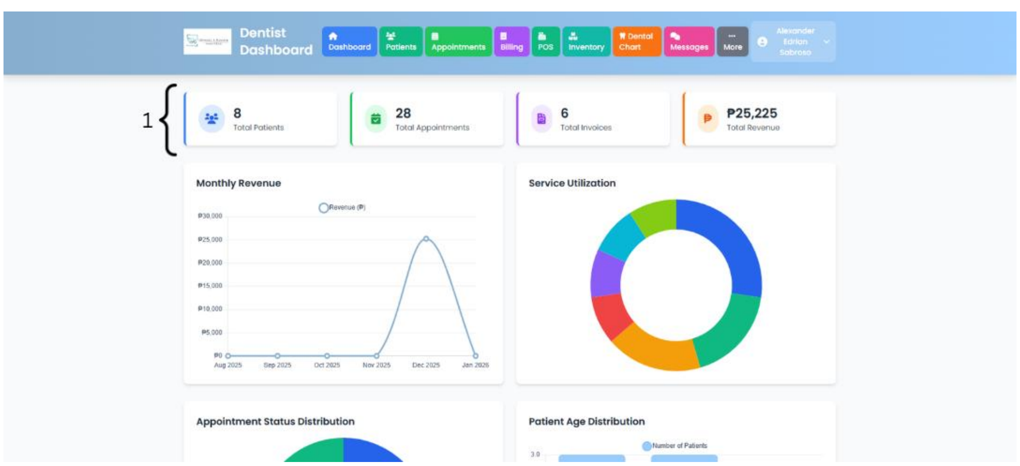
A. Scan or add items to the cart
B. Enter customer information (if needed)
C. Review total and apply discounts if applicable
D. Select payment method
E. Click the green payment button to complete transaction

Inventory Management Interface



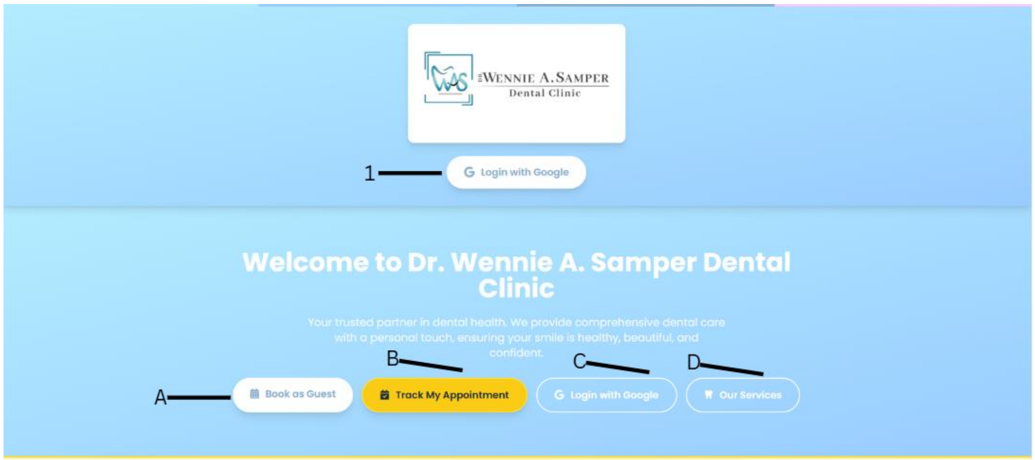
A. Click the purple **"Import/Export"** button to upload or download inventory data.
B. Click the blue **"Add New Item"** button to add medicines, supplies, or services.
Type item name in the search box to find specific inventory.
C. Select a branch to view inventory
E. Select Category
F. Select Status
G. Click the ***"Filter"*** *button to apply your search criteria*.

Business Analytics Dashboard



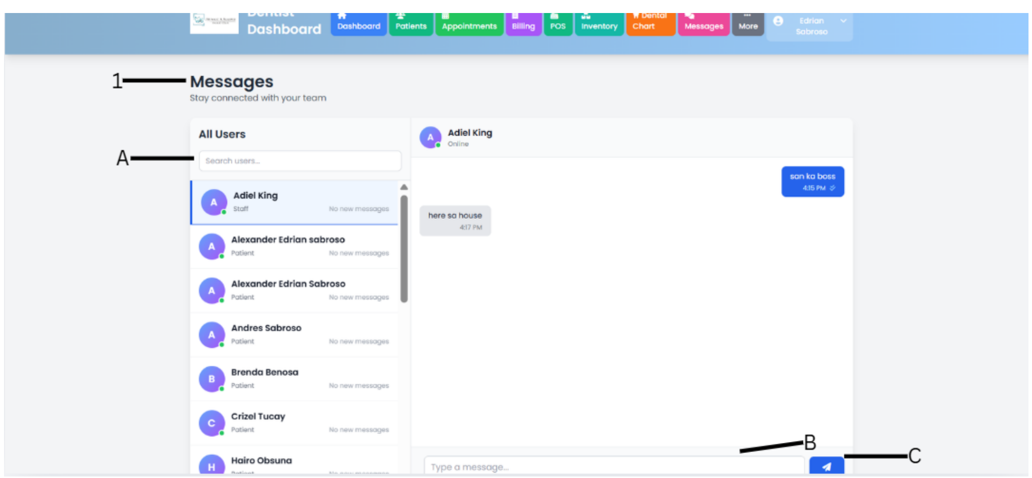
1. View four critical statistics at the top (Patients, Appointments, Invoices, Revenue)

Landing Page Interface



1. Click the **"Login with Google"** button in the top right to access your patient account.
A. Click the **"Book at Event"** button for special event bookings.
B. Click the yellow **"Book My Appointment"** button to schedule a regular dental visit.
C. Click **"Dental Services"** to view all available treatments and procedures.
D. Click **"Resources"** to access dental health information and clinic materials.

Communication and Messaging Interface



1. View all your conversations and stay connected with your team.
A. Type in the search box to find specific users or conversations quickly.
B. Click in the text box at the bottom to type your message.
C. Click the blue **send button** (paper plane icon) to send your message.

APPENDIX E

COMMUNICATION LETTER

Republic of the Philippines
Camarines Sur Polytechnic Colleges
Nabua, Camarines Sur
ISO 9001:2015 Certified
COLLEGE of COMPUTER STUDIES

DATE: MAY 22, 2025

DR. WENNNIE A. SAMPER-CAS
DENTIST
DR. WENNIE A. SAMPER DENTAL CLINIC

Dear Dr. Wennie,

Greetings!

The undersigned are Bachelor of Science in Information Technology (BSIT) students at Camarines Sur Polytechnic Colleges. As part of our academic curriculum, we are required to complete a capstone project that involves working with a real-world organization to address a relevant problem or develop an innovative solution.

We believe that **DR. WENNIE A. SAMPER DENTAL CLINIC** would be an excellent fit as a client for our capstone project. Our team is eager to contribute our skills and knowledge to support your organization while gaining valuable hands-on experience. We would be happy to fit our project to meet your organization's specific needs and challenges. The project timeline will span from March 2025 – December 2025, with key deliverables and milestones scheduled accordingly.

For questions, you can reach us at **alsabroso@my.cspc.edu.ph, jacomploma@my.cspc.edu.ph, matucay@my.cspc.edu.ph** or **09611115823, 09535710108, 09635917086**. Thank you for considering our request. We look forward to the possibility of working together and contributing to your organization.

Best regards,

ALEXANDER EDRIAN O. SABROSO
Project head/Programmer

JANINE R. COMPLOMA
Front-end/UI Designer

MA. CRIZEL TUCAY
Documentation Writer

Noted by:

PHILIP ALGER M. SERRANO, MIT
Capstone Project Adviser

MARICRIS L. RAMIZARES, MIT
Capstone Project Subject Adviser

APPENDIX F

NON-DISCLOSURE AGREEMENT FORM

Non-Disclosure Agreement Form

EFFECTIVE DATE: January 20, 2026
(TD submission date)

This Agreement sets forth the terms and conditions under which confidential, proprietary and other private information shall be disclosed between the College of Computer Studies-Camarines Sur Polytechnic Colleges and Brenda D. Benosa hereinafter referred to as "Expert."

By signing below, the parties acknowledge and accept the terms and conditions herein.

1. The Expert authorized to disclose and receive the confidential information is:
   BRENDA D. BENOSA, MIS – Panel Chairman
   Name and Title

   On behalf of the College of Computer Studies- Camarines Sur Polytechnic Colleges:
   ROSEL O. ONESA, MIT - Dean
   Name and Title

2. The confidential information disclosed under this Agreement is described as:
   Contents of the TCP by:

   Alexander Edrian O. Sabroso, Janine R. Comploma, Ma. Crizel H. Tucay

   which is entitled:

   Development of Multi-Branch Dental Practice Management System with Point-Of-Sale And Business Analytics for Dr. Wennie A. Samper Dental Clinic

3. The confidential information shall be used by the Expert only for the purpose of examination of TCP as part of the requirements of the Bachelor Program in which the student named above is enrolled.
4. This Agreement controls only confidential information, which is disclosed between the effective date and one year following the date of the TCP submission.
5. The obligations imposed upon an Expert hereunder shall apply only to information which at the time of disclosure is:
   a. marked as confidential if such information is disclosed in a physical form as the content of the TCP named above, and the oral defense, if any, of this same TCP, or
   b. if disclosed in some other form or manner is identified as confidential, and which identification is subsequently confirmed in a written notice delivered to the Expert specified in item 1 within thirty (30) days of disclosure.
6. The Expert agrees to take all action reasonably necessary to protect the confidentiality of the confidential information, including without limitation, implementing and enforcing operating procedures to minimize the possibility of unauthorized use or copying of the confidential information. Without limiting the foregoing, the Expert agrees to utilize the same degree of care, to avoid unauthorized disclosure or use of the confidential information of the discloser that the Expert would normally use with respect to its own confidential information.
7. The obligations imposed upon an Expert hereunder do not apply to information:
   a. which is or becomes publicly available without breach of this Agreement;
   b. which is already known to the Recipient prior to its disclosure hereunder;
   c. which is independently developed by the Expert.
8. The parties acknowledge that any technology, product, or other intellectual property identified as confidential information and provided hereunder is provided on an "as is" basis without warranty of any kind whether express or implied and that the implied warranties of merchantability and fitness for a particular purpose are expressly disclaimed. In particular,

the Expert shall not be liable for any direct, indirect, special, or consequential damages in connection with or arising out of the performance or use of any portion of the confidential information.

9. Nothing in this Agreement shall be construed to preclude the Expert from using, marketing, licensing, and/or selling any independently developed technology, product or other intellectual property that is similar or related to the confidential information disclosed hereunder.
10. Neither Party:
    a. acquires any intellectual property rights under this Agreement except the limited right to use the confidential information as specified in Paragraph 3;
    b. has an obligation hereunder to purchase or otherwise acquire any service or item from the other;
    c. has an obligation hereunder to commercially release any products or services using or incorporating the confidential information.
11. Upon the Camarines Sur Polytechnic Colleges written request, the Expert shall immediately return any Confidential Information and the physical media on which it was received or destroy all copies of the Confidential Information and certify in writing to the Camarines Sur Polytechnic Colleges that it has destroyed all copies made of the Confidential Information. Such certification shall be delivered within five (5) days of the Camarines Sur Polytechnic Colleges' request.
12. All modifications or amendments to this Agreement must be in writing and must be signed by both parties.
13. The parties are independent contractors, and this Agreement does not establish any relationship of agency, partnership or joint venture.
14. This Agreement shall be governed by the laws of the Nabua, Camarines Sur and the laws of the Philippines therein.

ACCEPTED BY:	CAMARINES SUR POLYTECHNIC COLLEGES
BRENDA D. BENOSA	**ROSEL O. ONESA, MIT**
Expert	**CSPC Representative**
(Print name and Signature)	(Print name and Signature)
DATE: January 20, 2026	DATE: January 21, 2026

Non-Disclosure Agreement Form

EFFECTIVE DATE: January 20, 2026
(TD submission date)

This Agreement sets forth the terms and conditions under which confidential, proprietary and other private information shall be disclosed between the College of Computer Studies-Camarines Sur Polytechnic Colleges and **Jayvee Niel S. Sias**, hereinafter referred to as "Expert."

By signing below, the parties acknowledge and accept the terms and conditions herein.

1. The Expert authorized to disclose and receive the confidential information is:
   JAYVEE NIEL S. SIAS – Panel Member 3
   Name and Title

   On behalf of the College of Computer Studies- Camarines Sur Polytechnic Colleges:
   ROSEL O. ONESA, MIT - Dean
   Name and Title

2. The confidential information disclosed under this Agreement is described as:
   Contents of the TCP by:

   Alexander Edrian O. Sabroso, Janine R. Comploma, Ma. Crizel H. Tucay

   which is entitled:

   Development of Multi-Branch Dental Practice Management System with Point-Of-Sale And Business Analytics for Dr. Wennie A. Samper Dental Clinic

3. The confidential information shall be used by the Expert only for the purpose of examination of TCP as part of the requirements of the Bachelor Program in which the student named above is enrolled.
4. This Agreement controls only confidential information, which is disclosed between the effective date and one year following the date of the TCP submission.
5. The obligations imposed upon an Expert hereunder shall apply only to information which at the time of disclosure is:
   a. marked as confidential if such information is disclosed in a physical form as the content of the TCP named above, and the oral defense, if any, of this same TCP, or
   b. if disclosed in some other form or manner is identified as confidential, and which identification is subsequently confirmed in a written notice delivered to the Expert specified in item 1 within thirty (30) days of disclosure.
6. The Expert agrees to take all action reasonably necessary to protect the confidentiality of the confidential information, including without limitation, implementing and enforcing operating procedures to minimize the possibility of unauthorized use or copying of the confidential information. Without limiting the foregoing, the Expert agrees to utilize the same degree of care, to avoid unauthorized disclosure or use of the confidential information of the discloser that the Expert would normally use with respect to its own confidential information.
7. The obligations imposed upon an Expert hereunder do not apply to information:
   a. which is or becomes publicly available without breach of this Agreement;
   b. which is already known to the Recipient prior to its disclosure hereunder;
   c. which is independently developed by the Expert.
8. The parties acknowledge that any technology, product, or other intellectual property identified as confidential information and provided hereunder is provided on an "as is" basis without warranty of any kind whether express or implied and that the implied warranties of merchantability and fitness for a particular purpose are expressly disclaimed. In particular,

the Expert shall not be liable for any direct, indirect, special, or consequential damages in connection with or arising out of the performance or use of any portion of the confidential information.

9. Nothing in this Agreement shall be construed to preclude the Expert from using, marketing, licensing, and/or selling any independently developed technology, product or other intellectual property that is similar or related to the confidential information disclosed hereunder.
10. Neither Party:
    a. acquires any intellectual property rights under this Agreement except the limited right to use the confidential information as specified in Paragraph 3;
    b. has an obligation hereunder to purchase or otherwise acquire any service or item from the other;
    c. has an obligation hereunder to commercially release any products or services using or incorporating the confidential information.
11. Upon the Camarines Sur Polytechnic Colleges written request, the Expert shall immediately return any Confidential Information and the physical media on which it was received or destroy all copies of the Confidential Information and certify in writing to the Camarines Sur Polytechnic Colleges that it has destroyed all copies made of the Confidential Information. Such certification shall be delivered within five (5) days of the Camarines Sur Polytechnic Colleges' request.
12. All modifications or amendments to this Agreement must be in writing and must be signed by both parties.
13. The parties are independent contractors, and this Agreement does not establish any relationship of agency, partnership or joint venture.
14. This Agreement shall be governed by the laws of the Nabua, Camarines Sur and the laws of the Philippines therein.

ACCEPTED BY:

JAYVEE NIEL S. SIAS
Expert
(Print name and Signature)

DATE: January 20, 2026

CAMARINES SUR POLYTECHNIC COLLEGES

ROSEL O. ONESA, MIT
CSPC Representative
(Print name and Signature)

DATE: January 21, 2026

Non-Disclosure Agreement Form

EFFECTIVE DATE: January 20, 2026
(TD submission date)

This Agreement sets forth the terms and conditions under which confidential, proprietary and other private information shall be disclosed between the College of Computer Studies-Camarines Sur Polytechnic Colleges and Mhelrose B. Prades, hereinafter referred to as "Expert."

By signing below, the parties acknowledge and accept the terms and conditions herein.

1. The Expert authorized to disclose and receive the confidential information is:
   MHELROSE B. PRADES, MIT – Panel Member 2
   Name and Title

   On behalf of the College of Computer Studies- Camarines Sur Polytechnic Colleges:
   ROSEL O. ONESA, MIT - Dean
   Name and Title

2. The confidential information disclosed under this Agreement is described as:
   Contents of the TCP by:

   Alexander Edrian O. Sabroso Janine R. Comploma, Ma. Crizel H. Tucay

   which is entitled:

   Development of Multi-Branch Dental Practice Management System with Point-Of-Sale And Business Analytics for Dr. Wennie A. Samper Dental Clinic

3. The confidential information shall be used by the Expert only for the purpose of examination of TCP as part of the requirements of the Bachelor Program in which the student named above is enrolled.
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   b. if disclosed in some other form or manner is identified as confidential, and which identification is subsequently confirmed in a written notice delivered to the Expert specified in item 1 within thirty (30) days of disclosure.
6. The Expert agrees to take all action reasonably necessary to protect the confidentiality of the confidential information, including without limitation, implementing and enforcing operating procedures to minimize the possibility of unauthorized use or copying of the confidential information. Without limiting the foregoing, the Expert agrees to utilize the same degree of care, to avoid unauthorized disclosure or use of the confidential information of the discloser that the Expert would normally use with respect to its own confidential information.
7. The obligations imposed upon an Expert hereunder do not apply to information:
   a. which is or becomes publicly available without breach of this Agreement;
   b. which is already known to the Recipient prior to its disclosure hereunder;
   c. which is independently developed by the Expert.
8. The parties acknowledge that any technology, product, or other intellectual property identified as confidential information and provided hereunder is provided on an "as is" basis without warranty of any kind whether express or implied and that the implied warranties of merchantability and fitness for a particular purpose are expressly disclaimed. In particular,

the Expert shall not be liable for any direct, indirect, special, or consequential damages in connection with or arising out of the performance or use of any portion of the confidential information.

9. Nothing in this Agreement shall be construed to preclude the Expert from using, marketing, licensing, and/or selling any independently developed technology, product or other intellectual property that is similar or related to the confidential information disclosed hereunder.
10. Neither Party:
    a. acquires any intellectual property rights under this Agreement except the limited right to use the confidential information as specified in Paragraph 3;
    b. has an obligation hereunder to purchase or otherwise acquire any service or item from the other;
    c. has an obligation hereunder to commercially release any products or services using or incorporating the confidential information.
11. Upon the Camarines Sur Polytechnic Colleges written request, the Expert shall immediately return any Confidential Information and the physical media on which it was received or destroy all copies of the Confidential Information and certify in writing to the Camarines Sur Polytechnic Colleges that it has destroyed all copies made of the Confidential Information. Such certification shall be delivered within five (5) days of the Camarines Sur Polytechnic Colleges' request.
12. All modifications or amendments to this Agreement must be in writing and must be signed by both parties.
13. The parties are independent contractors, and this Agreement does not establish any relationship of agency, partnership or joint venture.
14. This Agreement shall be governed by the laws of the Nabua, Camarines Sur and the laws of the Philippines therein.

ACCEPTED BY:	CAMARINES SUR POLYTECHNIC COLLEGES
MHELROSE B. PRADES, MIT	**ROSEL O. ONESA, MIT**
Expert	**CSPC Representative**
(Print name and Signature)	(Print name and Signature)
DATE: January 20, 2026	DATE: January 21, 2026

APPENDIX G

JOINT AFFIDAVIT OF UNDERTAKING

Joint Affidavit of Undertaking

We, (1) ALEXANDER EDRIAN O. SABROSO, of legal age, single/ married and a resident of BRGY. STA CLARA BUHI CAMARINES SUR, (2) JANINE R. COMPLOMA, of legal age, single/ married and a resident of MONTE CALVARIO BUHI CAMARINES SUR, (3) MA. CRIZEL H. TUCAY, of legal age, single/ married and a resident of BRGY. STA LOURDES BUHI CAMARINES SUR, after having been sworn to in accordance with law, do hereby take oath and state:

(i) That, we are officially enrolled for the thesis/capstone project on the topic titled DEVELOPMENT OF MULTI-BRANCH DENTAL PRACTICE MANAGEMENT SYSTEM WITH POINT-OF-SALE AND BUSINESS ANALYTICS FOR DR. WENNIE A. SAMPER DENTAL CLINIC in the COLLEGE OF COMPUTER STUDIES of CAMARINES SUR POLYTECHNIC COLLEGES.

(ii) That, the contents of our thesis/ capstone project submitted to the Camarines Sur Polytechnic Colleges, for award of BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY Degree are original and our own work, and is not plagiarized.

(iii) That, if, after completing the thesis/ capstone project, are found copied or come under plagiarism, we will be solely responsible for it and College shall have sole right to cancel my research work ab-initio.

(iv) That, this work has not been submitted for the award of any other Degree/Diploma in any other University/ Institute.

(v) That, we shall be responsible for any legal dispute/case(s) for violation of any provisions of the Copyright Act relating to our thesis/ capstone project.

IN WITNESS WHEREOF, I have hereunto set my name this ____ day of ________ 202__ in 02 FEB 2026

Nabua Camarines Sur, Philippines.

(1) ALEXANDER EDRIAN O. SABROSO
Affiant
221001500

(2) JANINE R. COMPLOMA
Affiant
2210015616

(3) MA. CRIZEL H. TUCAY
Affiant
221007152

SUBSCRIBED AND SWORN TO before me this 02 FEB 2026 day of ______ at Nabua Camarines Sur, Philippines, affiants exhibiting to me their competent proofs of identity above stated.

Doc. No.: 75
Page No.: 15
Book No.: YCVIII
Series of 2026

ATTY. JOSELITO F. FIGURACION
Notary Public
Until December 31, 2027
Roll of Attys. No. 52026
PTR No. 7641698 T 01-05-26 Nabua, Cam. Sur
IBP OR No. 596589 1/22/2026
MCLE Compliance No. VIII-0013316 Valid 04-14-28
San Francisco, Nabua, Camarines Sur

APPENDIX H

ROLE ACCEPTANCE FORM

ROLE ACCEPTANCE FORM
College of Computer Studies

Date: February 3, 2025

To: **MR. PHILIP ALGER M. SERRANO, MIT**

We, the third-year students of Camarines Sur Polytechnic Colleges pursuing a degree in Bachelor of Science in Information Technology, are currently enrolled in Capstone Project 1.

We are writing to humbly request your service and expertise to serve as our Adviser for our capstone project. We believe that your knowledge and experience will be essential to greatly enrich our work. Attached are our capstone tentative title proposals for your kind reference.

Thank you and looking forward to your favorable response of our request.

Respectfully,

Alexander Edrian O. Sabroso

Janine R. Comploma

Ma. Crizel H. Tucay

To: **MS. ROSEL O. ONESA**
OIC Dean, CCS

This formally signifies that I **ACCEPT/ REJECT** the request to serve as Adviser of the team Byblis Liniflora.

As Adviser, I agree to perform my duties and responsibilities stipulated in Section 2.6 of the TCP Guidebook from Capstone Project 1 until Capstone Project 2.

Furthermore, I agree to set the schedule for advising or consultation to help the students and ensure the success of the capstone project.

Conformed:

MR. PHILIP ALGER M. SERRANO, MIT
Name and Signature

ROLE ACCEPTANCE FORM
College of Computer Studies

Date: February 3, 2025

To: **MR. JEREMY JIREH S. NEO**

We, the third-year students of Camarines Sur Polytechnic Colleges pursuing a degree in Bachelor of Science in Information Technology, are currently enrolled in Capstone Project 1.

We are writing to humbly request your service and expertise to serve as our Consultant for our capstone project. We believe that your knowledge and experience will be essential to greatly enrich our work. Attached are our capstone tentative title proposals for your kind reference.

Thank you and looking forward to your favorable response of our request.

Respectfully,

Alexander Edrian O. Sabroso

Janine M. Compioma

Ma. Crizel H. Tucay

To: **MS. ROSEL O. ONESA**
OIC Dean, CCS

This formally signifies that I **ACCEPT/ REJECT** the request to serve as Consultant of the team Byblis Liniflora.

As Adviser, I agree to perform my duties and responsibilities stipulated in Section 2.6 of the TCP Guidebook from Capstone Project 1 until Capstone Project 2.

Furthermore, I agree to set the schedule for advising or consultation to help the students and ensure the success of the capstone project.

Conformed:

JEREMY JIREH S. NEO
Name and Signature

ROLE ACCEPTANCE FORM
College of Computer Studies

Date: February 3, 2025

To: **MR. NEL FRANCIS BUENA, MLL**

We, the third-year students of Camarines Sur Polytechnic Colleges pursuing a degree in Bachelor of Science in Information Technology, are currently enrolled in Capstone Project 1.

We are writing to humbly request your service and expertise to serve as our Grammarian for our capstone project. We believe that your knowledge and experience will be essential to greatly enrich our work. Attached are our capstone tentative title proposals for your kind reference.

Thank you and looking forward to your favorable response of our request.

Respectfully,

Alexander Edrian O. Sabroso

Janine R. Comploma

Ma. Crizel H. Tucay

To: **MS. ROSEL O. ONESA**
OIC Dean, CCS

This formally signifies that I **ACCEPT/ REJECT** the request to serve as Grammarian of the team Byblis Liniflora.

As Adviser, I agree to perform my duties and responsibilities stipulated in Section 2.6 of the TCP Guidebook from Capstone Project 1 until Capstone Project 2.

Furthermore, I agree to set the schedule for advising or consultation to help the students and ensure the success of the capstone project.

Conformed:

MR. NEL FRANCIS BUENA, MLL
Name and Signature

APPENDIX I

PROJECT TEAM ASSIGNMENT FORM

PROJECT TEAM ASSIGNMENTS FORM

Team Alias	BYBLIS LINIFLORA
Course Code	IT 3211
Subject adviser/ CaPSA	MARICRIS L. RAMIZARES, MIT

Name and Signature	Project Role	Email address and mobile#(s)
Alexander Edrian O. Sabroso	Project Head/Programmer	alsabroso@my.cspc.edu.ph 09511115823
Janine R. Comploma	Front-end/UI Desinger	jacomploma@my.cspc.edu.ph 09535710108
Maria Crizel H. Tucay	QA Tester/Documentation Writer	crtucay@my.cspc.edu.ph 09635917086

*** Accomplish in 2 copies

APPENDIX J

FINAL PROJECT TITLE FORM

FINAL PROJECT TITLE FORM

Proponents/Researchers:

1. Sabroso, Alexander Edrian O.
2. Comploma, Janine R.
3. Tucay, Ma. Crizel H.

Proposed Thesis/ Capstone Project Title:

DEVELOPMENT OF MULTI-BRANCH DENTAL PRACTICE MANAGEMENT SYSTEM WITH POINT-OF-SALE AND BUSINESS ANALYTICS FOR DR. WENNIE A. SAMPER DENTAL CLINIC

Submitted by: **ALEXANDER EDRIAN O. SABROSO** (Signature of Project Head over printed name) Date: 1/11/2026	**Noted by:** **MARICRIS L. RAMIZARES, MIT** (Signature of Subject adviser over printed name) Date: 1/20/2026
Recommending Approval: **PHILIP ALGER M. SERRANO, MIT** (Signature of Adviser over printed name) Date: 1/15/2026	**Approved:** **ROSEL O. ONESA, MIT** OIC DEAN, CCS Date: 1/15/2026

APPENDIX K

CAPSTONE HEARING FORM (TD, POD, FOD)

CAPSTONE PROJECT HEARING FORM

Date Filed: ~~April 28~~ MAY 9, 2025 [☒] Title Proposal [☐] Pre-Oral [☐] Final Oral

Date of Hearing: MAY 14, 2025 Time: 1:00- 3:00 Venue: Open Laboratory

Department: **COLLEGE OF COMPUTER STUDIES (CCS)**

Research Title:

DEVELOPMENT OF INTEGRATED DENTAL PRACTICE SYSTEM WITH POINT-OF-SALE INTEGRATION AND BUSINESS ANALYTICS

Proponent/s:

ALEXANDER EDRIAN O. SABROSO **JANINE R. COMPLOMA**

MA. CRIZEL H. TUCAY

Recommended by:

PHILIP ALGER M. SERRANO, MIT
Capstone Project Adviser

CERTIFICATION

The undersigned members comprising the panel for oral examination hereby agree to the schedule of hearing for the above research.

MHELROSE B. PRADES, MIT
PANEL MEMBER 1

JAYVEE NEIL S. SIAS
PANEL MEMBER 2

BRENDA D. BENOSA, ~~MSI~~ MIS
PANEL CHAIR

NOTED BY:

MARICRIS L. RAMIZARES, MIT
Subject Adviser

APPROVED:

ROSEL O. ONESA, MIT
OIC Dean, CCS

CAPSTONE PROJECT HEARING FORM

Date Filed: November 5, 2025 [☐] Title Proposal [☒] Pre-Oral [☐] Final Oral

Date of Hearing: November 7, 2025 Time: 1:00 - 4:00 PM Venue: AIRCoDe

Department: **COLLEGE OF COMPUTER STUDIES (CCS)**

Research Title:

DEVELOPMENT OF MULTI-BRANCHED DENTAL PRACTICE MANAGEMENT SYSTEM WITH POINT-OF-SALE AND BUSINESS ANALYTICS FOR DR. WENNIE A. SAMPER DENTAL CLINIC

Proponent/s:

ALEXANDER EDRIAN O. SABROSO **JANINE R. COMPLOMA**

MA. CRIZEL H. TUCAY

Recommended by:

PHILIP ALGER M. SERRANO, MIT
Capstone Project Adviser

CERTIFICATION

The undersigned members comprising the panel for oral examination hereby agree to the schedule of hearing for the above research.

MHELROSE B. PRADES, MIT
PANEL MEMBER 1

JAYVEE NIEL S. SIAS
PANEL MEMBER 2

BRENDA D. BENOSA, MIS
PANEL CHAIR

NOTED BY:

MARICRIS L. RAMIZARES, MIT
Subject Adviser

APPROVED:

ROSEL O. ONESA, MIT
OIC Dean, CCS

CAPSTONE PROJECT HEARING FORM

Date Filed: December 19, 2025 [☐] Title Proposal [☐] Pre-Oral [☒] Final Oral

Date of Hearing: December 20, 2025 Time: 11:00 am - 12:00 pm Venue: AIRCode

Department: COLLEGE OF COMPUTER STUDIES (CCS)

Research Title:

DEVELOPMENT OF MULTI-BRANCH DENTAL PRACTICE MANAGEMENT SYSTEM WITH POINT-OF-SALE AND BUSINESS ANALYTICS FOR DR. WENNIE A. SAMPER DENTAL CLINIC

Proponent/s:

ALEXANDER EDRIAN O. SABROSO

JANINE R. COMPLOMA

MA. CRIZEL H. TUCAY

Recommended by:

PHILIP ALGER M. SERRANO, MIT

Capstone Project Adviser

CERTIFICATION

The undersigned members comprising the panel for oral examination hereby agree to the schedule of hearing for the above research.

MHELROSE B. PRADES, MIT
PANEL MEMBER 1

JAYVEE NIEL S. SIAS
PANEL MEMBER 2

BRENDA D. BENOSA, MIS
PANEL CHAIR

NOTED BY:

MARICRIS L. RAMIZARES, MIT
Subject Adviser

APPROVED:

ROSEL O. ONESA, MIT
OIC Dean, CCS

APPENDIX L

PANEL RSC (TD, POD, FOD)

Republic of the Philippines
Camarines Sur Polytechnic Colleges
Nabua, Camarines Sur
ISO 9001:2015 Certified
COLLEGE *of* COMPUTER STUDIES

Title : **Development of Integrated Dental Practice System with Point-of-Sale and Business Analytics**
Alias : Byblis Liniflora
Date : MAY 29, 2025, 10:00 AM – 12:00PM ☒ TD(REDEFENSE) ☐ POD ☐ FOD
Secretary : JOHNMIR ISAAC

MANUSCRIPT			
CHAPTER	**PAGE NO.**	**RECOMMENDATIONS, SUGGESTIONS AND COMMENT (RSC)**	**ACTION TAKEN**
		• REVISE TITLE THAT INTEGRATES MULTI-BRANCH • Fix hanging pages • Flowchart – enlarge and revise	
NOTES	**ALL**	USE ACM FORMAT	
1	**1**	Project Context • Support with RRL • Add profile discussion of the client	• RRL-supported context is present • Client profile (Dr. Wennie A. Samper Dental Clinic) is now introduced clearly
	6	Objectives • Revise general objective • SO1 – add inventory and communication • SO3 – use functional and non-functional and have at least 2 tests each	• Updated to include multi-branch, inventory communication, and analytics focus • Include in 1.4 and 1. Under SO1 • Added unit & Integration (Functional) and Usability &Performance (Non-Functional) under SO3
	10-11	Scope and Limitation • 1 paragraph each for scope and limitation	• Rewrite into one clear paragraph each and includes functional/non-functional testing
	12	Project Dictionary • Include all technical terms with conceptual and operational definitions	• Includes definitions for modules like POS, Inventory, Dental Chart, etc.
2		• Make discussion coherent • Add topic exclusively about dental health care • Use updated RL's (5 years)	• Paragraphs and transitions improved

SOFTWARE PROGRAM		
MODULE NO. (DFD)	**RECOMMENDATIONS, SUGGESTIONS, AND COMMENTS (RSC)**	**ACTION TAKEN**
MODULES	• ADD INVENTORY	An **Inventory Management Module** has been added to the system. It tracks consumables and dental medicines, automatically updates stocks during POS transactions, and provides alerts for low inventory.
	• Include inventory POS for medicines (for pharmacy)	• The POS is now linked with inventory, allowing tracking of used medicines and supplies during treatments. Stock automatically updates after each transaction.

	• Research about POS system	• Updated RRL and system design incorporate key POS features such as real-time invoicing, multi-payment support, and financial reporting.
	• Include dental chart of patients (with image) and with before and after treatments	• The **Electronic Dental Chart** includes tooth-specific tracking and supports image uploads for before-and-after treatment comparisons.
	• Inventory for consumables	• Consumables such as syringes, gloves, and other tools are tracked per usage. Inventory levels are updated when treatments are recorded in the system.
	• 1 login for all	• Implemented a **single login system** with Role-Based Access Control (RBAC) for Admin, Dentist, Staff, and Patient accounts.
	• Quotation for price of each treatment	• System allows creation of itemized treatment quotations, pulling item prices from the inventory and service list.
	• Patients can request for appointment for treatment	• The system allows both registered and guest patients to request appointments through the scheduling module.
	• Option for patients to have an account after appointment without account	• Guest users are offered the option to register after their first appointment. Previous appointment data is linked to their new account.
	• Record to view of patient (treatment, bills, dental chart etc.)	• Patients can view their treatment history, bills, and dental chart via a **patient portal** (read-only access).
	• Notification for confirmation of appointment thru SMS or email for first timers)	• System sends **SMS/email confirmations** for first-time appointment requests via the Communication and Workflow Module.

NOTED BY:

Brenda D. Benosa, MIS
PANEL, CHAIRMAN

Mhelrose B. Prades, MIT
PANEL, MEMBER 1

Jayvee Niel S. Sias
PANEL, MEMBER 2

Republic of the Philippines
Camarines Sur Polytechnic Colleges
Nabua, Camarines Sur
ISO 9001:2015 Certified
COLLEGE of COMPUTER STUDIES

Title : **Development of Integrated Dental Practice System with Point-of-Sale and Business Analytics**

Alias : BYBLIS LINIFLORA

Date : November 21, 2025 ☐ TD ☒ POD ☒ FOD

Secretary : JOHN PETER G. ANDALIS

MANUSCRIPT			
CHAPTER	**PAGE NO.**	**RECOMMENDATIONS, SUGGESTIONS AND COMMENT (RSC)**	ACTION TAKEN
		• The title page has no page number	Title page number has been added
Notes	**ALL**	• Use ACM Format	Used ACM format
ALL	**ALL**	• Use proper formatting for all tables	Used proper formatting for all tables
ALL	**ALL**	• All figures must be clear and readable	
ALL	**ALL**	• Proper size of the footer and margins	Proper size of footer and margins has been edited
	1	**Project Context** • Support with RRL • Add a short profile discussion of the client	Supported with RRL and short profile discussion of client has been added
	2	**Project Context** • Fix citations, place them within the exact context	Fixed citation and placed with the exact context
	3	**Purpose and Description of the Project** • 3-5 sentences per paragraph only	Edited to 3-5 sentences
	4	**Objectives of the Project** • Fix the introduction discussion and mention the system for whose doctor	Fixed introduction and discussion and DR. Wennie A. Samper has been mentioned
	6	**Significance of the Study** • Combine the Clinic Owners and Managers and the Dr. Wennie A. Samper Sental Clinic Management	Combined the Clinic Owners and Managers and the Dr. Wennie A. Samper Sental Clinic Management
1	**7**	**Significance of the Study** • Remove the Policy Makers and Health Administrators Arrange according to: 1. Dr. Wennie A. Samper Dental Clinic Management 2. Administrative Staff 3. Patients 4. Dental Practitioners 5. Clinic Owners and Managers 6. Healthcare Providers and Private Clinics 7. Information Technology Students and Developers 8. Researchers 9. Future Researchers	Arranged and Removed the Policy makers and health administrators
	9	**Project Dictionary** • Conceptual definition first, and then operational meaning or technical terms only	Fixed Project Dictionary
3	**ALL Chapter 3**	• Update all content in Chapter 3 according to your actual system	All contents has been updated
	33 - 34	**Table 1 and Table 2** • Identify whether it's for development or deployment	Identified the development and deployment

	39 – 40 & 42	**Figure 1** • Include the processes • Use appropriate use of flowchart symbols • Proper naming of the label	Fixed Figure 1
	48	**Figure 5** • Fix figure title	Fixed Figure Title
4	All Chapter 4	• Fix all content in Chapter 4 according to the actual content of the system • Re-do the evaluation testing	Fixed Chapter 4 & re-do the testing
	54	**Testing** • Elaborate testing used	Testing used elaborated
	55	**Data Gathering Procedures** • Shorten the paragraph, direct it to the point	Shorted and direct to the point
	58 - 59	**Table 4** • Fix the table **Statistical Analysis Tool** • Put the formula	Fixed table and formula added
	66	**Figure 6** • Revise the fishbone diagram • The head of the fishbone diagram is the main problem; elaborate on it	Revised Fishbone Diagram
	76	**Figure 9** • Revise the Context Diagram	Revised the Context Diagram
	78	**Figure 10** • Fix and make sure it's consistent with other diagrams	Fixed Figure 10
5	138	**Findings** • Discuss the results	Result Discussed
	140	**Conclusion** • Discuss based on the findings	Conclusion based on findings

SOFTWARE PROGRAM		
MODULE NO. (DFD)	**RECOMMENDATIONS, SUGGESTIONS, AND COMMENTS (RSC)**	**ACTION TAKEN**
	• Research about POS	Did a research and created a new UI for the POS
	• Inventory for consumables	There is an existing inventory
	• Quotation for the price of each treatment	Added an quotation of prices
	• Option for patients to have an account after the appointment without an account	Added an option for the patient to create an account after booking
	• Notification for confirmation of appointment through SMS or email for first timers	Email works properly now
	• No email	Email works properly now
	• Error creating new account	Fixed the error when creating a new account
	• Add sign up	Added an option for the users who don't have an account to create an account
	• Add history of the patient	Added patient history

	• No settings for multi-branch	Added multi-branch option
	• Calendar view the appointment	There is an existing calendar view
	• Add an upload of a picture of before and after the treatment	Added a function where the dentist or staff can upload the before and after image of the patients teeth
	• Add a filter in services, inventory, patient, and invoices	There is an existing filter
	• Fix the computation in the payment	Fixed the computation where the invoices shows only the price of the service

NOTED BY:

BRENDA D. BENOSA, MIS
PANEL, CHAIRMAN

MHELROSE B. PRADES, MIT
PANEL, MEMBER 1

JAYVEE NIEL S. SIAS
PANEL, MEMBER 2

Title : **Development of Integrated Dental Practice System with Point-of-Sale and Business Analytics**
Alias : BYBLIS LINIFLORA
Date : December 15, 2025 ☐ TD ☒ POD ☐ FOD
Secretary : JOHN PETER G. ANDALIS

MANUSCRIPT			
CHAPTER	**PAGE NO.**	**RECOMMENDATIONS, SUGGESTIONS AND COMMENT (RSC)**	**ACTION TAKEN**
		• On every first page of the chapters, the page number is not visible	
Notes	**ALL**	• Remove the hyperlink	Hyperlink Removed
ALL	**ALL**	• Use square brackets [ ], not parentheses ( )	• ( ) replaced with [ ]
ALL	**ALL**	• Fix all the hanging pages	Fixed hanging pages
ALL	**ALL**	• Use ACM Format	The others are in ACM format, and others are not because we could not find the full names of the authors.
ALL	**ALL**	• Use proper formatting for all tables	Fix table format
ALL	**ALL**	• All figures must be clear and readable	All figures are clear
2	**21**	**System Testing using Black Box Testing** • Convert into 2 paragraph the 4th to 8th paragraph and fix the discussion	Converted into 2 paragraph and discussion fixed
	23	**Gap Bridged by the Study** • After identification of the similarities and differences, identify the gaps and how it is able to address them in the study	Fixed
3	**ALL Chapter 3**	• Update all content in Chapter 3 according to your actual system	Chapter 3 Updated
	33 - 34	**Table 1 and Table 2** • Identify whether it's for development or deployment	Table 1 is for deployment and table 2 is for development
	39 – 41	**Figure 1 - 3** • Include the processes • Use appropriate use of flowchart symbols • Proper naming of the label • Update according to the actual system and diagrams	Updated figure 1-3
	48	**Figure 5** • Fix figure title	Fixed Figure title
4	**All Chapter 4**	• Fix all content in Chapter 4 according to the actual content of the system • Re-do the evaluation testing • All content of Chapter 4 aligns with the chosen SDM	
	51	**Methodology, Results and Discussion**	Citation added

		• Add reference/citation for Agile Scrum Methodology	
	54	**Respondents** • Remove the first paragraph **Table 5** • Add at least 5 IT experts. The sampling technique is purposive, and then identify the criteria for selection. • For the dentist and staff, the sampling technique should be total enumeration. For the patient, at least 30 respondents should be used, employing convenience sampling. • Justify the use of each sampling technique.	Respondent's first paragraph has been removed.
	55	• Remove Table 6	Removed
	61	**Table 9** • The result is 1 - 0	Fixed Basis Result
	63	**Table 10** • Add another column for the problems	As per Ma'am Mhel's suggestion, just add 1 row and base it on the objectives of the project under 1. Identify the challenges encountered in managing dental clinic operations.
	68	**Figure 6** • The main problem is "Inefficient Clinic Operations"	Revised Fishbone
	70	**Figure 7** • Update according to actual system	Revised Functional decomposition diagram
	75	• The designing and development are under the Execution Phase	Fixed
	77	**Figure 9** • Revise the Context Diagram	Revised context diagram
	79	**Figure 10** • The process in DFD must be noun • Fix and make sure it's consistent with other diagrams	Revised DFD
	87	**Figure 14** • Adjust according to the process	Fixed Figure 14
	98	**Output and User-Interface Design** • This is under the Execution Phase (Development)	Under na ss nalang
		Testing • Elaborate testing used	Elaborated testing used
5	133	**Summary** • Add the overall findings **Findings** • Discuss the results	Fixed summary and findings
	136	**Conclusion** • Discuss based on the findings	Fix conclusion
	142	• Remove the Notes	Removed

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Camarines Sur Polytechnic Colleges
Nabua, Camarines Sur
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COLLEGE of COMPUTER STUDIES

SOFTWARE PROGRAM		
MODULE NO. (DFD)	RECOMMENDATIONS, SUGGESTIONS, AND COMMENTS (RSC)	ACTION TAKEN
	• Option for patients to have an account after the appointment without an account	Added an option for patients to have an account after booking an appointment
	• Notification for confirmation of appointment through SMS or email for first timers	Added a notification through email
	• Error creating new account	Fixed the error on creating new account
	• Change the complete button to save	Changed the button into "Save" to avoid repetitions
	• View of before and after treatment in the image	Fixed the error where the before and after image on patient history does not show
	• Multiple services per appointment	Added a feature where the user can add multiple services on booking
	• Change status options for services in inventory (available or active), not in stock	Changed the status option into (Available or not available) for services
	• Error in creating an account after the appointment	Fixed the error on creating an account
	• Test SMS notification	No actions taken
	• Report generation (medicines and services), daily, weekly, monthly, and yearly revenue	Added a new feature where the Dentist/Admin has options to show a daily, weekly, monthly and yearly report revenue
	• 1 – 2 reports that can be used as a basis for decision-making	added reports for decision making
	• Import and export products/services to/from the branch or add a product/service for multiple branches	Added an option to add services or medicines/items to multiple branches
	• Connected appointment – treatment - POS	The Connection is already active
	• Add in multi-branch in settings update and delete	Added a setting where the branch can be updated
	• In edit appointment, add before and after treatment	This is currently active
	• Username or phone number in creating a new account	Added Username or phone number option in creating new account
	• In POS, it will show the booked services	Shown Booked services

NOTED BY:

BRENDA D. BENOSA, MIS
PANEL, CHAIRMAN

MHELROSE B. PRADES, MIT
PANEL, MEMBER 1

JAYVEE NIEL S. SIAS
PANEL, MEMBER 2

Republic of the Philippines
Camarines Sur Polytechnic Colleges
Nabua, Camarines Sur
ISO 9001:2015 Certified
COLLEGE of COMPUTER STUDIES

Title : DEVELOPMENT OF INTEGRATED DENTAL PRACTICE SYSTEM WITH POINT-OF-SALE AND BUSINESS ANALYTICS

Alias : TEAM BYBLIS LINIFLORA

Date : DECEMBER 20, 2025 ☐ TD ☐ POD ☑ FOD

Secretary : MARIETON L. RAMIZARES

MANUSCRIPT			
CHAPTER	PAGE NO.	RECOMMENDATIONS, SUGGESTIONS AND COMMENT (RSC)	ACTION TAKEN
2	23	**Gap Bridged by the Study** • After identification of the similarities and differences, identify the gaps and how it is able to address them in the study.	Fixed Gap Bridged by the Study
3	33-34	**Table 1 and Table 2** • Identify whether it's for development or deployment. • If possible, use a single table where you can identify whether each specification is for development or deployment.	Fixed Table 1 and 2 and we identified whether it's for development or deployment in a single table
3	39-41	**Figures 1-3** • Include the processes. • Use appropriate use of flowchart symbols. • Proper naming of label. • Update according to the actual system and diagrams. • Increase size of images.	Fixed Figured 1-3
4	61	**Table 9** • Include a reference for the range used.	Added a reference
4	63	**Table 11** • Add 1 more column according to your SOs.	Added 1 Column to Table 11, the table number adjusted, that's why now on the table 10
4	ALL	The UI presentation in the manuscript should be organized by module, based on the stated SOs.	UI organized by module
4	112	**Table 12** • Replace "PASS" with "PASSED".	Fixed Table 12, the table number adjusted, that's why it's on the table 11
4	113	• Place the unit testing results first. • Specify the value that indicates a passing result.	
4		• Problem with unit testing (system error) • Problem with testing (evaluation process)	

SOFTWARE PROGRAM		
MODULE NO. (DFD)	RECOMMENDATIONS, SUGGESTIONS, AND COMMENTS (RSC)	ACTION TAKEN
	No SMS notification.	
	Report generation for medicines and services should include daily, weekly, monthly, and yearly revenue. Currently, no PDF report is generated; it should also be printable.	Added a feature where the reports generation includes daily, weekly, monthly and yearly revenue and added a printable PDF report
	Add inventory item – should have the same UI.	Added the inventory the the main UI
	When adding a new branch, there should be a feature that allows them to just import their existing products.	Added an Import and export option

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Nabua, Camarines Sur
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	Add in multi-branch in settings update and delete.	Added update and delete
	The system was not deployed and was running on localhost during FOD.	It was deployed during the FOD but we chose to present it on localhost due to webhost issue that time and internet problem

NOTED BY:

BRENDA D. BENOSA, MIS
PANEL, CHAIRMAN

MHELROSE B. PRADES, MIT
PANEL, MEMBER 1

JAYVEE NIEL S. SIAS
PANEL, MEMBER 2

APPENDIX M
CONSULTATION LOG FORM

CONSULTATION LOGS FORM

Capstone Project Title:	Development of Integrated Dental Practice System with Point-of-Sale Integration and Business Analytics
Proponents:	1. Alexander Edrian O. Sabroso 2. Janine R. Comploma 3. Ma. Crizel H. Tucay
Alias:	Byblis Liniflora

Total # of Modules:				as approved by the CAPSA /TSA:			
PROTOTYPE	Date of Consultation	# of Modules Fully Implemented	# of Modules Partially Implemented	Running Score	Percentage	Project Manager's Signature	CAPA/CAPSA/ Consultant/ Grammarian Signature
Chapter 1 Draft	2/14/23					[signature]	[signature]
	Remarks: - Double Check the content of Chapter 1 Introduction. - Project Context is AI Generated - Please try to read completed / finished Capstone Project so that you will have an idea on how each part should look like						
Deadline:							
Chapter 1 Draft							
	Remarks:						
Deadline:							

CONSULTATION LOGS FORM

Capstone Project Title:	Development of Integrated Dental Practice System with Point-of-Sale and Business Analytics						
Proponents:	1. Alexander Edrian O. Sabroso 2. Janine R. Comploma 3. Ma. Crizel H. Tucay						
Alias:	Byblis Liniflora						
Total # of Modules:				**as approved by the CAPSA /TSA:**			
PROTOTYPE	**Date of Consultation**	**# of Modules Fully Implemented**	**# of Modules Partially Implemented**	**Running Score**	**Percentage**	**Project Manager's Signature**	**~~CAPA~~/CAPSA/ Consultant/ Grammarian Signature**
Chapter 1	4/22/25						
	Remarks: - Purpose & Description looks AI Generated						
Deadline:							
Chapter 1							
	Remarks:						
Deadline:							

CONSULTATION LOGS FORM

Capstone Project Title:	Development of Integrated Dental Practice System with Point-of-Sale Integration and Business Analytics						
Proponents:	1. Alexander Edrian O. Sabroso 2. Janine R. Comploma 3. Ma. Crizel H. Tucay						
Alias:	Byblis Liniflora						
Total # of Modules:				**as approved by the CAPSA /TSA:**			
PROTOTYPE	**Date of Consultation**	**# of Modules Fully Implemented**	**# of Modules Partially Implemented**	**Running Score**	**Percentage**	**Project Manager's Signature**	**~~CAPA~~/CAPSA/ Consultant/ Grammarian Signature**
Chapter 1	5/28/25						
	Remarks: - New module was added - Client was changed						
Deadline:							
	Remarks:						
Deadline:							

*** Must attach with this form your chosen Data and Process Model (IT) or Algorithm Model (CS)

CONSULTATION LOGS FORM

Capstone Project Title:	Development of Integrated Dental Practice System with Point-of-Sale Integration and Business Analytics
Proponents:	1. Alexander Edrian O. Sabroso 2. Janine R. Comploma 3. Ma. Crizel H. Tucay
Alias:	Byblis Liniflora

Total # of Modules:				**as approved by the CAPSA /TSA:**			
PROTOTYPE	**Date of Consultation**	**# of Modules Fully Implemented**	**# of Modules Partially Implemented**	**Running Score**	**Percentage**	**Project Manager's Signature**	**CAPA/CAPSA/ Consultant/ Grammarian Signature**
Chapter 1	4/23/25					[signature]	[signature]
	Remarks: Good to go						
Deadline:							
	Remarks:						
Deadline:							

*** Must attach with this form your chosen Data and Process Model (IT) or Algorithm Model (CS)

CONSULTATION LOGS FORM

Capstone Project Title:	Development of Integrated Dental Practice System with Point-of-Sale Integration and Business Analytics						
Proponents:	1. Alexander Edrian O. Sabroso 2. Janine R. Comploma 3. Ma. Crizel H. Tucay						
Alias:	Byblis Liniflora						
Total # of Modules:				**as approved by the CAPSA /TSA:**			
PROTOTYPE	**Date of Consultation**	**# of Modules Fully Implemented**	**# of Modules Partially Implemented**	**Running Score**	**Percentage**	**Project Manager's Signature**	**CAPA/CAPSA/ ~~Consultant~~/ Grammarian Signature**
Chapter 2 Draft	3/12/25					[signature]	[signature]
	Remarks: - The system / study will add more functionalities, therefore you must also revise the RRL, synthesis & SAP						
Deadline:							
Chapter 2 Draft							
	Remarks:						
Deadline:							

CONSULTATION LOGS FORM

Capstone Project Title:	Development of Integrated Dental Practice System with Point-of-Sale Integration and Business Analytics
Proponents:	1. Alexander Edrian O. Sabroso 2. Janine R. Comploma 3. Ma. Crizel H. Tucay
Alias:	Byblis Liniflora

Total # of Modules:				**as approved by the CAPSA /TSA:**			
PROTOTYPE	**Date of Consultation**	**# of Modules Fully Implemented**	**# of Modules Partially Implemented**	**Running Score**	**Percentage**	**Project Manager's Signature**	**CAPA/CAPSA/ ~~Consultant~~/ Grammarian Signature**
Research Istrument	4/22/25						
	Remarks: Good to go						
Deadline:							
	Remarks:						
Deadline:							

*** Must attach with this form your chosen Data and Process Model (IT) or Algorithm Model (CS)

CONSULTATION LOGS FORM

Capstone Project Title:	Development of Integrated Dental Practice System with Point-of-Sale Integration and Business Analytics						
Proponents:	1. Alexander Edrian O. Sabroso 2. Janine R. Comploma 3. Ma. Crizel H. Tucay						
Alias:	Byblis Liniflora						
Total # of Modules:				**as approved by the CAPSA /TSA:**			
PROTOTYPE	**Date of Consultation**	**# of Modules Fully Implemented**	**# of Modules Partially Implemented**	**Running Score**	**Percentage**	**Project Manager's Signature**	**CAPA/CAPSA/ Consultant/ Grammarian Signature**
Chapter 2	4/23/25						
	Remarks: Revise Synthesis & Gap						
Deadline:							
	Remarks:						
Deadline:							

*** Must attach with this form your chosen Data and Process Model (IT) or Algorithm Model (CS)

CONSULTATION LOGS FORM

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Proponents:	1. Alexander Edrian O. Sabroso 2. Janine R. Comploma 3. Ma. Crizel H. Tucay						
Alias:	Byblis Liniflora						
Total # of Modules:				**as approved by the CAPSA /TSA:**			
PROTOTYPE	**Date of Consultation**	**# of Modules Fully Implemented**	**# of Modules Partially Implemented**	**Running Score**	**Percentage**	**Project Manager's Signature**	**CAPA/CAPSA/ Consultant/ Grammarian Signature**
Chapter 2	4/28/25						
	Remarks: Good [illegible]						
Deadline:							
	Remarks:						
Deadline:							

*** Must attach with this form your chosen Data and Process Model (IT) or Algorithm Model (CS)

CONSULTATION LOGS FORM

Capstone Project Title:	Development of Integrated Dental Practice System with Point-of-Sale Integration and Business Analytics						
Proponents:	1. Alexander Edrian O. Sabroso 2. Janine R. Comploma 3. Ma. Crizel H. Tucay						
Alias:	Byblis Liniflora						
Total # of Modules:				**as approved by the CAPSA /TSA:**			
PROTOTYPE	**Date of Consultation**	**# of Modules Fully Implemented**	**# of Modules Partially Implemented**	**Running Score**	**Percentage**	**Project Manager's Signature**	**CAPA/CAPSA/ Consultant/ Grammarian Signature**
Chapter 3	9/28/25						
	Remarks: -Good to Go						
Deadline:							
	Remarks:						
Deadline:							

*** Must attach with this form your chosen Data and Process Model (IT) or Algorithm Model (CS)

CONSULTATION LOGS FORM

Capstone Project Title:	Development of Integrated Dental Practice System with Point-of-Sale Integration and Business Analytics						
Proponents:	1. Alexander Edrian O. Sabroso 2. Janine R. Comploma 3. Ma. Crizel H. Tucay						
Alias:	Byblis Liniflora						
Total # of Modules:				**as approved by the CAPSA /TSA:**			
PROTOTYPE	**Date of Consultation**	**# of Modules Fully Implemented**	**# of Modules Partially Implemented**	**Running Score**	**Percentage**	**Project Manager's Signature**	**CAPA/CAPSA/ ~~Consultant~~/ Grammarian Signature**
Modules	4/29/25						
	Remarks: Good to Go						
Deadline:							
	Remarks:						
Deadline:							

*** Must attach with this form your chosen Data and Process Model (IT) or Algorithm Model (CS)

CONSULTATION LOGS FORM

Capstone Project Title:	Development of Integrated Dental Practice System with Point-of-Sale Integration and Business Analytics
Proponents:	1. Alexander Edrian O. Sabroso 2. Janine R. Comploma 3. Ma. Crizel H. Tucay
Alias:	Byblis Liniflora

Total # of Modules:				**as approved by the CAPSA /TSA:**			
PROTOTYPE	**Date of Consultation**	**# of Modules Fully Implemented**	**# of Modules Partially Implemented**	**Running Score**	**Percentage**	**Project Manager's Signature**	**CAPA/CAPSA/~~Consultant~~/ Grammarian Signature**
UI	4/29/26						
	Remarks: Good to go						
Deadline:							
	Remarks:						
Deadline:							

*** Must attach with this form your chosen Data and Process Model (IT) or Algorithm Model (CS)

CONSULTATION LOGS FORM

Capstone Project Title:	Development of Integrated Dental Practice System with Point-of-Sale Integration and Business Analytics						
Proponents:	1. Alexander Edrian O. Sabroso 2. Janine R. Comploma 3. Ma. Crizel H. Tucay						
Alias:	Byblis Liniflora						
Total # of Modules:				**as approved by the CAPSA /TSA:**			
PROTOTYPE	**Date of Consultation**	**# of Modules Fully Implemented**	**# of Modules Partially Implemented**	**Running Score**	**Percentage**	**Project Manager's Signature**	**CAPA/CAPSA/ Consultant/ Grammarian Signature**
Research Intrument	5/6/25					[signature]	[signature]
	Remarks: - Good to go						
Deadline:							
	Remarks:						
Deadline:							

*** Must attach with this form your chosen Data and Process Model (IT) or Algorithm Model (CS)

CONSULTATION LOGS FORM

Capstone Project Title:	Development of Integrated Dental Practice System with Point-of-Sale Integration and Business Analytics						
Proponents:	1. Alexander Edrian O. Sabroso 2. Janine R. Comploma 3. Ma. Crizel H. Tucay						
Alias:	Byblis Liniflora						
Total # of Modules:				**as approved by the CAPSA /TSA:**			

PROTOTYPE	**Date of Consultation**	**# of Modules Fully Implemented**	**# of Modules Partially Implemented**	**Running Score**	**Percentage**	**Project Manager's Signature**	**CAPA/CAPSA/ ~~Consultant~~/ Grammarian Signature**
Chapter 3	5/9/2025					[signature]	[signature]
	Remarks: GOOD TO GO						
Deadline:							
	Remarks:						
Deadline:							

*** Must attach with this form your chosen Data and Process Model (IT) or Algorithm Model (CS)

CONSULTATION LOGS FORM

Capstone Project Title:	Development of Integrated Dental Practice System with Point-of-Sale Integration and Business Analytics						
Proponents:	1. Alexander Edrian O. Sabroso 2. Janine R. Comploma 3. Ma. Crizel H. Tucay						
Alias:	Byblis Liniflora						
Total # of Modules:				**as approved by the CAPSA /TSA:**			
PROTOTYPE	**Date of Consultation**	**# of Modules Fully Implemented**	**# of Modules Partially Implemented**	**Running Score**	**Percentage**	**Project Manager's Signature**	**~~CAPA/~~CAPSA/ Consultant/ Grammarian Signature**
Module	6/20/24					[signature]	[signature]
	Remarks: - New Module was added						
Deadline:							
	Remarks:						
Deadline:							

*** Must attach with this form your chosen Data and Process Model (IT) or Algorithm Model (CS)

CONSULTATION LOGS FORM

Capstone Project Title:	Development of Integrated Dental Practice System with Point-of-Sale Integration and Business Analytics						
Proponents:	1. Alexander Edrian O. Sabroso 2. Janine R. Comploma 3. Ma. Crizel H. Tucay						
Alias:	Byblis Liniflora						
Total # of Modules:				**as approved by the CAPSA /TSA:**			
PROTOTYPE	**Date of Consultation**	**# of Modules Fully Implemented**	**# of Modules Partially Implemented**	**Running Score**	**Percentage**	**Project Manager's Signature**	**CAPA/CAPSA/ Consultant/ Grammarian Signature**
System	10/6/2025				302	[signature]	[signature]
	Remarks: ADD LOGIN ADD INVENTORY						
Deadline:							
System	10/17/25					[signature]	[signature]
	Remarks: GOOD TO GO						
Deadline:							

*** Must attach with this form your chosen Data and Process Model (IT) or Algorithm Model (CS)

CONSULTATION LOGS FORM

Capstone Project Title:	Development of Integrated Dental Practice System with Point-of-Sale Integration and Business Analytics						
Proponents:	1. Alexander Edrian O. Sabroso 2. Janine R. Comploma 3. Ma. Crizel H. Tucay						
Alias:	Byblis Liniflora						
Total # of Modules:				**as approved by the CAPSA /TSA:**			
PROTOTYPE	**Date of Consultation**	**# of Modules Fully Implemented**	**# of Modules Partially Implemented**	**Running Score**	**Percentage**	**Project Manager's Signature**	**CAPA/CAPSA/ Consultant/ Grammarian Signature**
DFP	10/17/25						
	Remarks: Good to go						
Deadline:							
	Remarks:						
Deadline:							

*** Must attach with this form your chosen Data and Process Model (IT) or Algorithm Model (CS)

CONSULTATION LOGS FORM

Capstone Project Title:	Development of Integrated Dental Practice System with Point-of-Sale Integration and Business Analytics						
Proponents:	1. Alexander Edrian O. Sabroso 2. Janine R. Comploma 3. Ma. Crizel H. Tucay						
Alias:	Byblis Liniflora						
Total # of Modules:				**as approved by the CAPSA /TSA:**			
PROTOTYPE	**Date of Consultation**	**# of Modules Fully Implemented**	**# of Modules Partially Implemented**	**Running Score**	**Percentage**	**Project Manager's Signature**	**CAPA/CAPSA/ Consultant/ Grammarian Signature**
Chapter 4	10/22/25						
	Remarks: - Content of Chapter 4 is not the same as the SPM						
Deadline:							
Chapter 5	10/22/25						
	Remarks: - NO RESULT						
Deadline:							

***Must attach with this form your chosen Data and Process Model (IT) or Algorithm Model (CS)

CONSULTATION LOGS FORM

Capstone Project Title:	Development of Integrated Dental Practice System with Point-of-Sale Integration and Business Analytics
Proponents:	1. Alexander Edrian O. Sabroso 2. Janine R. Comploma 3. Ma. Crizel H. Tucay
Alias:	Byblis Liniflora
Total # of Modules:	as approved by the CAPSA /TSA:

PROTOTYPE	Date of Consultation	# of Modules Fully Implemented	# of Modules Partially Implemented	Running Score	Percentage	Project Manager's Signature	CAPA/CAPSA/ Consultant/ Grammarian Signature
Chapter 4	10/23/28					Alexander	[illegible]
	Remarks: - Good to go						
Deadline:							
Chapter 5	10/23/20					Alexander	[illegible]
	Remarks: - [illegible] based from what we discussed						
Deadline:							

*** Must attach with this form your chosen Data and Process Model (IT) or Algorithm Model (CS)

CONSULTATIONLOGSFORM

Capstone ProjectTitle:	DEVELOPMENT OF MULTI-BRANCH DENTAL PRACTICE MANAGEMENT SYSTEM WITH POINT-OF-SALE AND BUSINESS ANALYTICS FOR DR. WENNIE A. SAMPER DENTAL CLINIC
Proponents:	1. Alexander Edrian O.Sabroso 2. Janine R.Comploma 3. Ma.Crizel H. Tucay
Alias:	Byblis Liniflora

Total#of Modules:				**As approved by the CAPSA /TSA:**			
PROTOTYPE	**Date of Consultation**	**# of Modules Fully Implemented**	**# of Modules Partially Implemented**	**Running Score**	**Percentage**	**Project Manager's Signature**	**CAPA/CAPSA/ Consultant/ Grammarian Signature**
for Bluebook	1/20/24						
	Remarks: - For Bluebook						
Deadline:							
	Remarks:						
Deadline:							

CONSULTATION LOGS FORM

Capstone Project Title:	**DEVELOPMENT OF MULTI-BRANCHED DENTAL PRACTICE SYSTEM WITH POINT-OF-SALE INTEGRATION AND BUSINESS ANALYTICS**						
Proponents:	1. Aelexander Edrian O. Sabroso 2. Janine R. Comploma 3. Ma. Crizel H. Tucay						
Alias:	Byblis Liniflora						
Total # of Modules:				**as approved by the CAPSA /TSA:**			
PROTOTYPE	**Date of Consultation**	**# of Modules Fully Implemented**	**# of Modules Partially Implemented**	**Running Score**	**Percentage**	**Project Manager's Signature**	**CAPA/CAPSA/ Consultant/ Grammarian Signature**
Bluebook Checking	01/29/26						
	Remarks:						
	Recommended for Bluebook						
Deadline:							
	Remarks:						

*** Must attach with this form your chosen Data and Process Model (IT) or Algorithm Model (CS)

APPENDIX N

SECRETARY CERTIFICATE

SECRETARY'S CERTIFICATION

This is to certify that the undersigned has provided true and correct recommendations, suggestions, and comments unanimously agreed and approved by the panel of examiners during the oral examination of the capstone project entitled "**DEVELOPMENT OF MULTI-BRANCH DENTAL PRACTICE MANAGEMENT SYSTEM WITH POINT-OF-SALE AND BUSINESS ANALYTICS FOR DR. WENNIE A. SAMPER DENTAL CLINIC**" prepared and submitted by Janine R. Comploma, Alexander Edrian O. Sabroso, Ma. Crizel H. Tucay and that the same have not been amended, modified or obliterated.

Signed:

JOHNMIR R. ISAAC

TD Secretary

Date signed: 1/14/25

JOHN PETER G. ANDALIS

POD Secretary

Date signed: 1/20/2026

MARIETON L. RAMIZARES

FOD Secretary

Date signed: 02/03/2026

APPENDIX O
GRAMMARIAN CERTIFICATE

GRAMMARIAN'S CERTIFICATION

This is to certify that the undersigned has reviewed and went through all the pages of the capstone project entitled "**DEVELOPMENT OF MULTI-BRANCH DENTAL PRACTICE MANAGEMENT SYSTEM WITH POINT-OF-SALE AND BUSINESS ANALYTICS FOR DR. WENNIE A. SAMPER DENTAL CLINIC**" as against the set of structural rules that govern research writing in accord with the composition of *sentences*, *phrases*, and *words* in the English language.

Signed:

NEL FRANCIS B. BUENA, MLL
Grammarian

Date signed: 02/02/2026

APPENDIX P

CERTIFICATE OF TRANSFER

CERTIFICATE

OF TRANSFER

This to certify that

DEVELOPMENT OF MULTI-BRANCH DENTAL PRACTICE MANAGEMENT SYSTEM WITH POINT-OF-SALE AND BUSINESS ANALYTICS FOR DR.WENNIE A. SAMPER DENTAL CLINIC

Developed by Janine R. Comploma, Alexander Edrian O. Sabroso and Ma. Crizel H. Tucay, Camarines Sur Polytechnic Colleges (CSPC) - College of Computer Studies (CCS) students, as their undergraduate thesis project was transfer to the Dr. Wennie A. Samper Dental Clinic for further research.

This certification is issued this Feb 16, 2026 at College Of Computer Studies, CSPC for whatever legitimate purpose it may serve.

ROSEL O. ONESA, MIT
OIC DEAN, CCS

APPENDIX Q
CERTIFICATE OF UTILIZATION

Dr. Wennie Samper
Dental Clinic

CERTIFICATE
OF UTILIZATION

This to certify that

DEVELOPMENT OF MULTI-BRANCH DENTAL PRACTICE MANAGEMENT SYSTEM WITH POINT-OF-SALE AND BUSINESS ANALYTICS FOR DR.WENNIE A. SAMPER DENTAL CLINIC

Developed by Janine R. Comploma, Alexander Edrian O. Sabroso and Ma. Crizel H. Tucay, Camarines Sur Polytechnic Colleges (CSPC) - College of Computer Studies (CCS) students, as their undergraduate thesis project was adopted and utilized by Wennie A. Samper from May 22, 2025 to present date.

This certification is issued this 15th day of January 2026 at Dr. Wennie Samper Dental Clinic for whatever legitimate purpose it may serve.

Wennie A. Samper
Dentist

APPENDIX R
ACM FORMAT

Development of a Multi-Branch Dental Practice Management System with Point-of-Sale and Business Analytics for Dr. Wennie A. Samper Dental Clinic

DWAS Dental System

A Web-Based Multi-Branched Dental Clinic Management with POS and Analytics

Sabroso, Alexander Edrian O.

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Comploma, Janine R.

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Tucay, Ma. Crizel H.

Camarines Sur **Polytechnic** Colleges, crtucay@my.cspc.edu.ph

ABSTRACT

The increasing demand for efficient dental healthcare services highlights the need for integrated information systems that support both clinical and administrative operations. Dr. Wennie A. Samper Dental Clinic, a growing multi-branch dental practice, previously relied on manual and fragmented processes for patient records, appointment scheduling, and billing, which resulted in operational inefficiencies and limited access to business insights. To address these challenges, this study developed a web-based Multi-Branch Dental Practice Management System with integrated Point-of-Sale (POS) and Business Analytics. The system was developed using the Agile Scrum methodology and includes modules for patient management, electronic dental charting, appointment scheduling, POS billing, inventory control, role-based access, and analytical reporting. System evaluation employed Black Box Testing techniques covering functionality, usability, performance, and security. Findings indicate that the system is reliable, user-friendly, and capable of supporting daily clinic operations. The implementation improved workflow efficiency, minimized administrative errors, and enabled data-driven decision-making across clinic branches.

CCS CONCEPTS • Health information systems • Dental clinic management • Point-of-sale systems • Business analytics

Additional Keywords and Phrases: Additional Keywords and Phrases: Dental Practice Management System, POS Integration, Business Analytics, Agile Scrum, Black Box Testing

ACM Reference Format:

Alexander Edrian O. Sabroso, Janine R. Comploma, Ma.Crizel H. Tucay. 2025. Development of a Multi-Branch Dental Practice Management System with Point-of-Sale and Business Analytics for Dr. Wennie A. Samper Dental Clinic.

1. INTRODUCTION

Dental clinics depend on accurate patient records, efficient scheduling, and reliable billing to deliver quality care. Despite advances in health information technology, many small to medium-sized dental practices continue to rely on paper-based or partially digital processes. These practices lead to delayed transactions, inconsistent records, and limited visibility into business performance. The challenges become more pronounced when clinics expand into multiple branches, where coordination and centralized reporting are critical.

Dr. Wennie A. Samper Dental Clinic, established in 2019, experienced these operational limitations as patient volume increased and additional branches were opened. Manual record-keeping and separate billing processes made it difficult to track patient histories, monitor revenue, and manage appointments efficiently. Commercial dental systems were evaluated but found to be either cost-prohibitive or misaligned with the clinic's workflow.

This study addresses these gaps by developing an integrated dental practice management system that consolidates clinical, administrative, and financial functions into a single platform. By incorporating POS and business analytics features, the system supports informed decision-making and improves operational efficiency across branches.

2. BACKGROUND

Digital management systems have been shown to improve efficiency and accuracy in healthcare and dental settings. Prior studies report that POS integration reduces billing errors and improves transaction transparency, while analytics tools enable organizations to monitor performance trends and resource utilization.

Existing dental systems typically focus on patient records and appointment scheduling but often lack comprehensive support for multi-branch operations and integrated analytics. Other research emphasizes the importance of security and access control in healthcare systems, noting that weak safeguards may compromise patient confidentiality. The present study builds on these findings by delivering a unified system that integrates patient management, POS billing, analytics, and workflow coordination within a secure, role-based environment.

3. METHODOLOGY

This study adopted the Agile Scrum methodology to guide system development. The process included planning, design, development, sprint review, sprint retrospective, and deployment. Requirements were gathered through interviews and observation of clinic workflows to ensure alignment with real-world operations.

3.1 Software Development Methodology

Software Development Methodology Agile Scrum was selected for its iterative approach and ability to accommodate evolving requirements. Development was divided into sprints, each producing a functional increment of the system. Continuous stakeholder feedback ensured that each iteration addressed practical clinic needs.

3.2 SYSTEM ARCHITECTURE

The system follows a centralized web-based architecture with a relational database that stores patient records, appointments, billing transactions, and inventory data. Role-based access control restricts system functions according to user roles such as administrator, dentist, and staff. Security mechanisms include authentication, encrypted data handling, and audit logging. Embedded analytics provide visual reports on revenue, patient volume, and service utilization.

3.3 Sampling and Evaluation

Sampling and Evaluation System evaluation involved clinic staff and administrators who interacted with the system in a controlled environment. Participants were selected based on their roles and familiarity with clinic operations to ensure relevant feedback.

3.4 Instrument

A researcher-developed questionnaire based on software quality attributes was used to evaluate functionality, usability, performance, reliability, and security. Responses were measured using a criterion scale to determine system readiness.

3.5 Testing Plan

Black Box Testing was employed to assess the system without reference to internal code structure. Testing included unit testing to validate individual modules, user acceptance testing to confirm compliance with user requirements, and performance testing to evaluate stability under concurrent access.

3.6 Statistical Treatment of Data

Descriptive statistics, specifically the arithmetic mean, were used to analyze evaluation results and determine overall system acceptance.

Table 1: Level of Performance

RESULT	PERFORMANCE LEVEL	DESCRIPTION
1.00	Optimal Performance	100% – Optimal Performance: Response times are instantaneous; resources fully utilized; system handles maximum load flawlessly.
0.67 - 0.99	High Performance	67–99% – High Performance: Excellent response and resource use; minor optimizations possible; production-ready.
0.34 – 0.66	Moderate Performance	34–66% – Moderate Performance: Acceptable under normal load; performance degrades under stress; optimization needed.
0.01 – 0.33	Poor Performance	1–33% – Poor Performance: Slow responses; frequent timeouts or crashes; major improvements required.
0.00	No Performance	0% – No Performance: System fails to execute; unresponsive under any load; complete redesign required.

4. RESULTS AND DISCUSSION

Testing confirmed that all major modules functioned as intended. Users successfully managed patient records, scheduled appointments, processed billing transactions, and generated reports without system errors. Usability results indicated that the interface was intuitive and required minimal training.

4.1 Testing

This section presents the evaluation results of the *Multi-Branch Dental Practice Management System with Point-of-Sale and Business Analytics for Dr. Wennie A. Samper Dental Clinic*. Since the system plays a critical role in managing patient records, appointments, billing transactions, and analytical reporting across multiple clinic branches, its operational reliability and correctness were validated through systematic testing. Black box testing was conducted to assess whether the system's modules performed according to their specified functions without examining internal code structures.

Table 2: System's Level of Functionality Result on Unit Testing

UNIT TESTING	PASSED

As presented in Table 2, the results of the unit testing indicate that the system successfully passed all required functionality checks, confirming that each core module operated according to its intended design. The overall functionality score of 0.99, classified as Highly Functional, demonstrates that the system meets the essential operational requirements for deployment in a multi-branch dental clinic environment. Modules related to patient records management, POS billing, and user access control achieved complete functionality, reflecting the system's ability to handle critical clinical and financial processes without errors. Although minor variations were observed in modules involving scheduling, inventory updates, and analytics reporting, these did not compromise system stability or correctness. Overall, the unit testing results verify that the system is reliable, cohesive, and capable of supporting daily clinic operations, thereby indicating that the system has successfully passed unit testing and is ready for subsequent acceptance and performance evaluations.

5. CONCLUSION

The study identified operational inefficiencies and data management challenges in the existing processes of Dr. Wennie A. Samper Dental Clinic, particularly those arising from manual record-keeping, fragmented appointment scheduling, and separate billing workflows across multiple branches. These limitations resulted in delayed transactions, data inconsistencies, and restricted access to timely business information. To address these concerns, the study developed a **Multi-Branch Dental Practice Management System with integrated Point-of-Sale and Business Analytics**, consolidating clinical, administrative, and financial functions into a single web-based platform.

System testing confirmed that all major modules operated according to their intended purpose. Unit testing results indicated that the system is **Highly Functional**, demonstrating reliable performance in patient records management, appointment scheduling, POS billing, inventory control, and analytical reporting. The results validate that the system supports daily clinic operations effectively, enhances workflow efficiency, reduces administrative errors, and enables data-driven decision-making. These findings confirm that the developed system is suitable for real-world deployment in a multi-branch dental clinic environment.

For future development, the study recommends extending the system through mobile application support to improve accessibility, integrating cloud-based deployment for scalability, and incorporating advanced analytics features to further enhance decision support. Additional improvements may also include insurance claim processing, predictive reporting, and further security enhancements to strengthen data protection.

ACKNOWLEDGMENTS

The researchers would like to express their sincere gratitude to everyone who contributed to the successful completion of this study. Special thanks are extended to Dr. Wennie A. Samper, for granting permission to conduct the study and for providing valuable insights into the clinic's operational processes. The researchers are deeply grateful to their Capstone Project Adviser and Subject Adviser for their continuous guidance, constructive feedback, and unwavering support throughout the development of the system. Appreciation is also given to the panel members for their valuable suggestions and technical expertise, which greatly improved the quality of this research. Finally, the researchers would like to thank their families, friends, and classmates for their moral encouragement and support, and above all, they give thanks to Almighty God for the wisdom and strength to complete this study.

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[8] Republic of the Philippines. (2012). *Data Privacy Act of 2012 (Republic Act No. 10173)*. Official Gazette.

APPENDIX S

CERTIFICATE OF PLAGIARISM CHECK

REPUBLIC OF THE PHILIPPINES
CAMARINES SUR POLYTECHNIC COLLEGES
NABUA, CAMARINES SUR
RESEARCH AND DEVELOPMENT SERVICES OFFICE
RESEARCH@CSPC.EDU.PH CSPC.RESEARCHOFFICE

CERTIFICATION

Date of Release: January 28, 2026

Submission ID: 23367:127025568

Output Title: DEVELOPMENT OF MULTI-BRANCH DENTAL PRACTICE MANAGEMENT SYSTEM WITH POINT-OF-SALE AND BUSINESS ANALYTICS FOR DR. WENNIE A. SAMPER DENTAL CLINIC

Author(s): Janine R. Comploma
Alexander Edrian O. Sabroso
Ma. Crizel H. Tucay

Program: Bachelor of Science in Information Technology

Authenticity Report:

Similarity Index Report: **8%**

Internet Sources: 2%
Publications: 0%
Student Papers: 7%

- *Interpretation:* The similarity index report means that **8%** of the output is similar to the sources in Turnitin's (authenticity software) online repository and databases.

AI-Generated Report: ***32%***

AI-generated only: 32%
AI-generated text that was AI-paraphrased: 0%

- *Interpretation:* The AI generated report means that **32%** of the output indicates a *likely* AI-generated text and *likely* AI-paraphrased text, which is ***within the acceptable AI usage*** in the field of Engineering.

HAROLD JAN R. TERANO, Ph.D.
Director, Research & Development Services Office